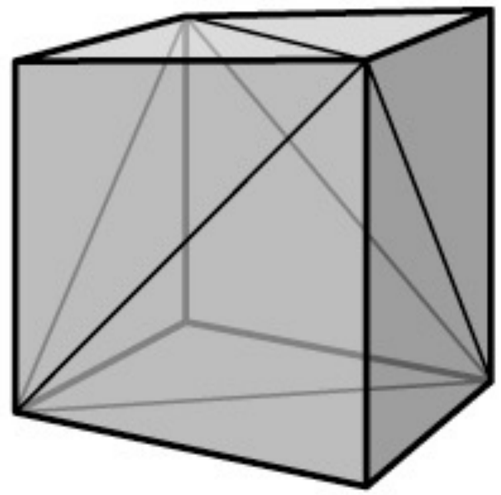
Lecture 8:

Mesh Representations & Geometry Processing

Computer Graphics and Imaging
UC Berkeley CS184/284A

Mesh Examples

A Small Triangle Mesh



8 vertices, 12 triangles

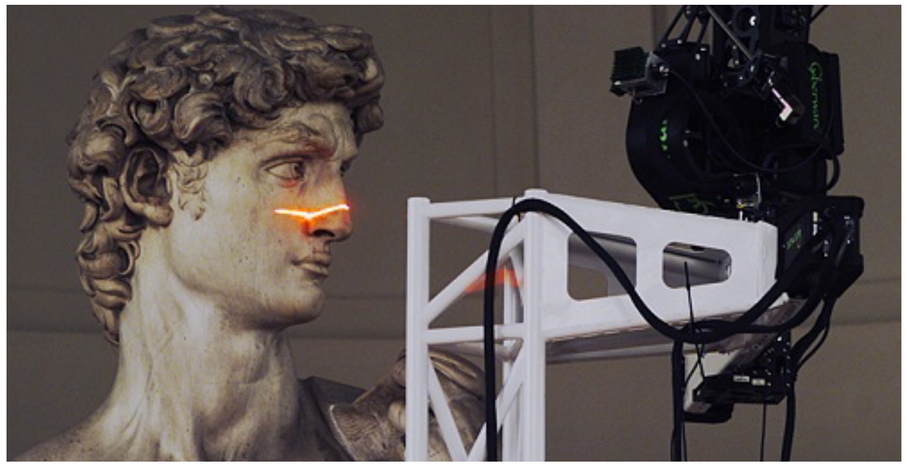
A Large Triangle Mesh

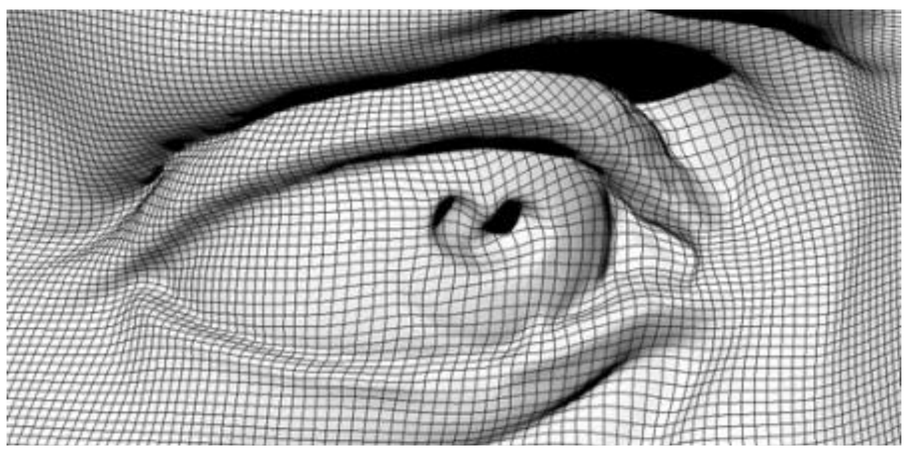
David

Digital Michelangelo Project

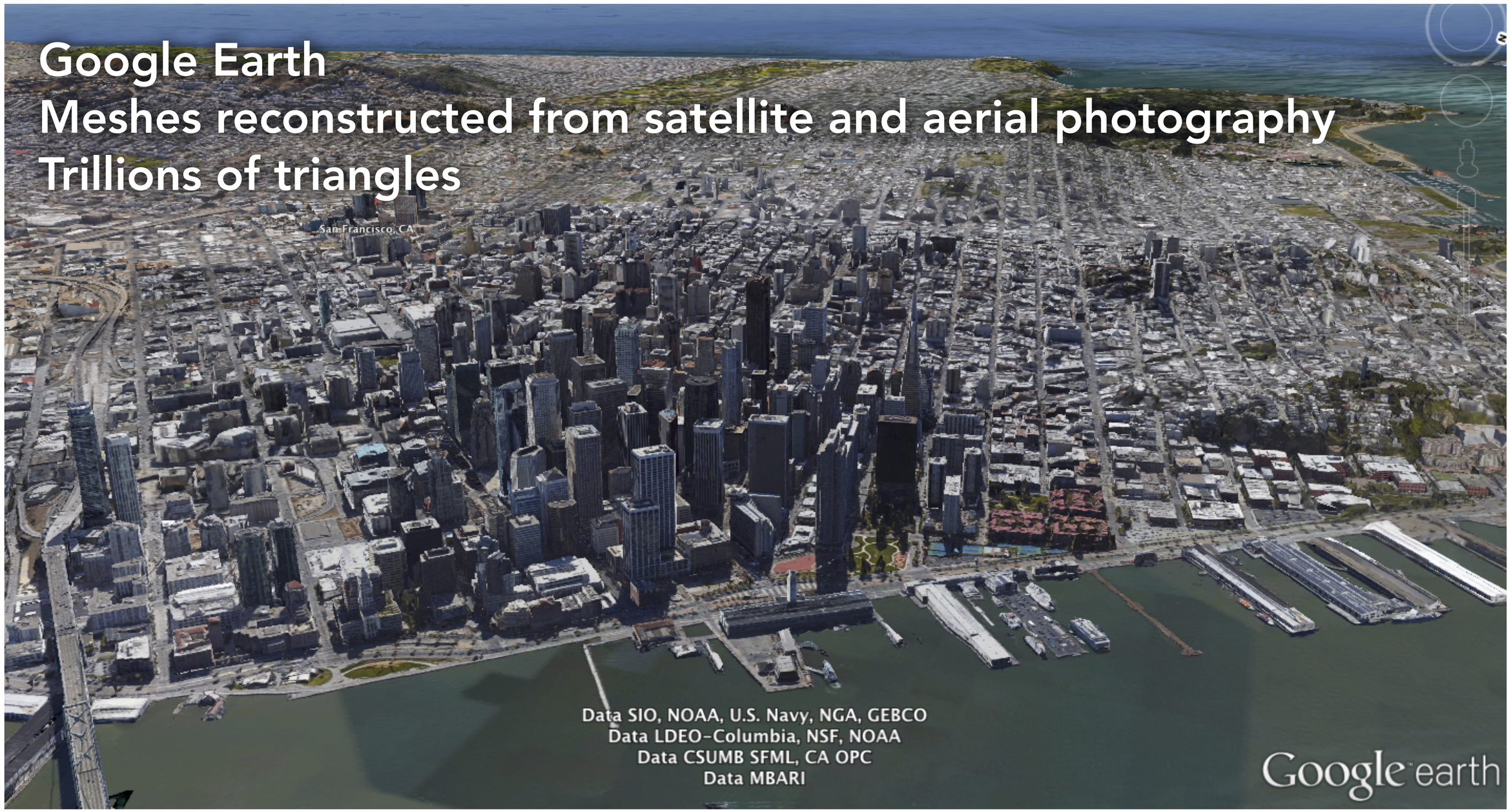
28,184,526 vertices

56,230,343 triangles

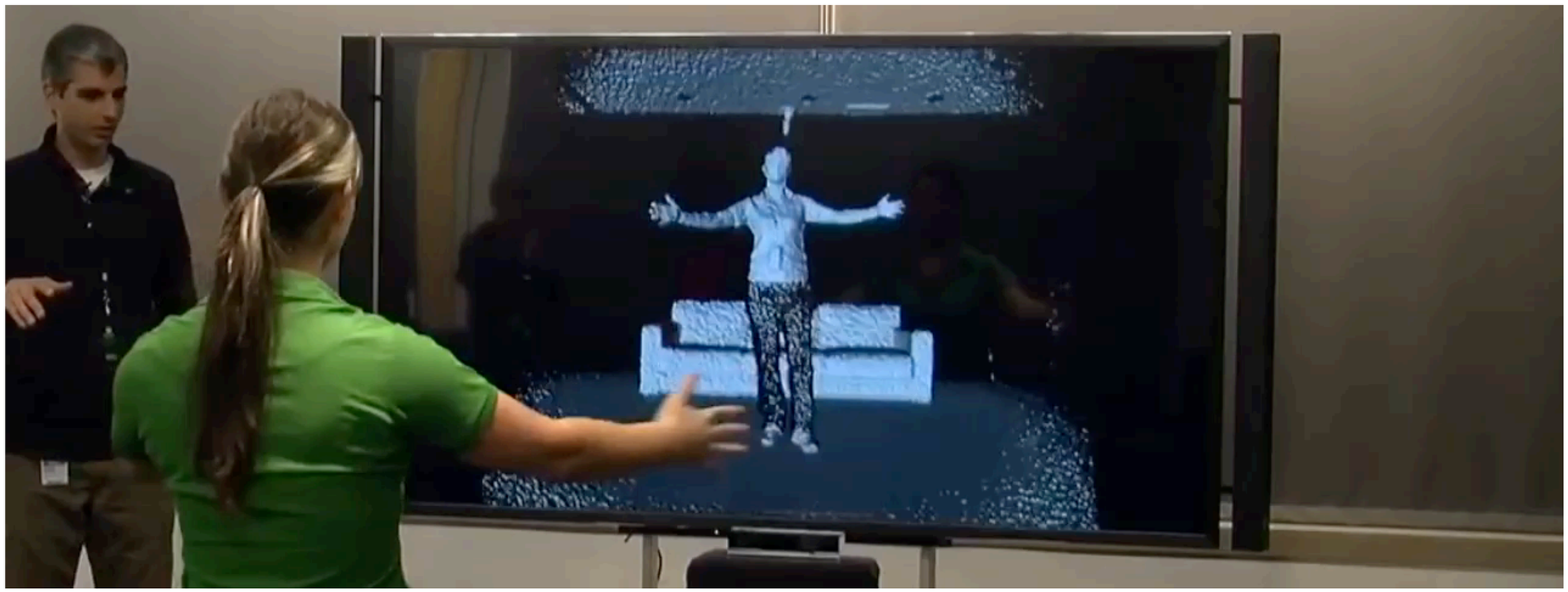




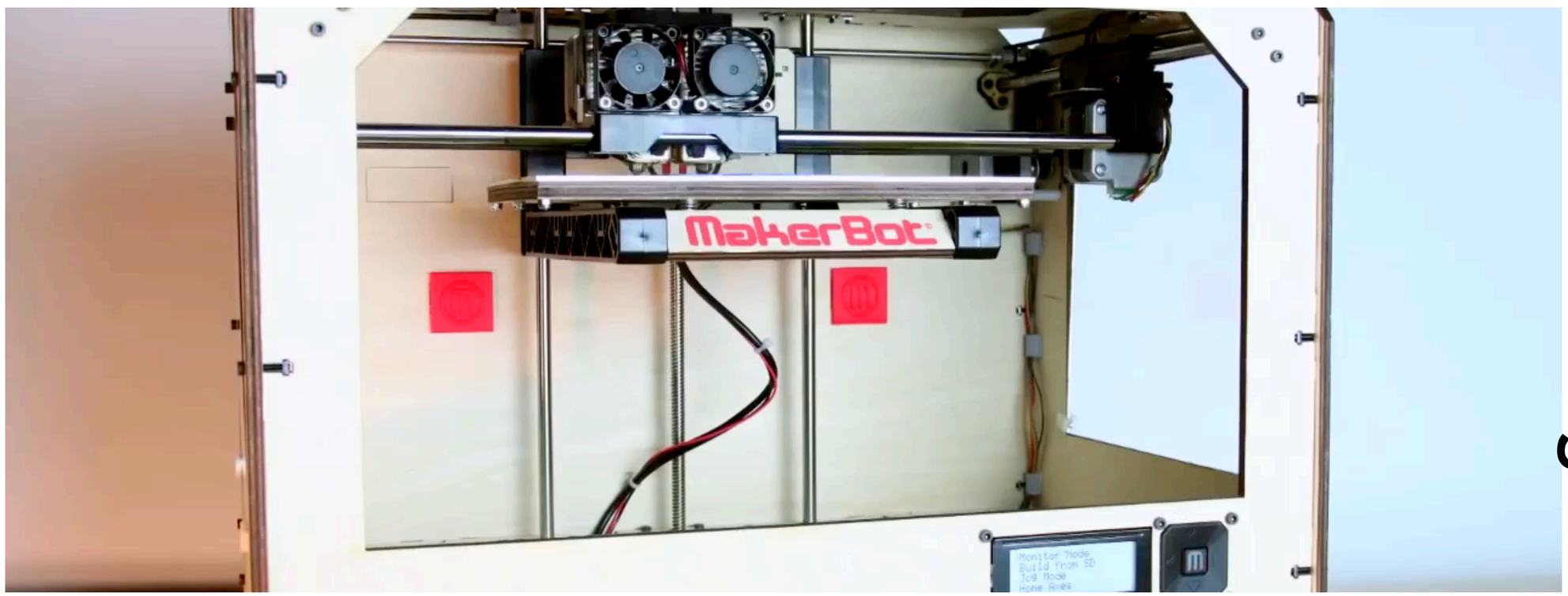
A Very Large Triangle Mesh



Digital Geometry Processing

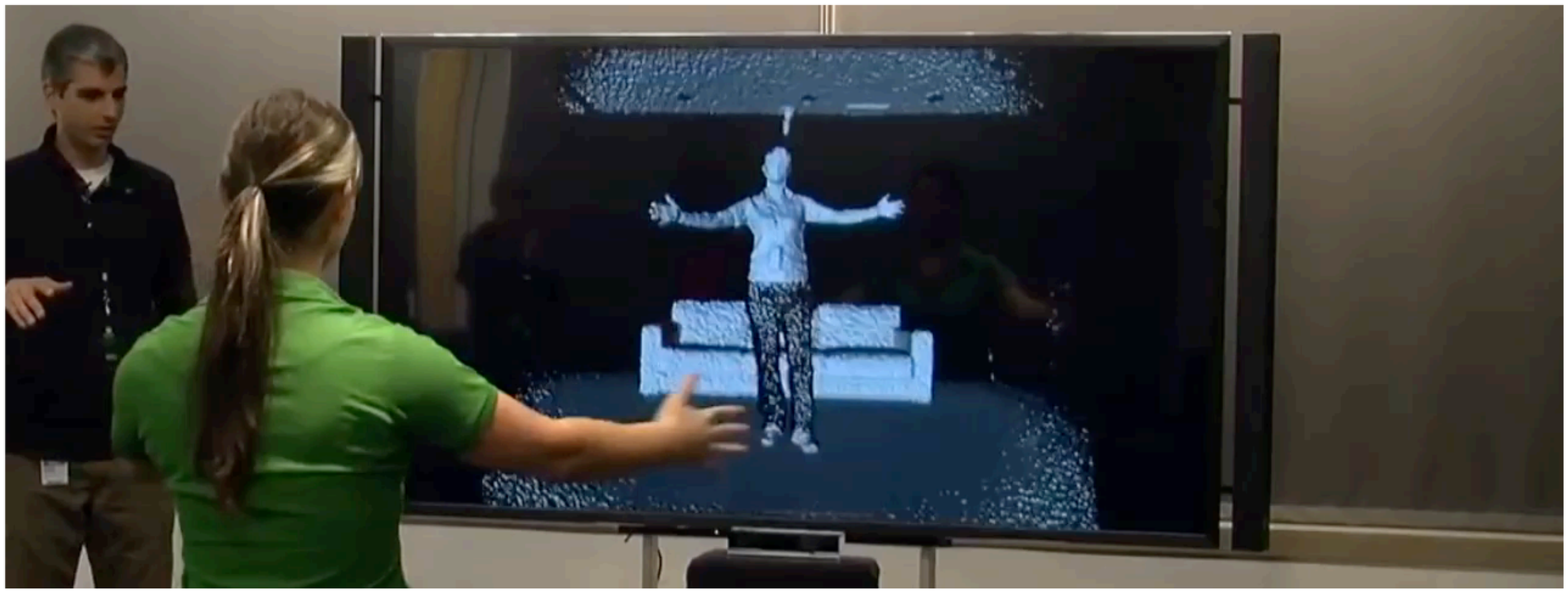


3D Scanning

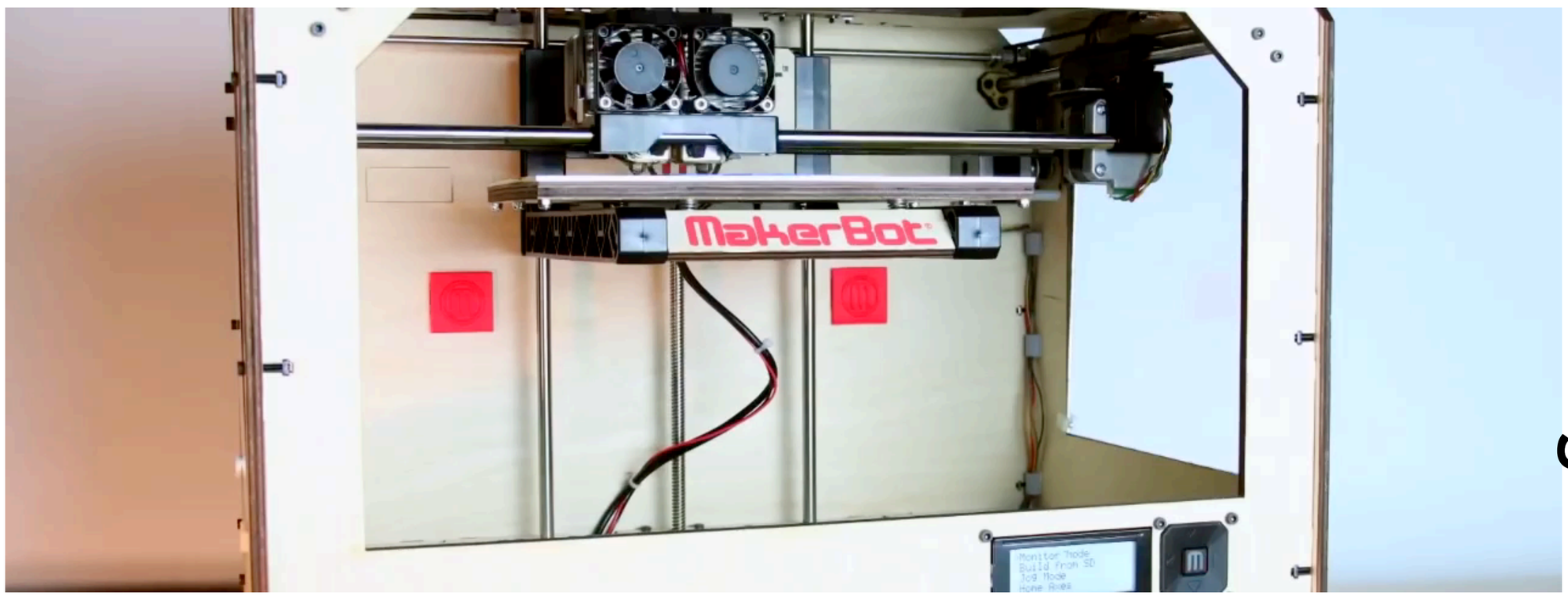


3D Printing

Digital Geometry Processing



3D Scanning

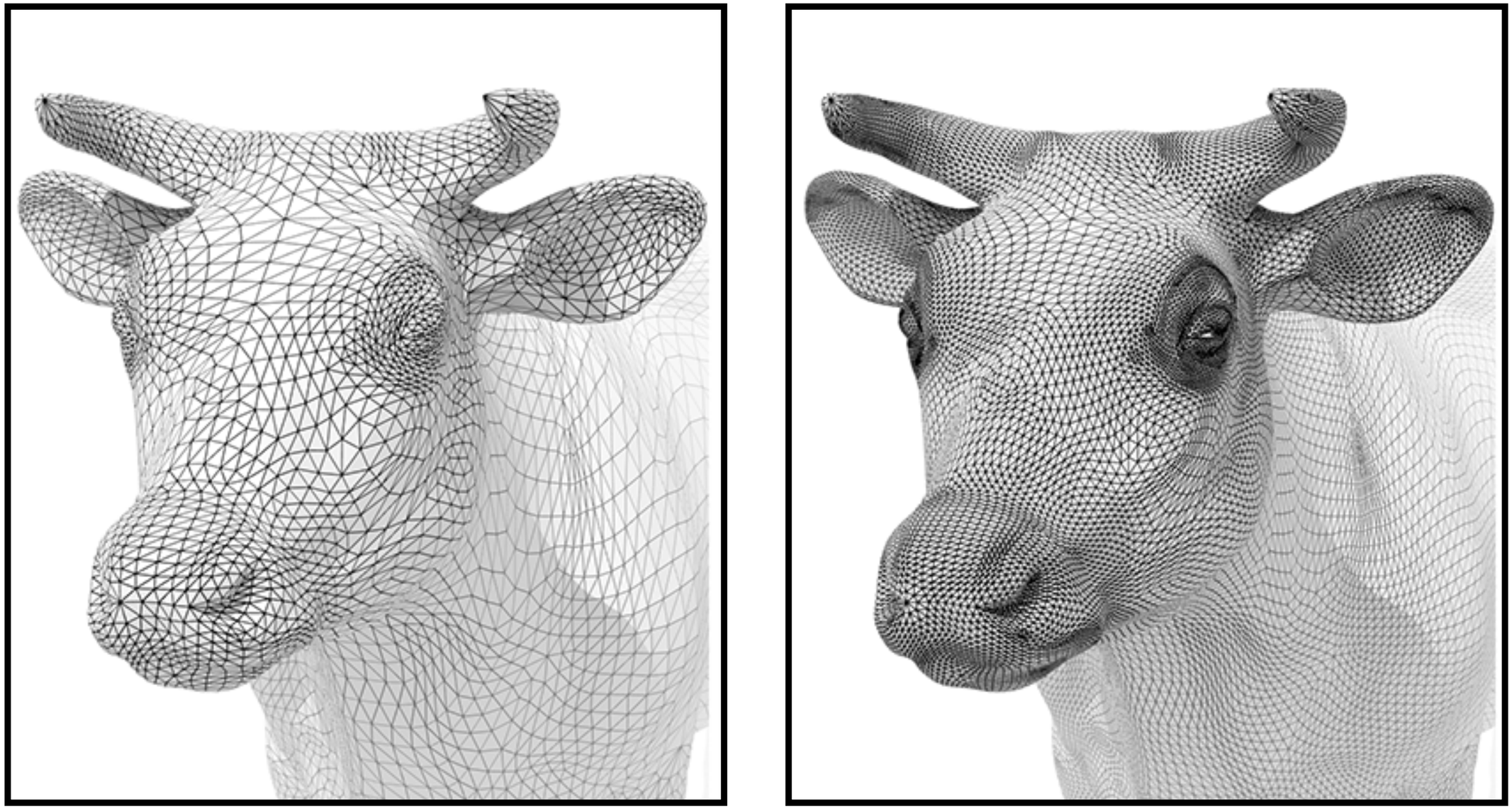


3D Printing

Geometry Processing

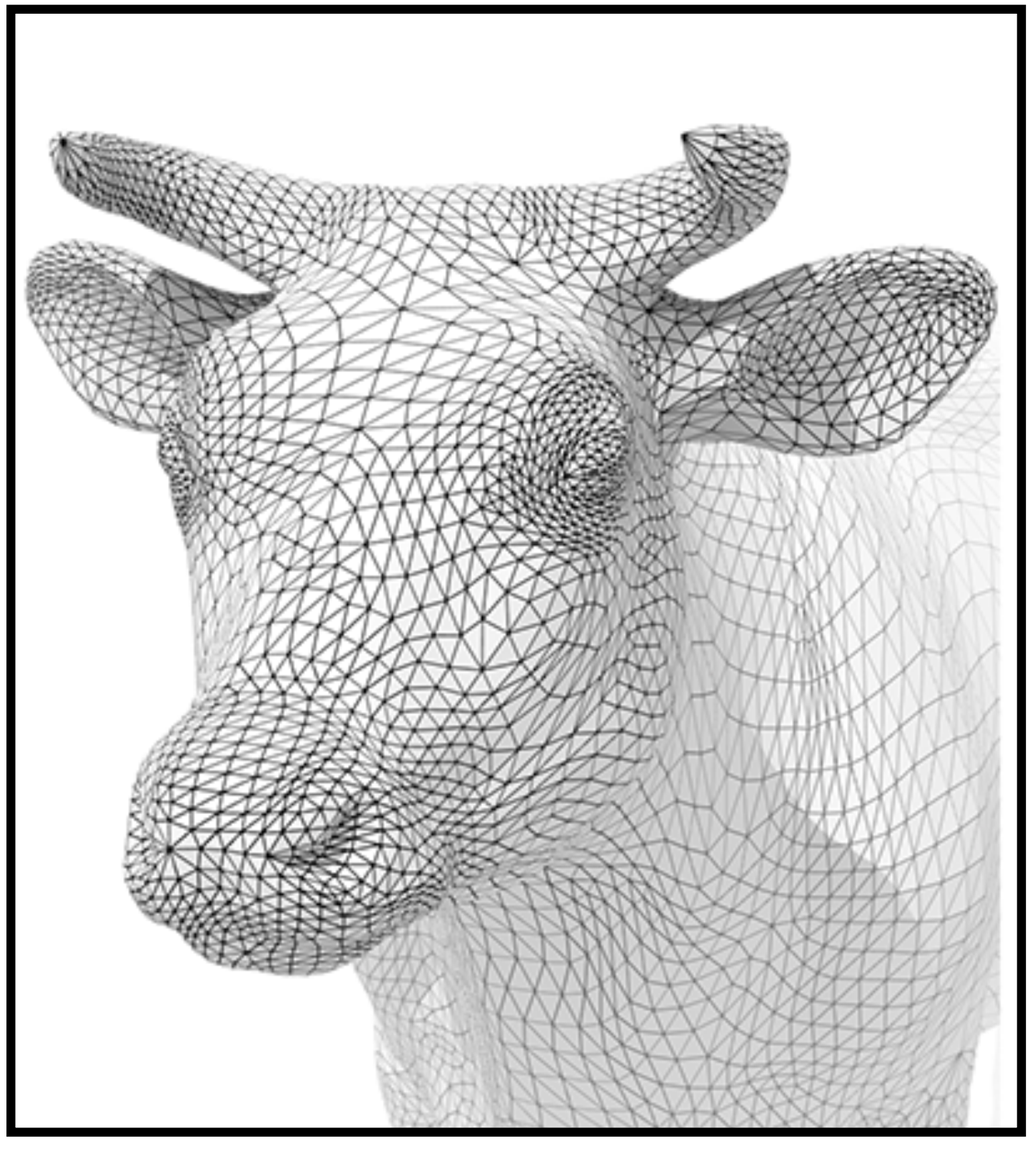
Tasks: 3 Examples

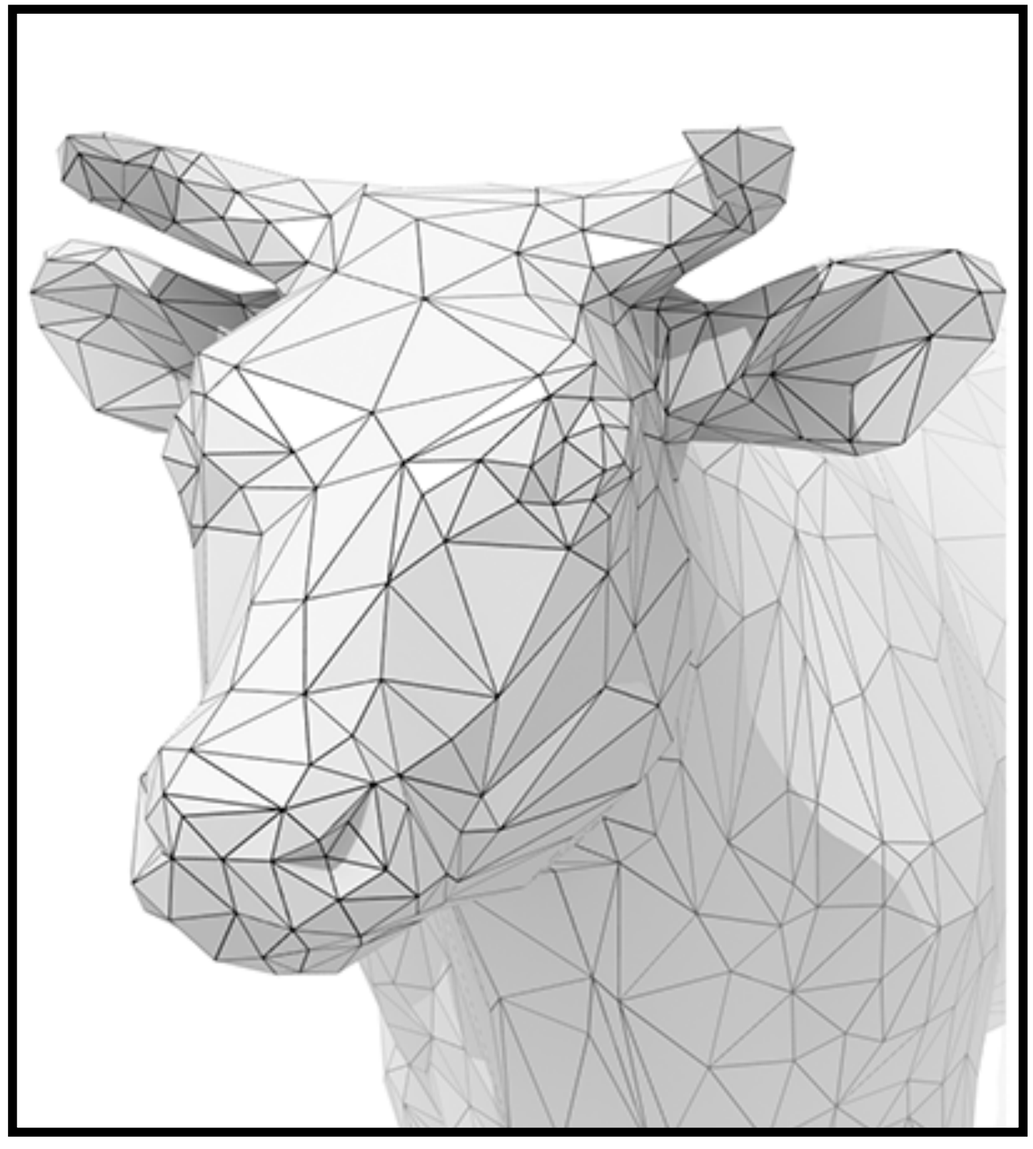
Mesh Upsampling – Subdivision



Increase resolution via subdivision

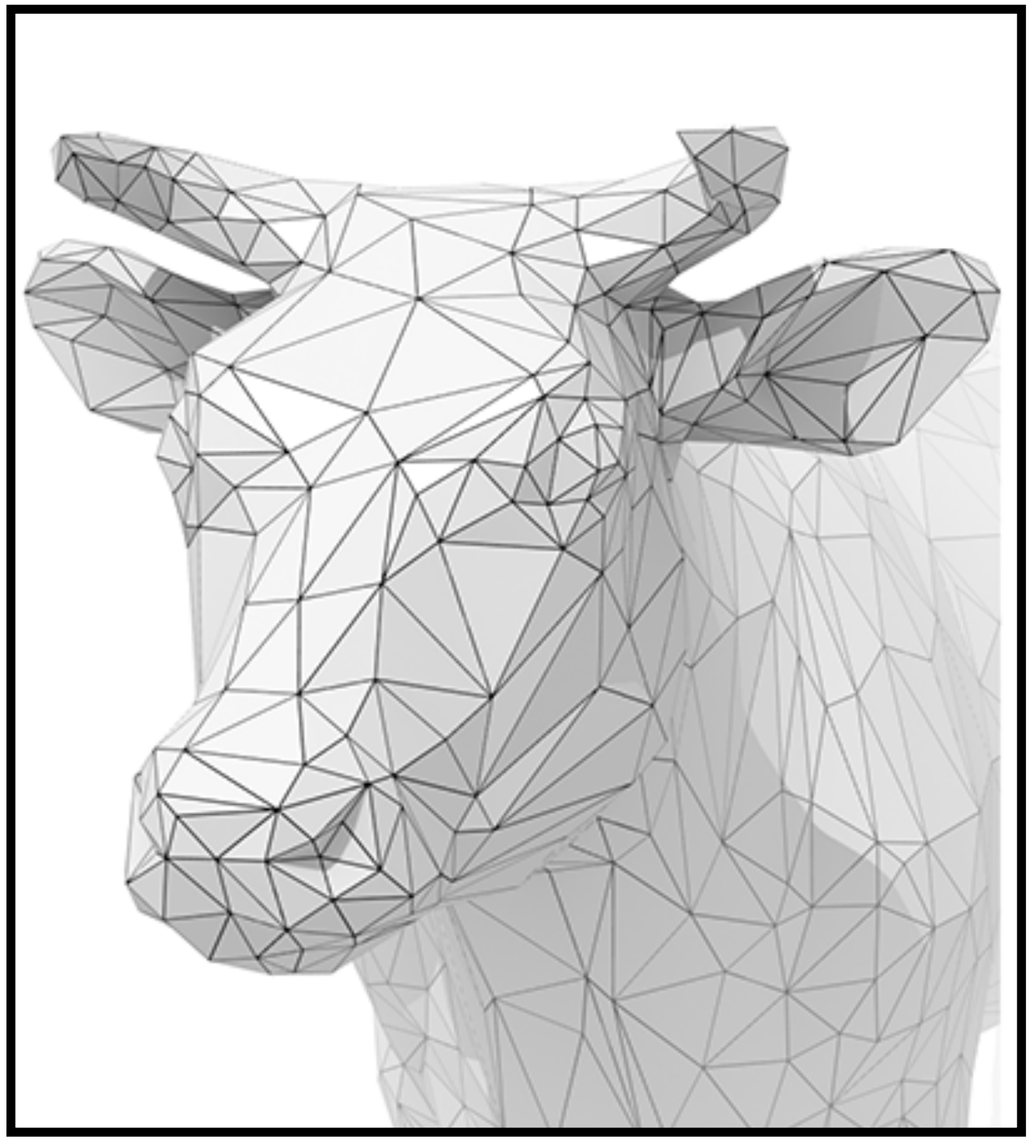
Mesh Downsampling – Simplification

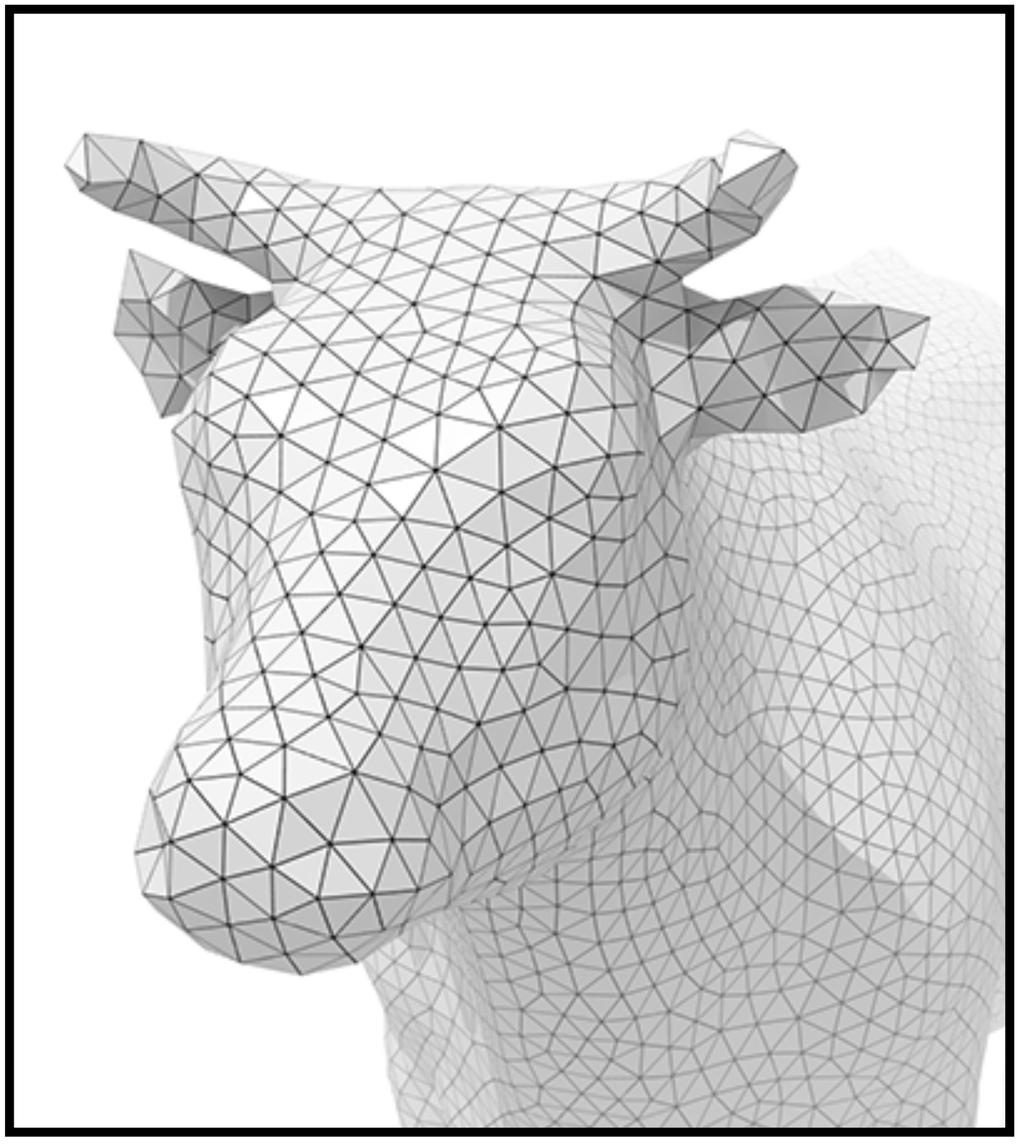




Decrease resolution; try to preserve shape/appearance

Mesh Regularization

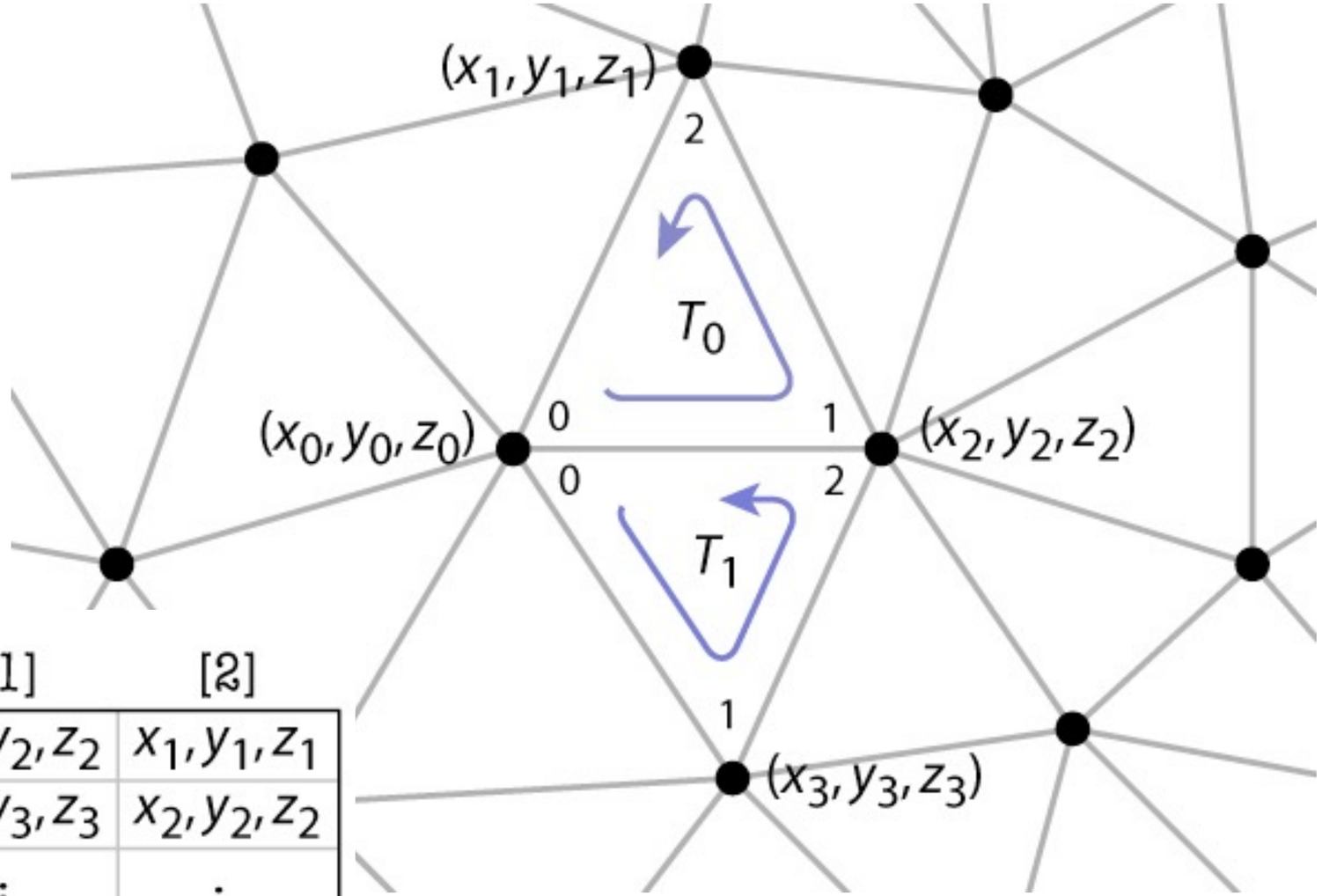




Modify sample distribution to improve quality

Mesh Representations

List of Triangles

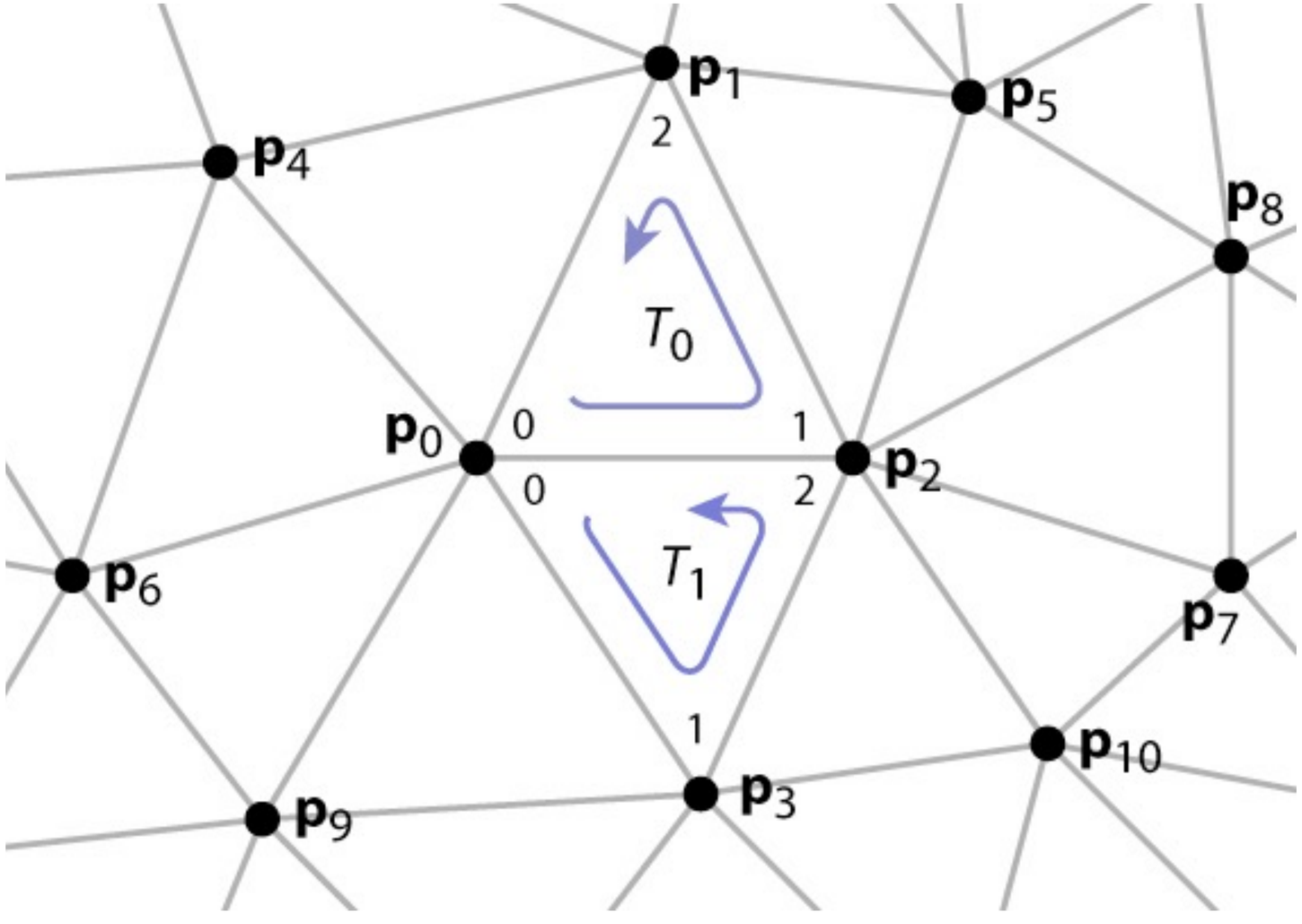


	[0]	[1]	[2]
tris[0]	x_0, y_0, z_0	x_2, y_2, z_2	x_1, y_1, z_1
tris[1]	x_0, y_0, z_0	x_3, y_3, z_3	x_2, y_2, z_2
	⋮	⋮	⋮

Lists of Points / Indexed Triangle

verts[0]	x_0, y_0, z_0
verts[1]	x_1, y_1, z_1
	x_2, y_2, z_2
	x_3, y_3, z_3
	⋮

tInd[0]	0, 2, 1
tInd[1]	0, 3, 2
	⋮



Comparison

Triangles

+ Simple

– Redundant information

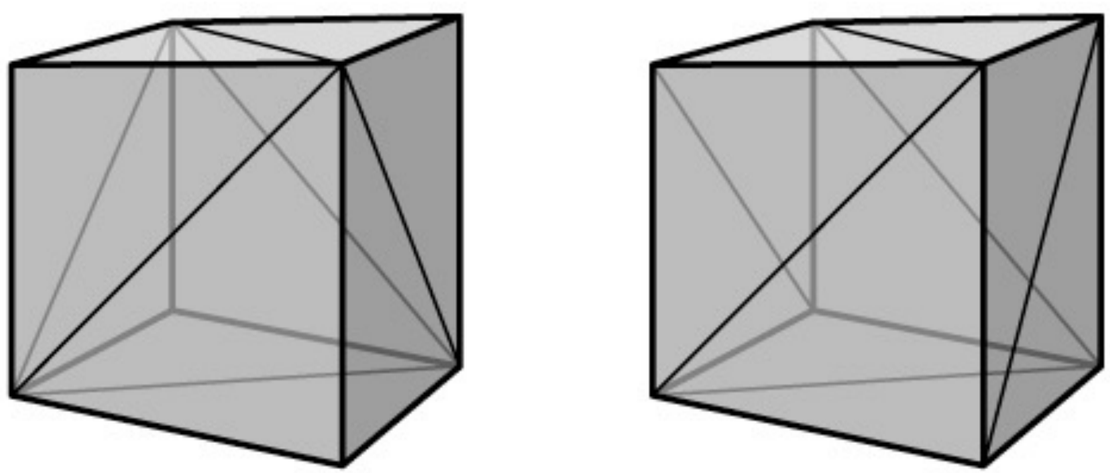
Points + Triangles

+ Sharing vertices reduces memory usage

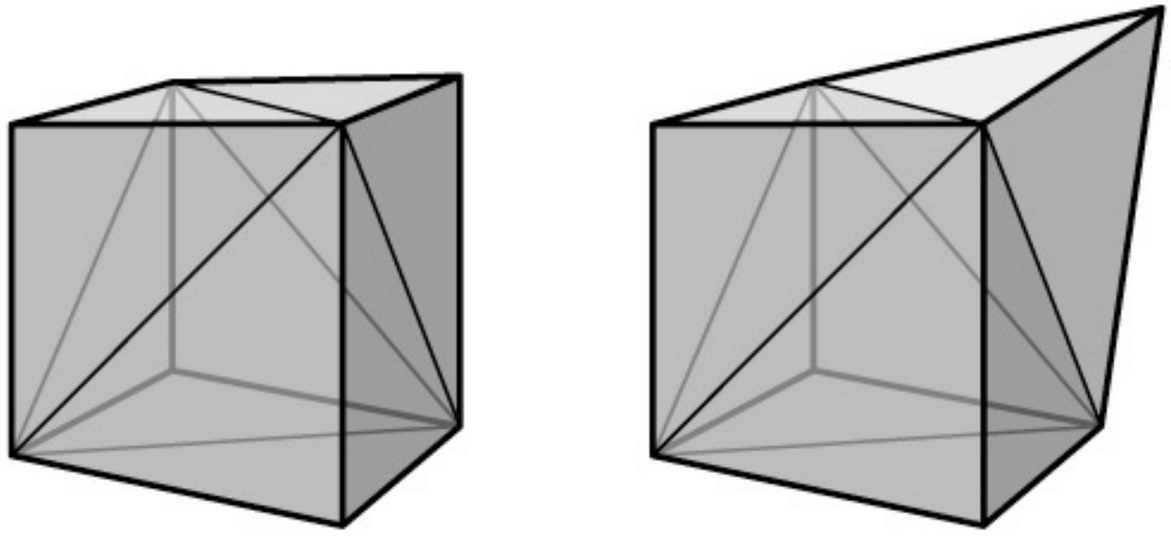
+ Ensure integrity of the mesh (moving a vertex causes that vertex in all the polygons to move)

Topology vs Geometry

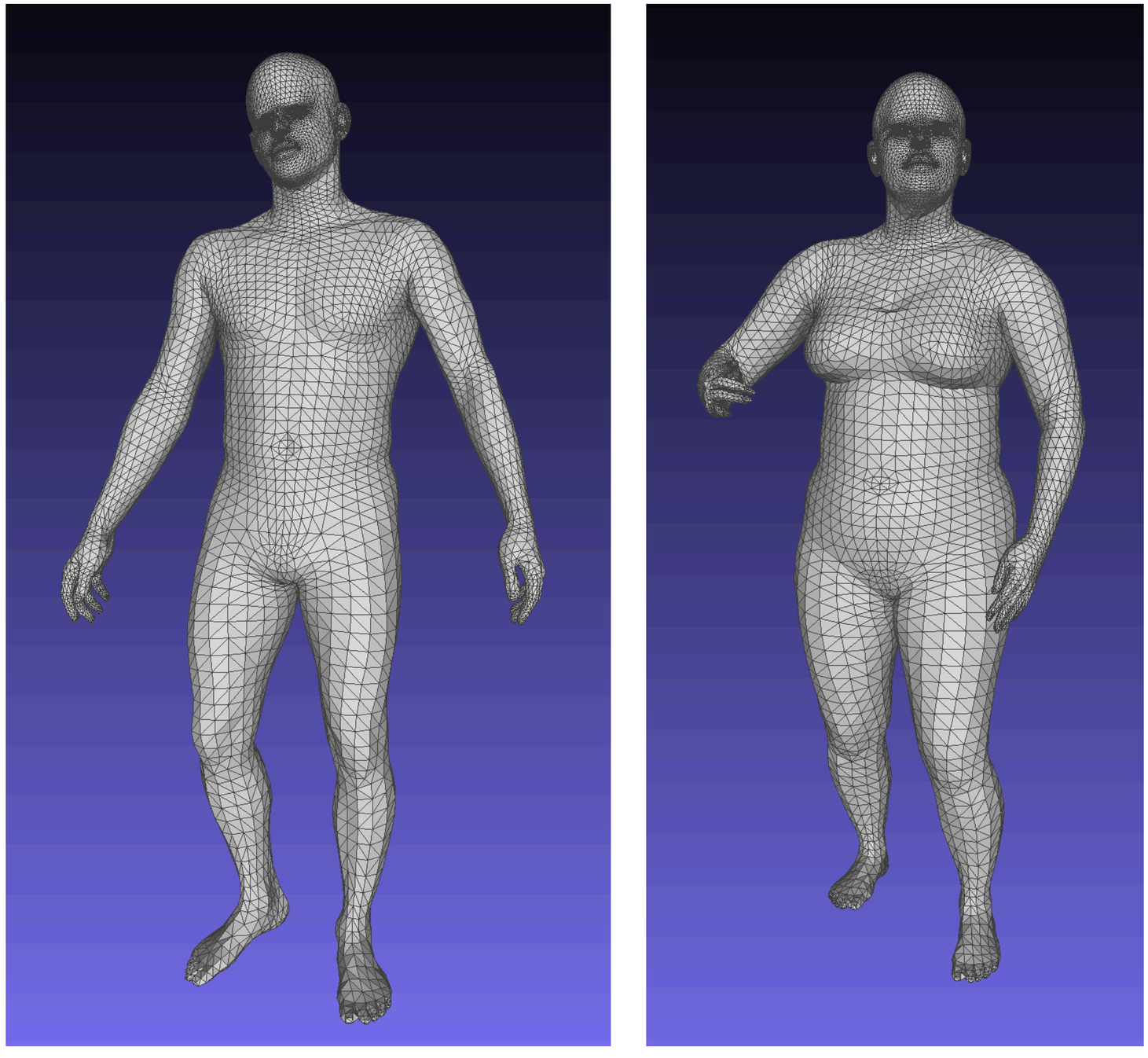
Same geometry, different mesh topology



Same mesh topology, different geometry



Topology vs Geometry



Same topology / Different Geometry

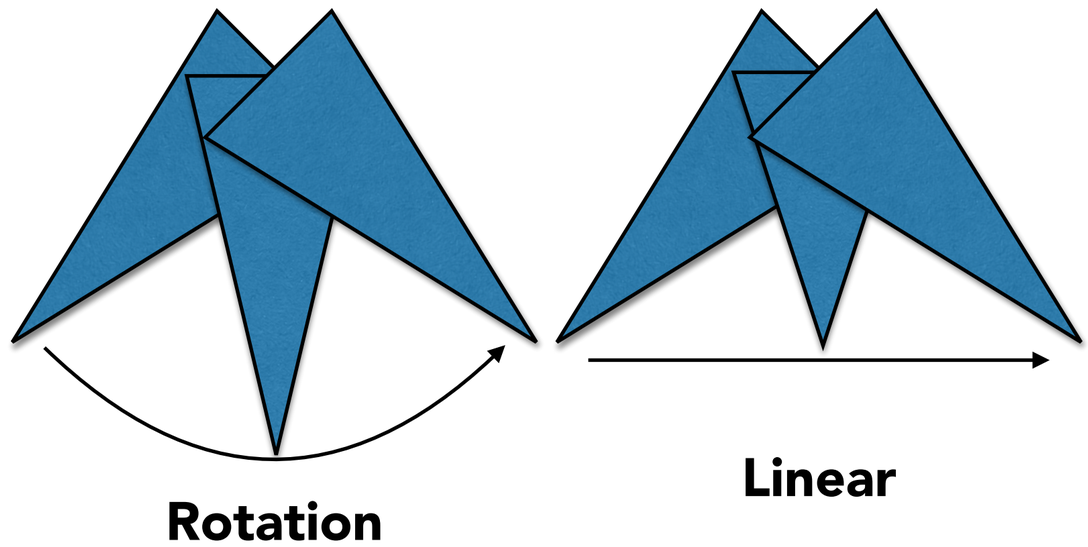
Topology vs Geometry

Same topology / Different Geometry

Meshes with same topology allow easy interpolation.

$$V_{\textbf{new}} = \alpha V_1 + (1-\alpha) V_2$$

Note that basic linear interpolation may not be semantically correct.



Topology vs Geometry

Same topology / Different Geometry

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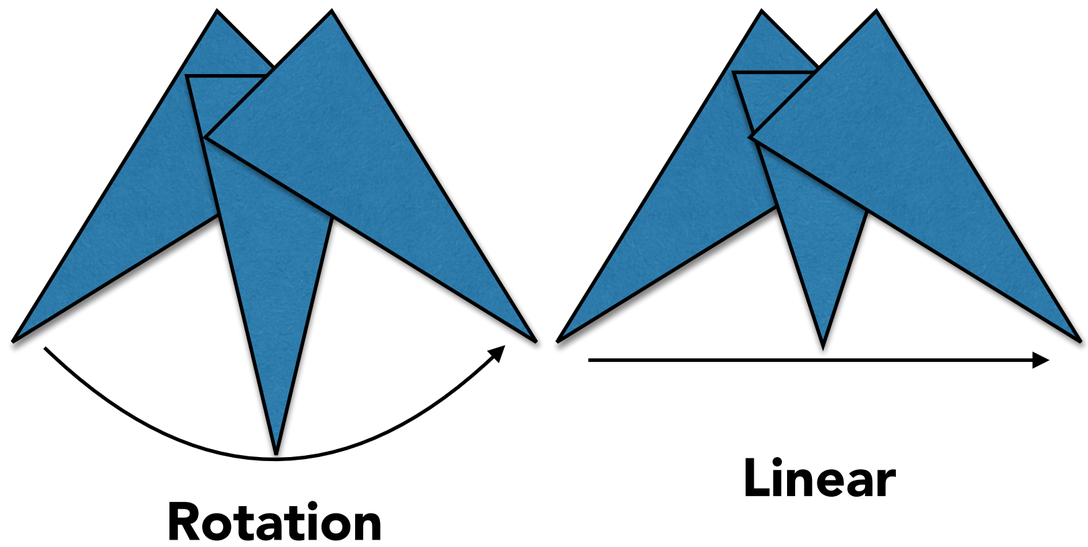


Topological Mesh Information

Applications:

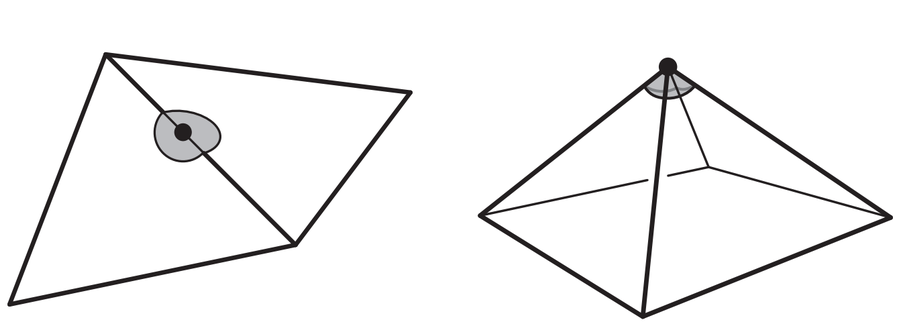
- Constant time access to neighbors
  e.g. surface normal calculation, subdivision
- Editing the geometry
  e.g. adding/removing vertices, faces, edges, etc.

Solution: Topological data structures

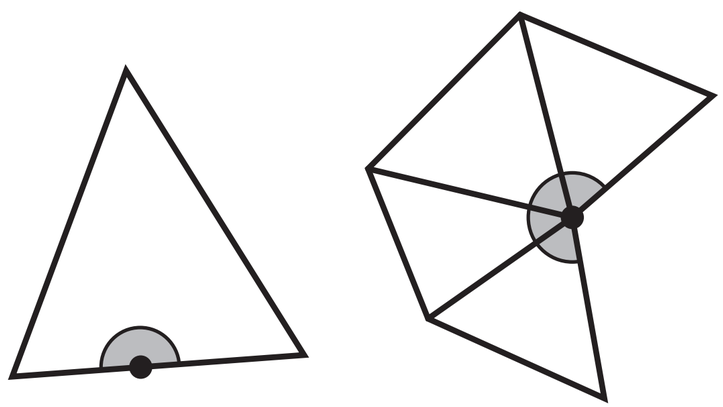
Topological Validity: Manifold

Definition: a 2D manifold is a surface that when cut with a small sphere always yields a disk.

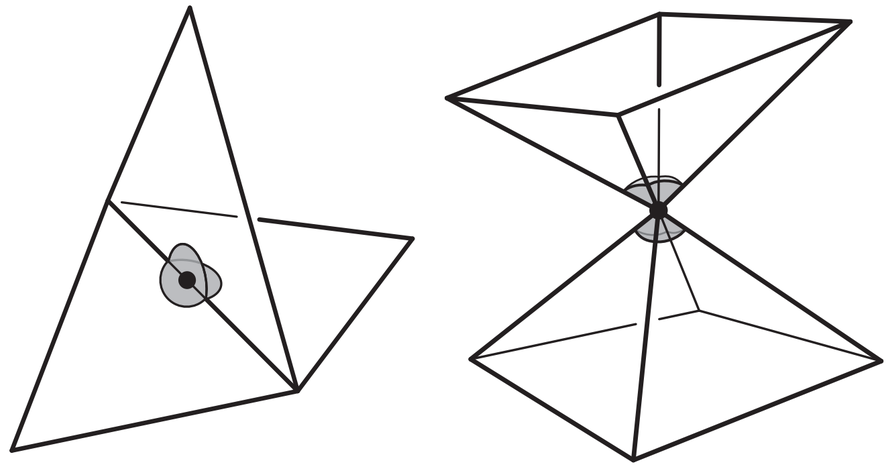
Manifold



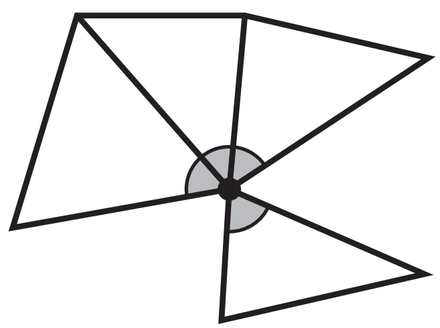
With border



Not manifold



With border



Topological Validity: Manifold

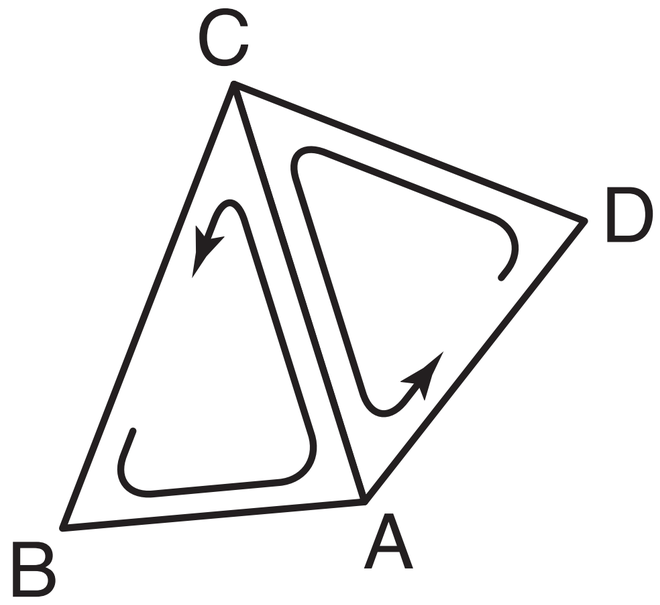
Definition: a 2D manifold is a surface that when cut with a small sphere always yields a “disk”.

If a mesh is manifold we can rely on these useful properties:

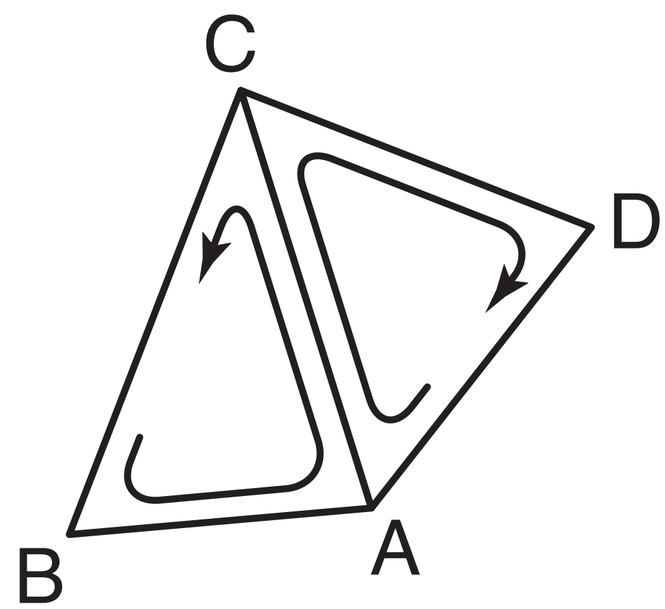
- An edge connects exactly two faces
- An edge connects exactly two vertices
- A face consists of a ring of edges and vertices
- A vertex consists of a ring of edges and faces
- Euler’s polyhedron formula holds: #f – #e + #v = 2
  (for a surface topologically equivalent to a sphere)
  (Check for a cube: 6 – 12 + 8 = 2)

Topological Validity: Orientation Consistency

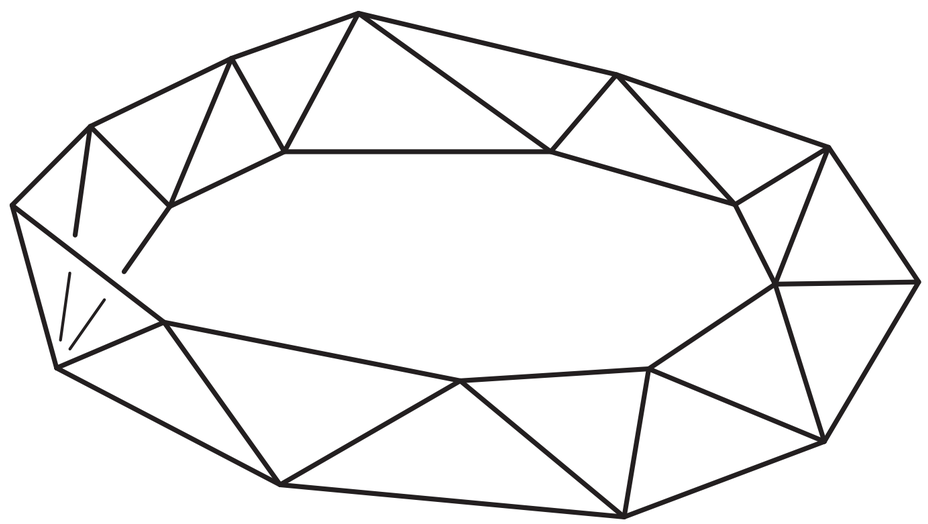
Both facing front



Inconsistent orientations



Non-orientable

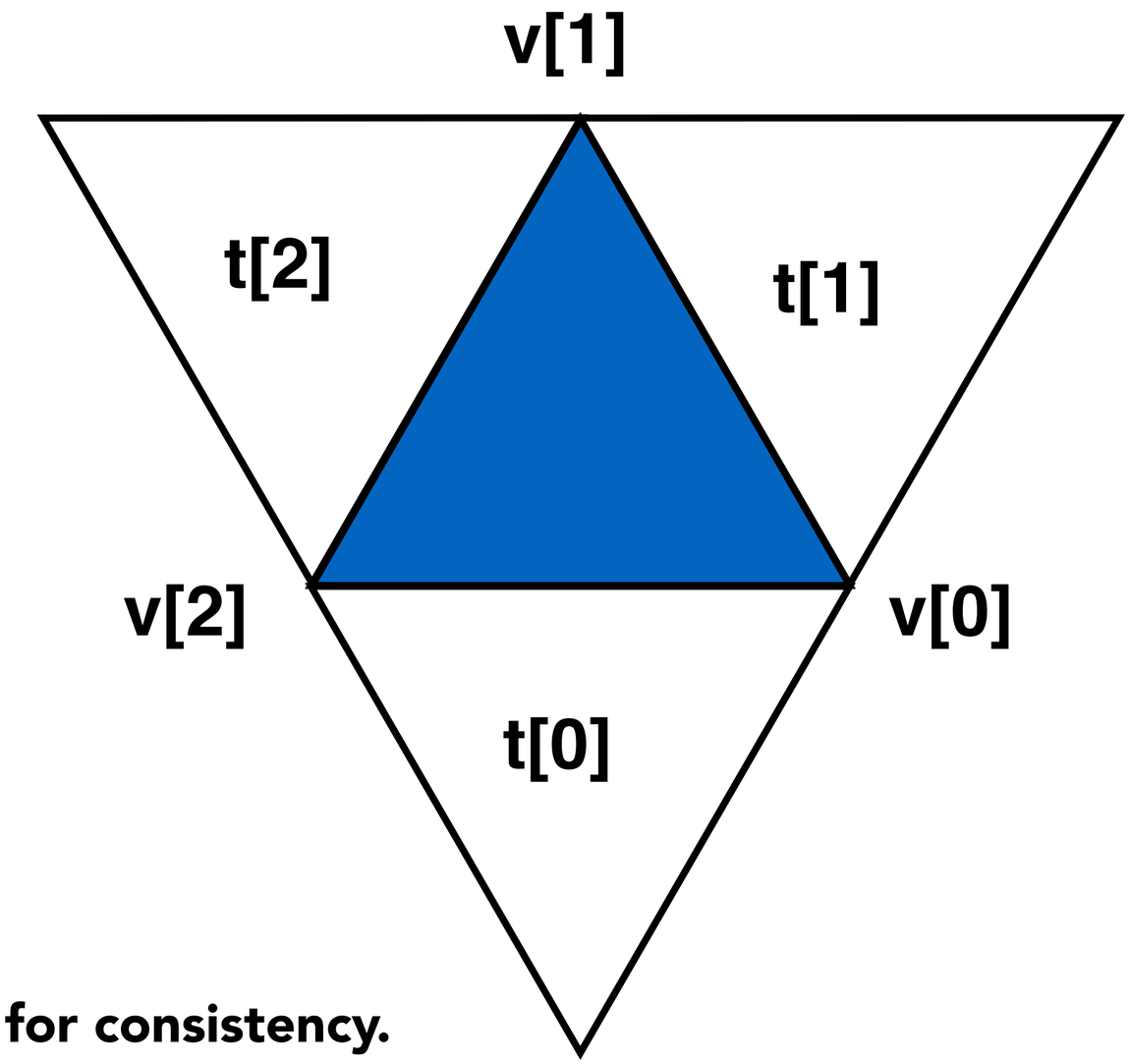


Mesh Data Structures

Triangle-Neighbor Data Structure

```
struct Tri {
      Vert       * v[3];
      Tri         * t[3];
}

struct Vert {
      Point     pt;
      Tri        *t[];
}
```

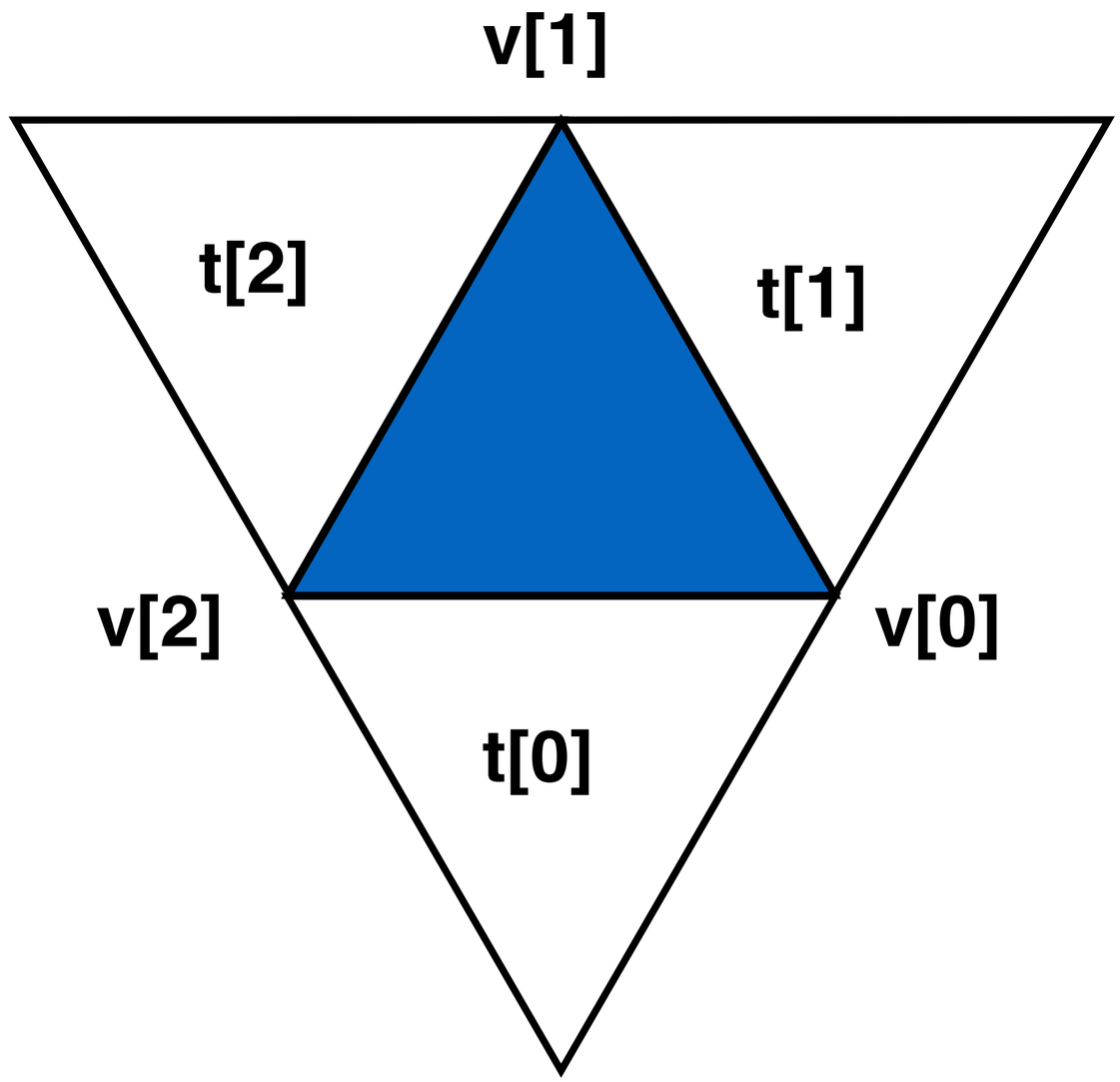


Note the various ways one can check a mesh for consistency. This is an excellent debug tool.

Triangle-Neighbor – Mesh Traversal

Find next triangle counter-clockwise around vertex v from triangle t

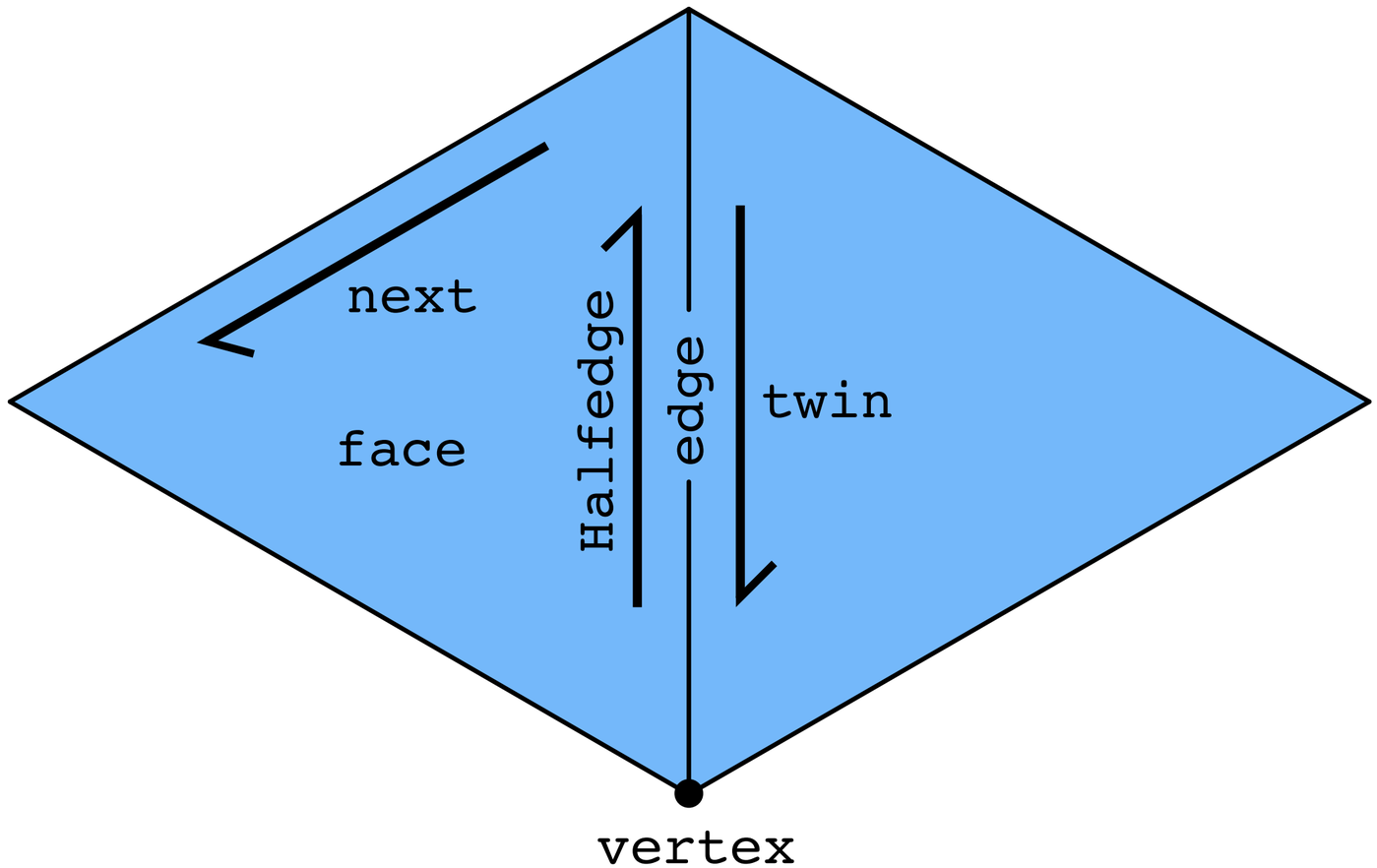
```
Tri *tccwvt(Vert *v, Tri *t)
{
    if (v == t->v[0])
        return t->t[0];
    if (v == t->v[1])
        return t->t[1];
    if (v == t->v[2])
        return t->t[2];
}
```



Half-Edge Data Structure

```
struct Halfedge {
   Halfedge *twin,
   Halfedge *next;
   Vertex *vertex;
   Edge *edge;
   Face *face;
}
struct Vertex {
   Point pt;
   Halfedge *one_halfedge;
   …
}
struct Edge {
   Halfedge *one_halfedge;
   …
}
struct Face {
   Halfedge *one_halfedge;
   …
}
```

Key idea: two half-edges act as "glue" between mesh elements



Each vertex, edge and face points to one of its half edges

*One arb. selected halfedge

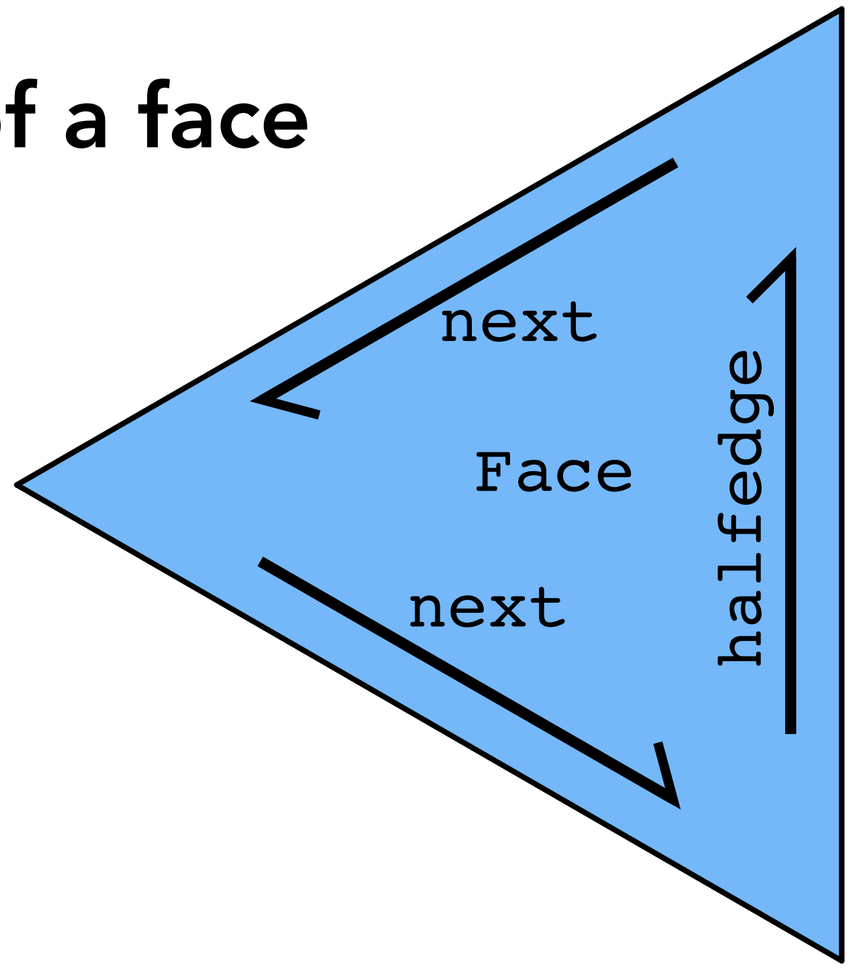
Half-Edge Facilitates Mesh Traversal

Use twin and next pointers to move around mesh

Process vertex, edge and/or face pointers

Example 1: process all vertices of a face

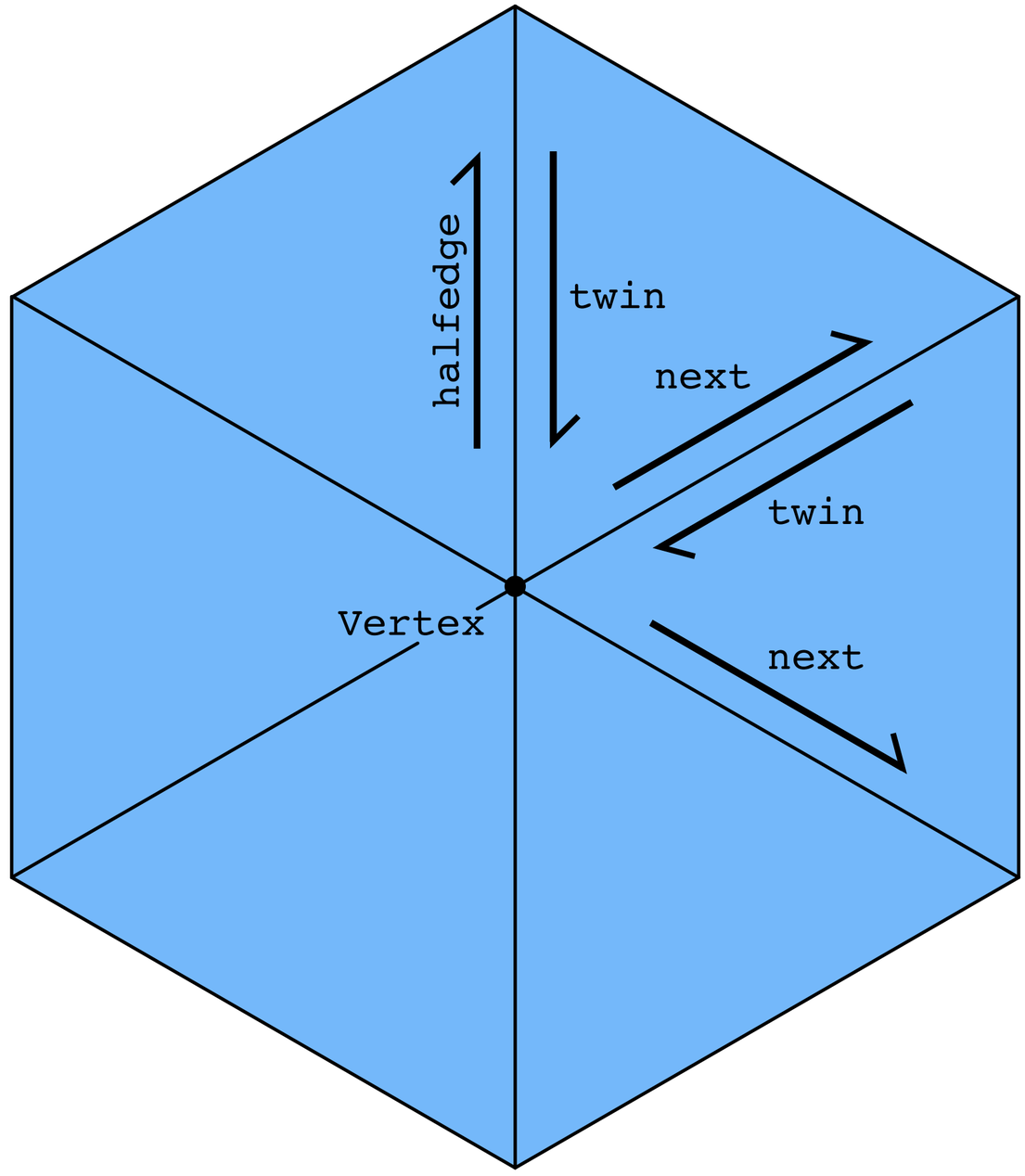
```
Halfedge* h = f->halfedge;
do {
   process(h->vertex);
   h = h->next;
}
```



Half-Edge Facilitates Mesh Traversal

Example 2: process all edges around a vertex

```
Halfedge* h = v->halfedge;
do {
   process(h->edge);
   h = h->twin->next;
}
```

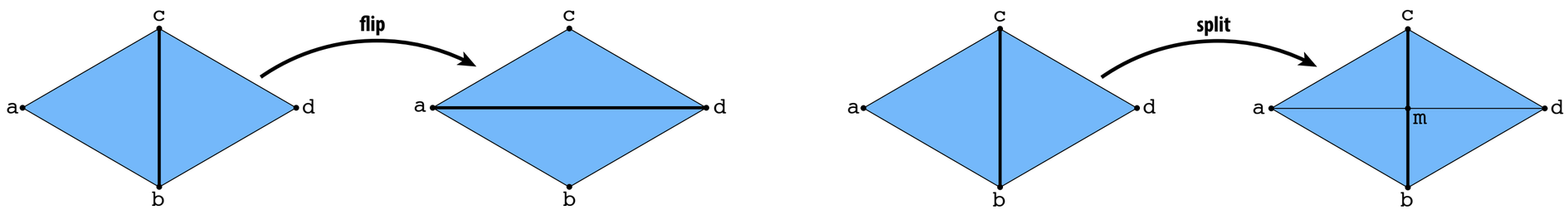


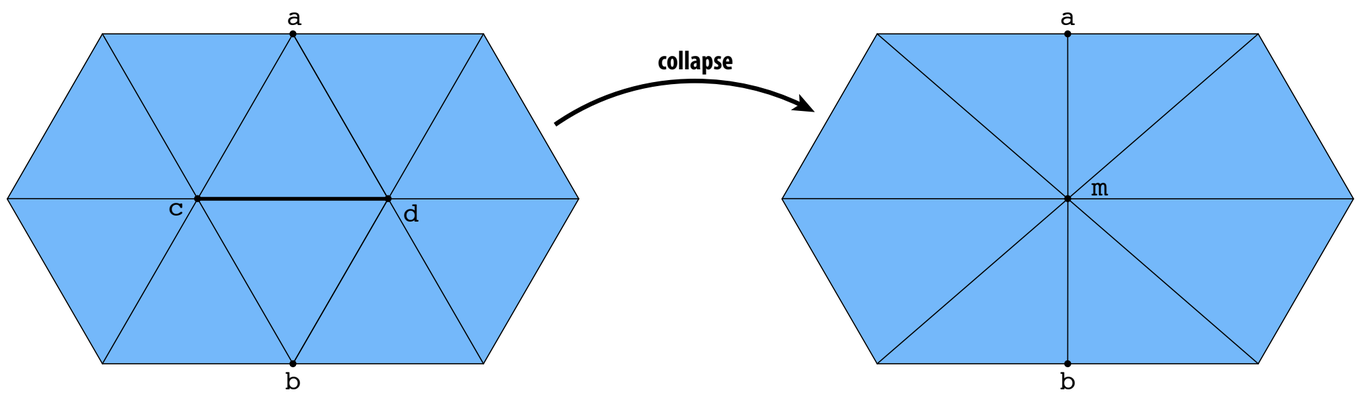
Local Mesh Operations

Half-Edge – Local Mesh Editing

Basic operations for linked list: insert, delete

Basic ops for half-edge mesh: flip, split, collapse edges



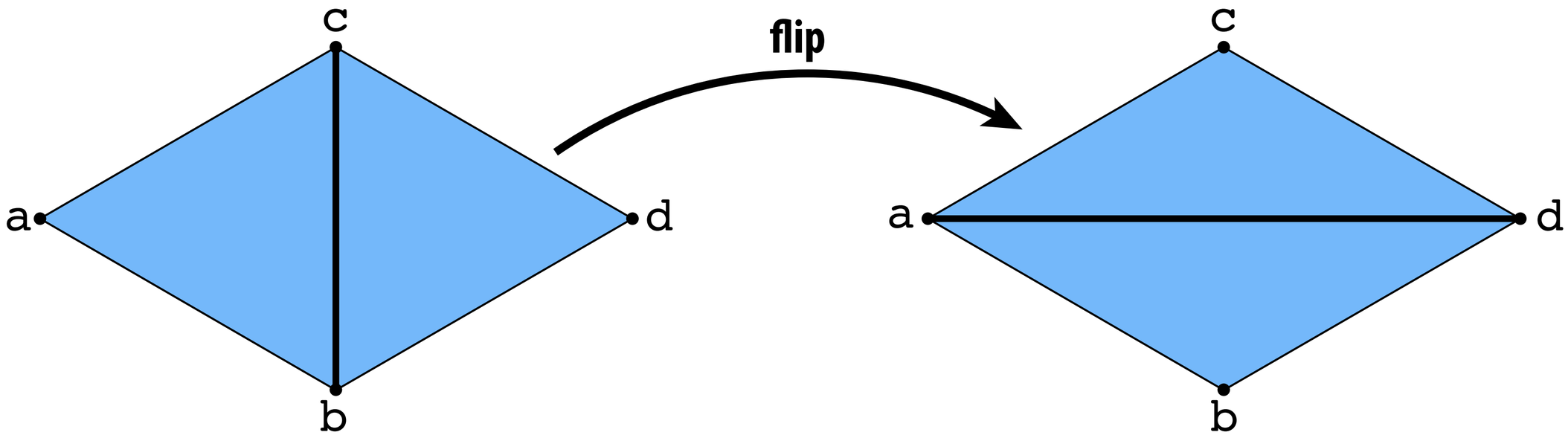


Allocate / delete elements; reassign pointers

(Care needed to preserve mesh manifold property)

Half-Edge – Edge Flip

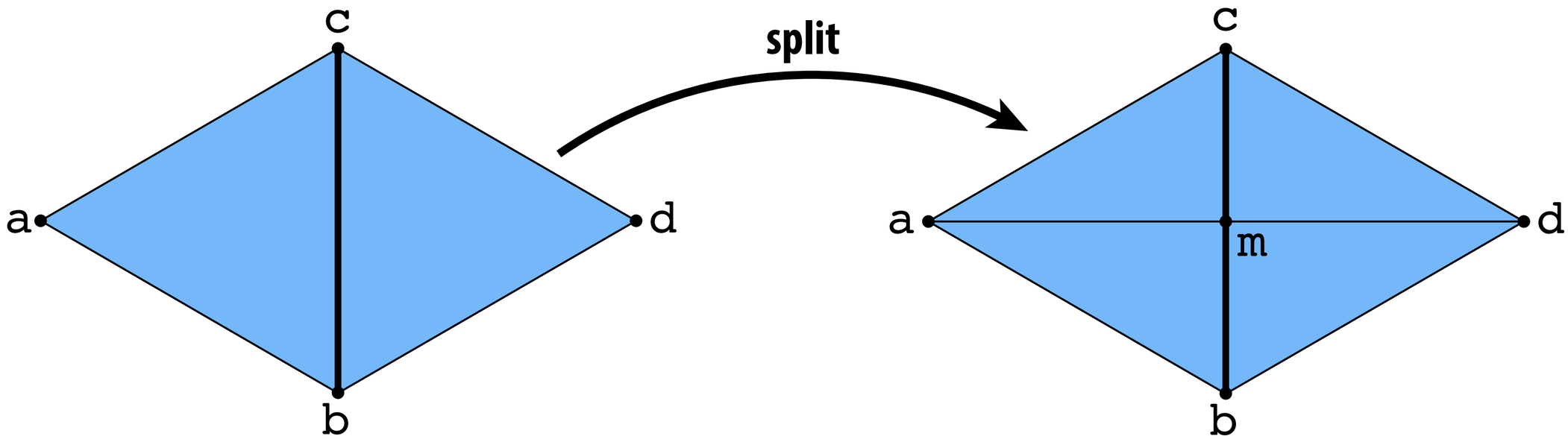
- Triangles (a,b,c), (b,d,c) become (a,d,c), (a,b,d):



- Long list of pointer reassignments
- However, no elements created/destroyed.

Half-Edge – Edge Split

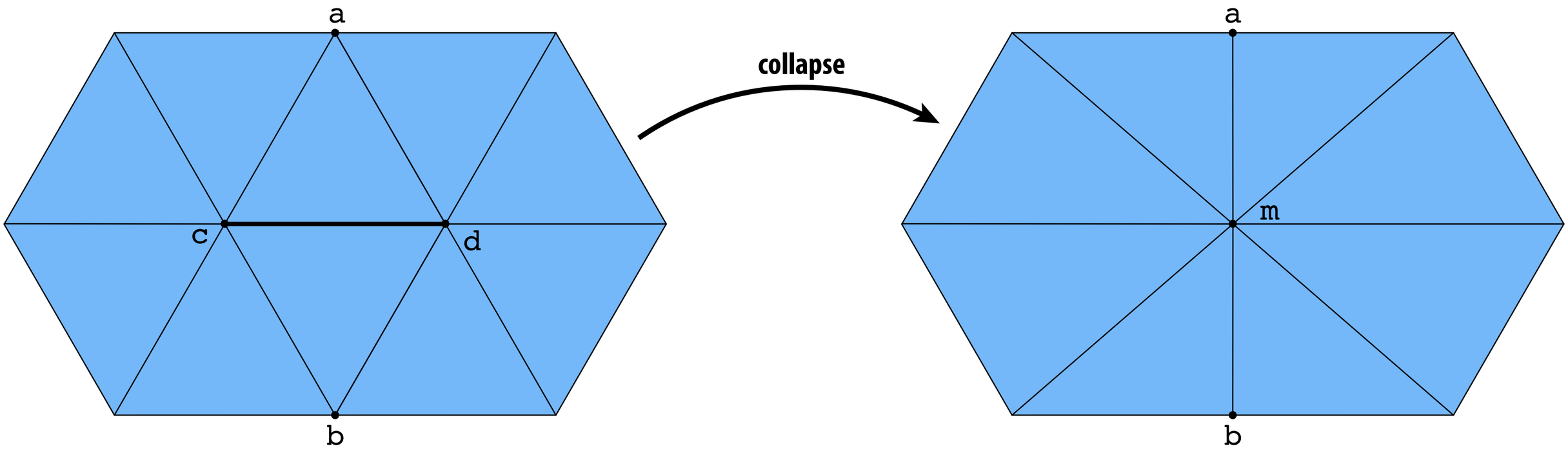
- Insert midpoint m of edge (c,b), connect to get four triangles:



- This time have to add elements
- Again, many pointer reassignments

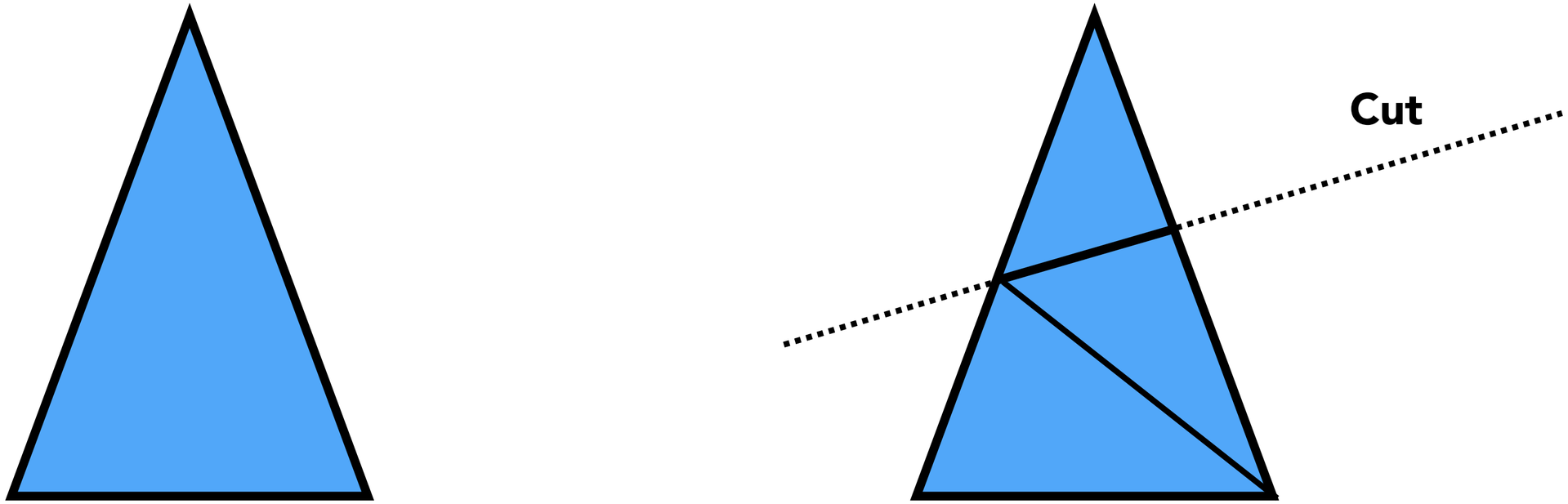
Half-Edge – Edge Collapse

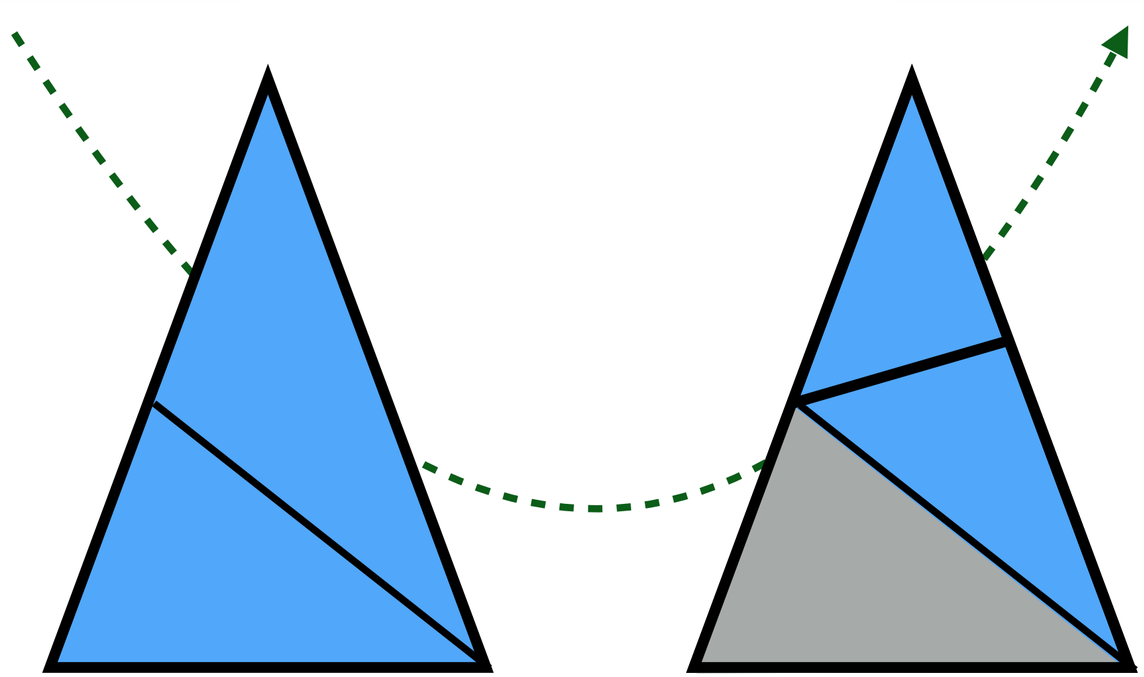
- Replace edge (c,d) with a single vertex m:



- This time have to delete elements
- Again, many pointer reassignments

Composite Opperations

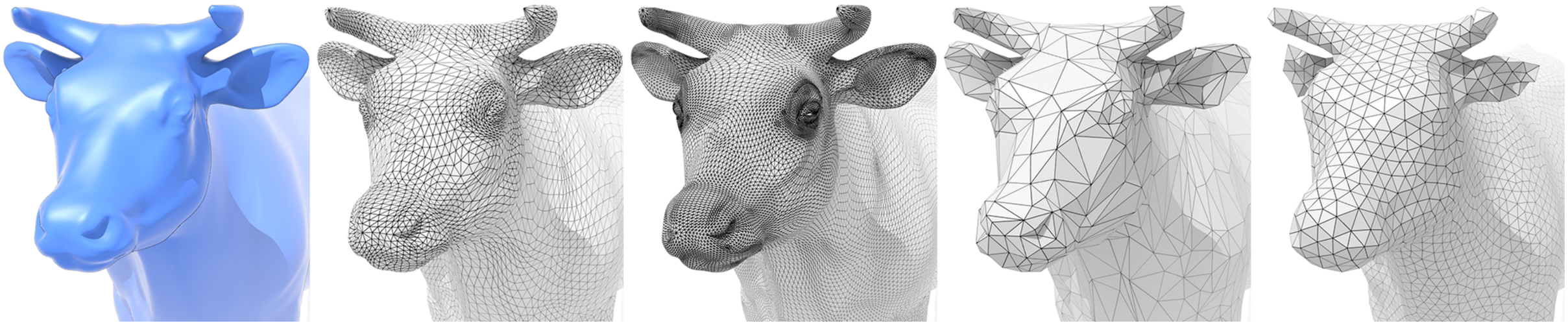




Global Mesh Operations

Global Mesh Operations: Geometry Processing

- Mesh subdivision
- Mesh simplification
- Mesh regularization



Subdivision Surfaces

Subdivision Surfaces

Start with coarse polygon mesh (“control cage”)

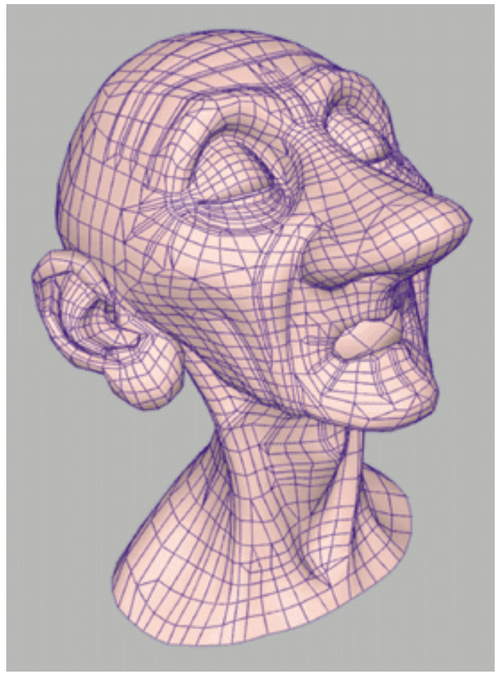
- Subdivide each element
- Update vertices via local averaging

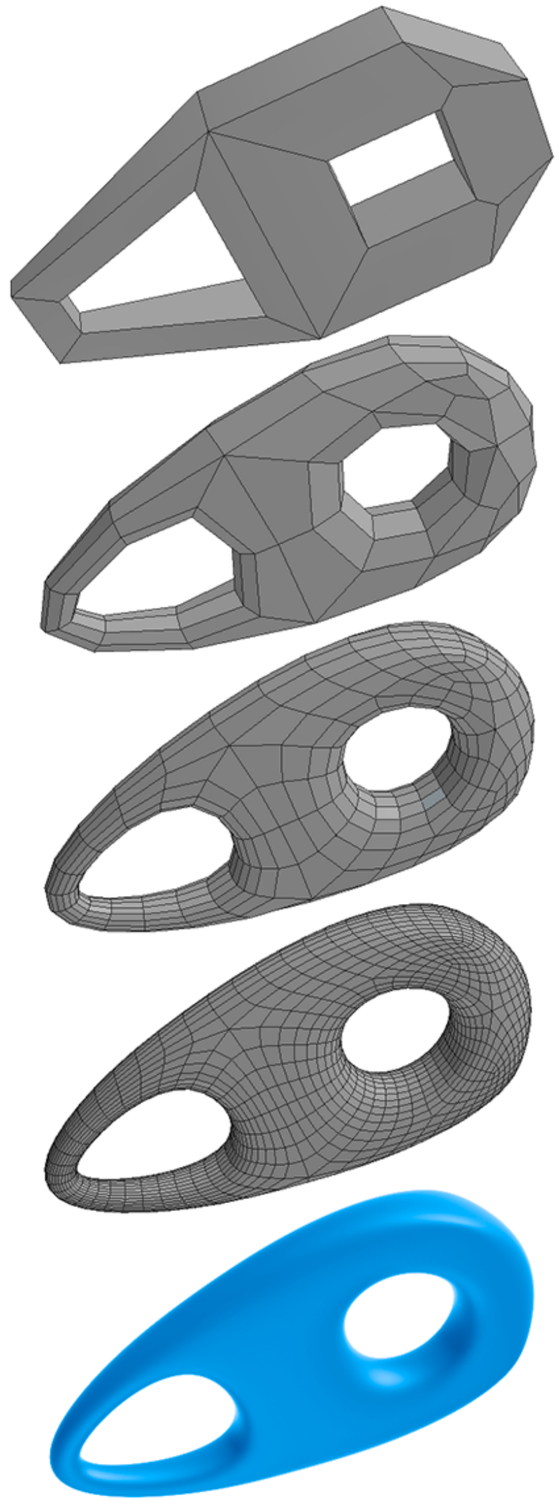
Many possible rules:

- Catmull-Clark (quads)
- Loop (triangles)
- ...

Common issues:

- interpolating or approximating?
- continuity at vertices?





Relatively easy for modeling; harder to guarantee continuity

Core Idea: Let Subdivision Define The Surface

In Bezier curves, we saw:

- Evaluation by subdivision (de Casteljau algorithm)
- Or evaluation by algebra (Bernstein polynomials)

Insight that leads to subdivision surfaces:

- Free ourselves from the algebraic evaluation
- Let subdivision fully define the surface

Many possible subdivision rules – different surfaces

- Technical challenge shifts to designing rules and proving properties (e.g. convergence and continuity)
- Applying rules to compute surface is procedural

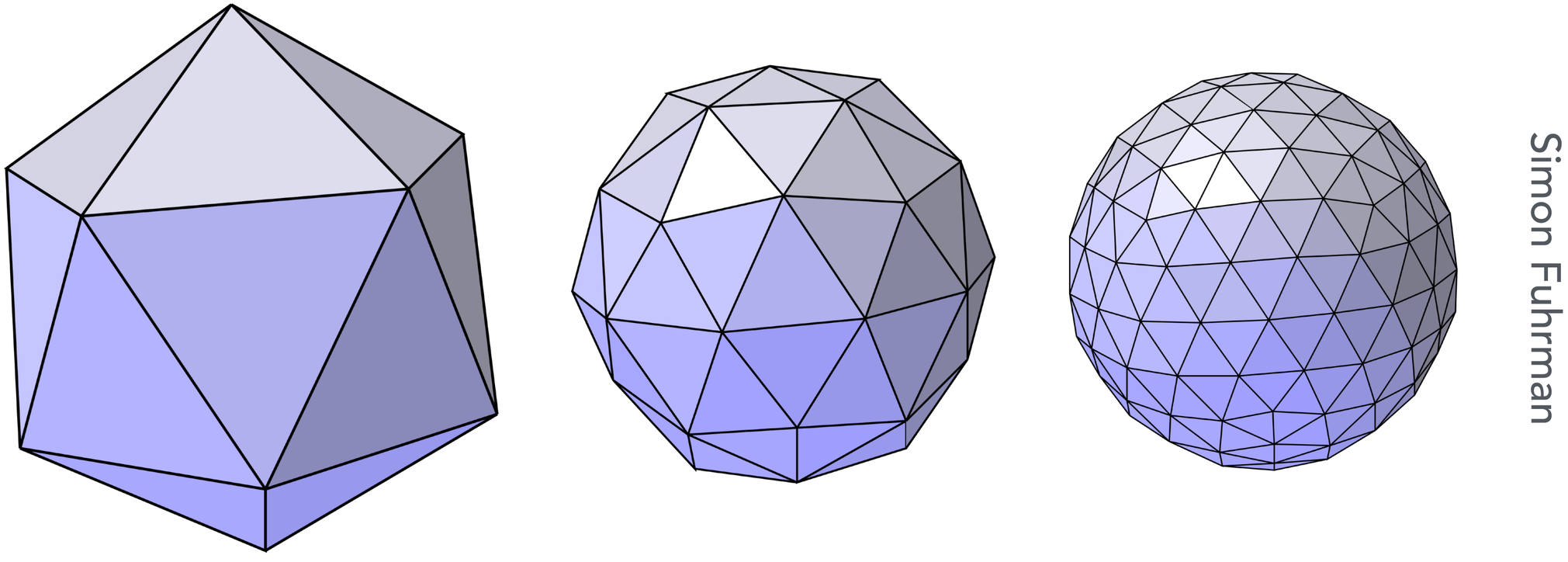
Loop Subdivision

Loop Subdivision

Common subdivision rule for triangle meshes

“C2” smoothness away from irregular vertices

Approximating, not interpolating



Simon Fuhrman

Loop Subdivision

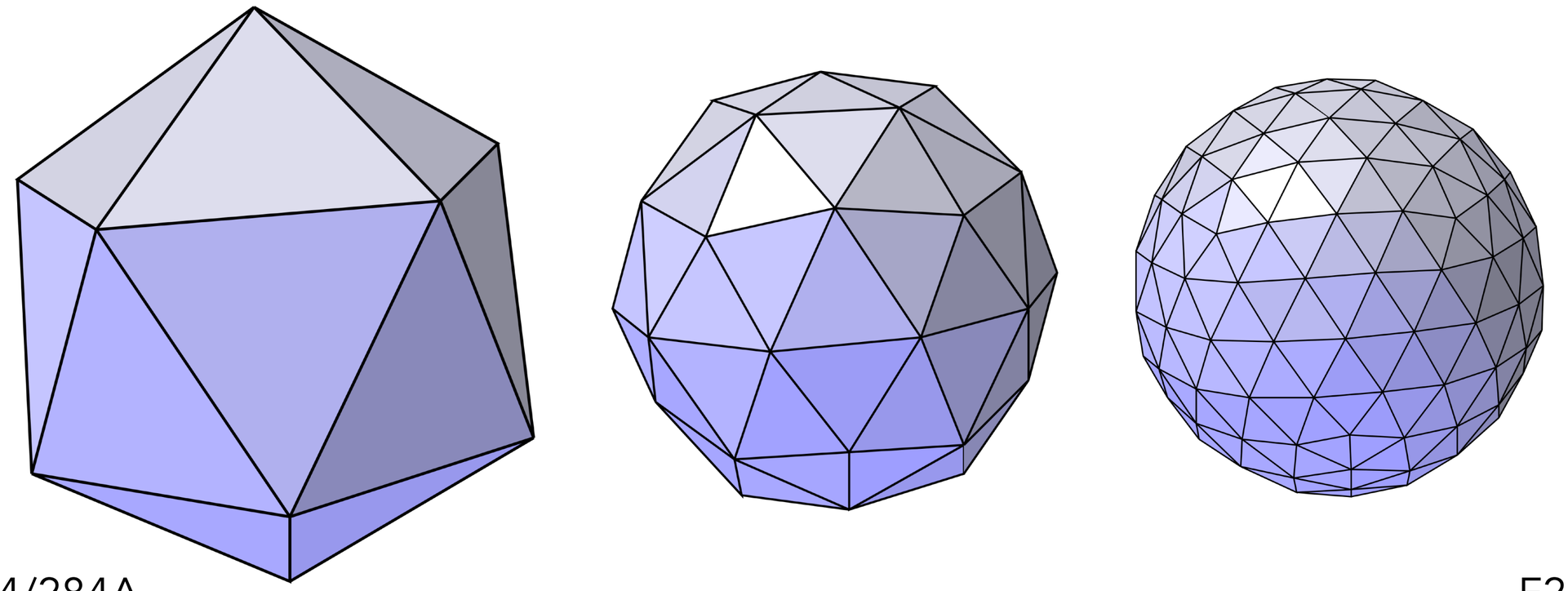
Common subdivision rule for triangle meshes

“C2” smoothness away from extraordinary vertices

Approximating, not interpolating

Generalization of Box Splines

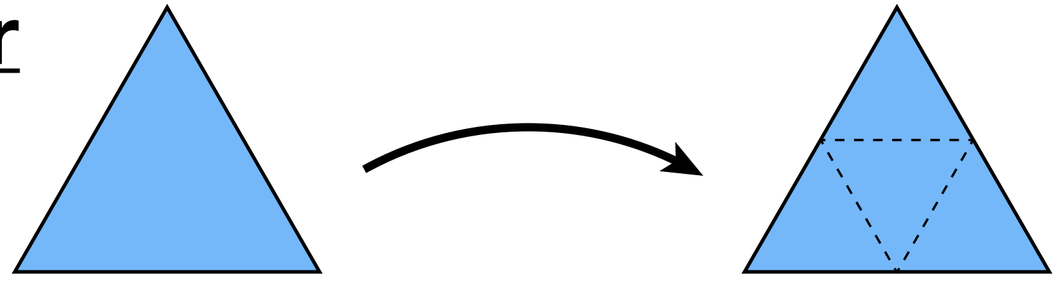
Basis functions are projections of hypercubes!



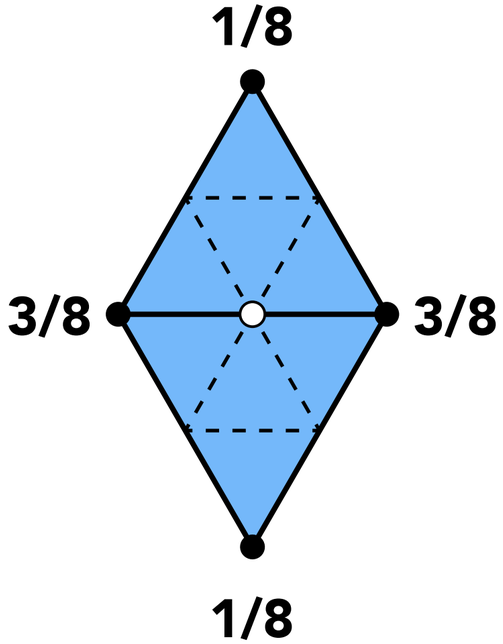
Simon Fuhrman

Loop Subdivision Algorithm

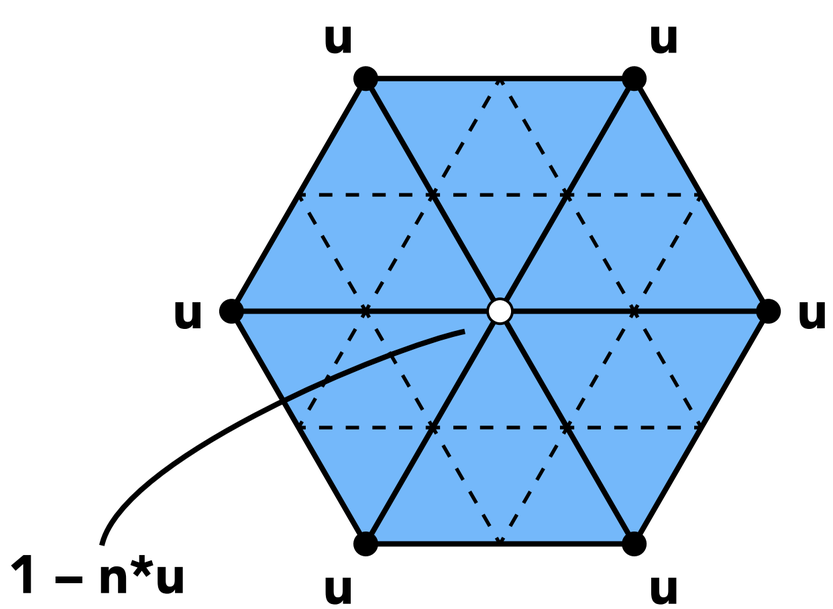
- Split each triangle into four



- Assign new vertex positions according to weights:



New vertices



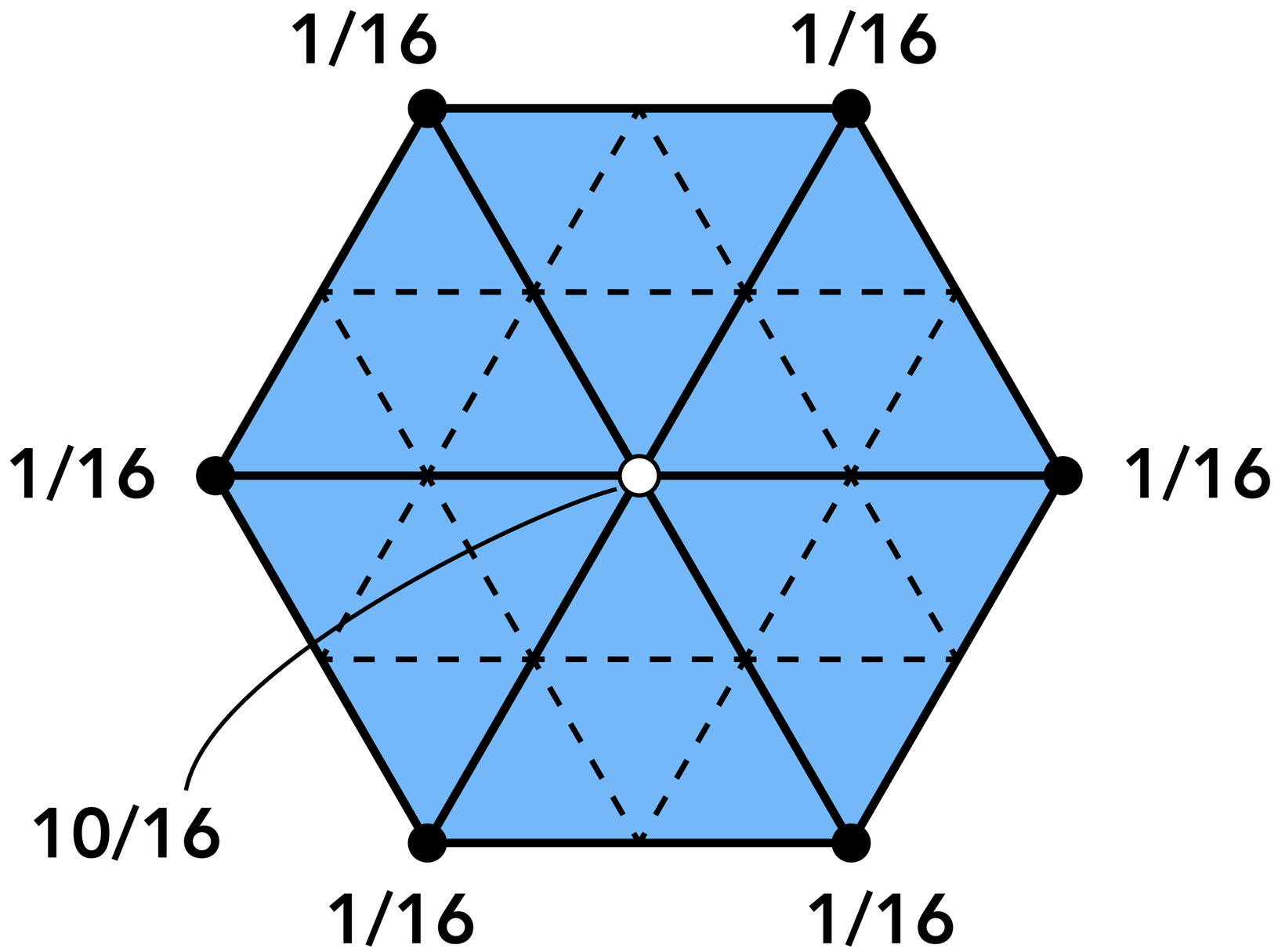
Old vertices

n: vertex degree

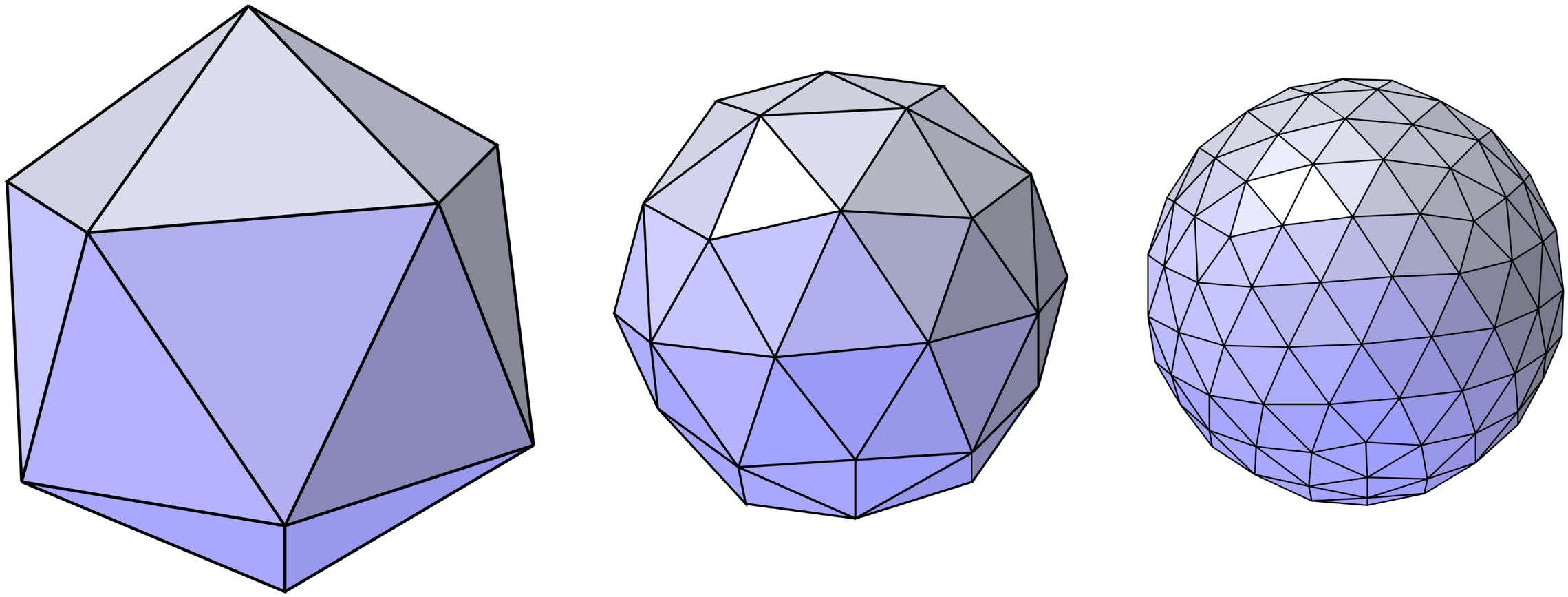
u: 3/16 if n=3, 3/(8n) otherwise

Loop Subdivision Algorithm

Example, for degree 6 vertices



Loop Subdivision Algorithm



Simon Fuhrman

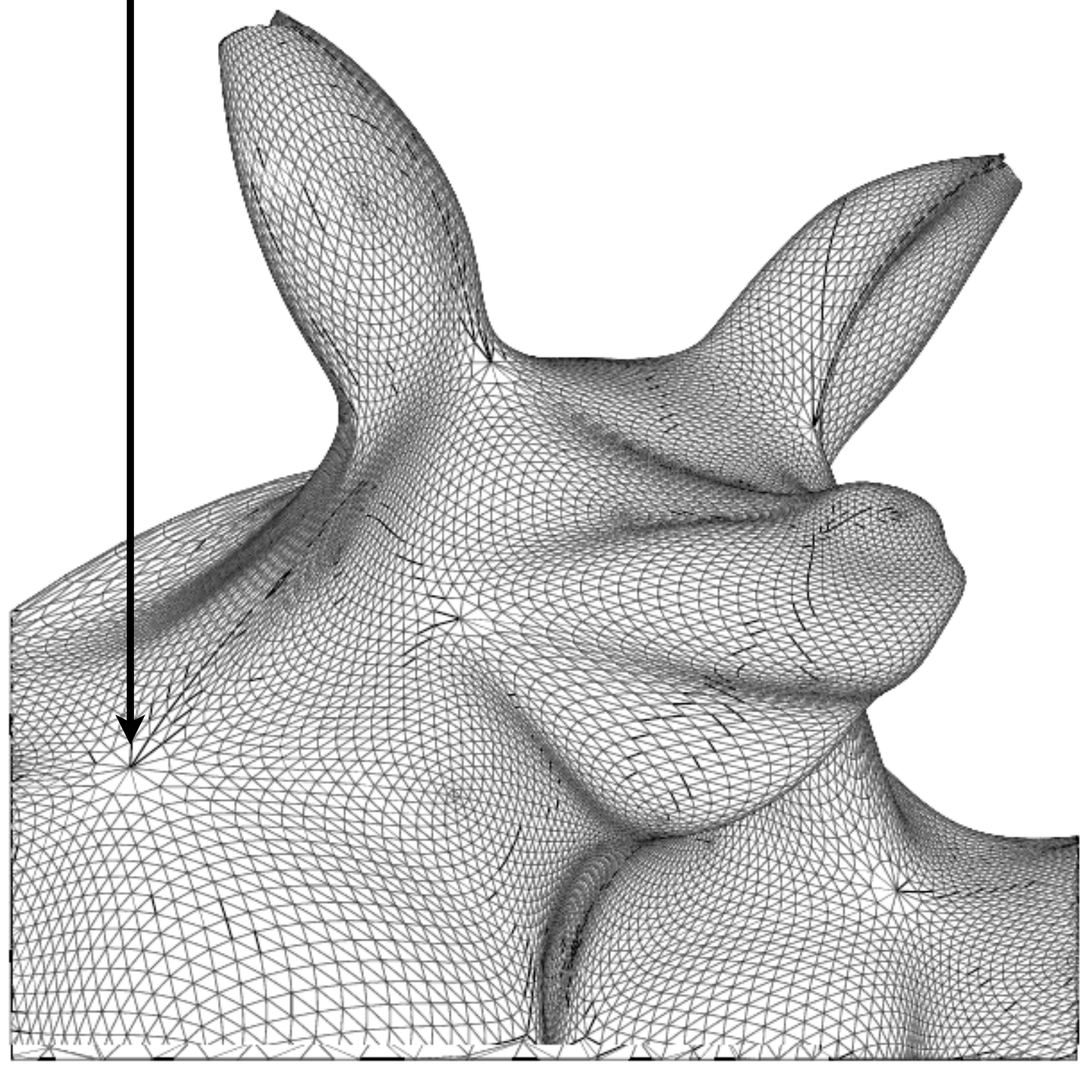
Semi-Regular Meshes

Most of the mesh has vertices with degree 6

But if the mesh is topologically equivalent to a sphere, then not all the vertices can have degree 6

Must have a few extraordinary points (degree not equal to 6)

Extraordinary point



Proof: Always an Extraordinary Vertex

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Our mesh (topologically equivalent to sphere) has V vertices, E edges, and T triangles

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Our mesh (topologically equivalent to sphere) has V vertices, E edges, and T triangles

E = 3/2 T

- There are 3 edges per triangle, and each edge is part of 2 triangles
- Therefore E = 3/2T

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- Therefore E = 3/2T

T = 2V – 4

- Euler Convex Polyhedron Formula: T – E + V = 2
- => V = 3/2 T – T + 2 => T = 2V – 4

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If all vertices had 6 triangles, T = 2V

- There are 6 edges per vertex, and every edge connects 2 vertices
- Therefore, E = 6/2V => 3/2T = 6/2V => T = 2V

Proof: Always an Extraordinary Vertex

Our mesh (topologically equivalent to sphere) has V vertices, E edges, and T triangles

E = 3/2 T

- There are 3 edges per triangle, and each edge is part of 2 triangles
- Therefore E = 3/2T

T = 2V – 4

- Euler Convex Polyhedron Formula: T – E + V = 2
- => V = 3/2 T – T + 2 => T = 2V – 4

If all vertices had 6 triangles, T = 2V

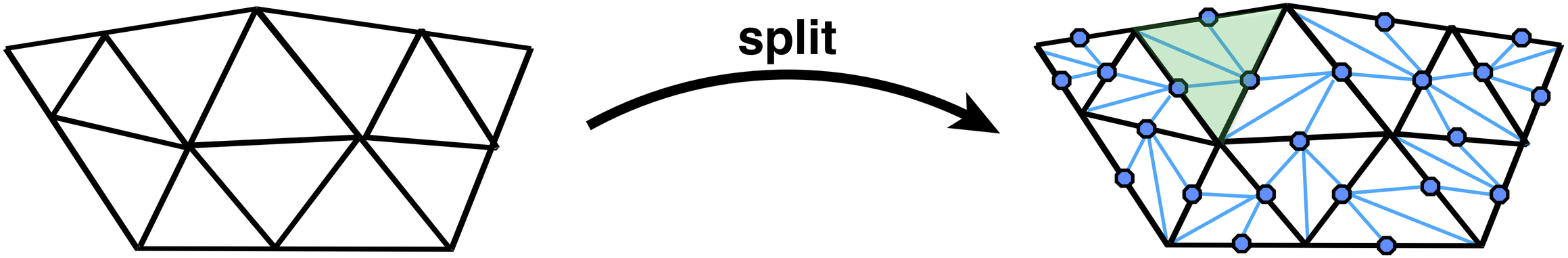
- There are 6 edges per vertex, and every edge connects 2 vertices
- Therefore, E = 6/2V => 3/2T = 6/2V => T = 2V

T cannot equal both 2V – 4 and 2V, a contradiction

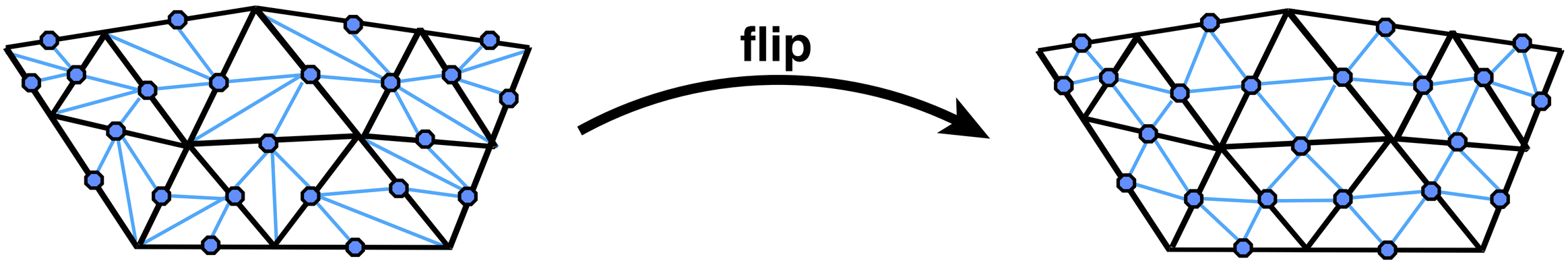
- Therefore, the mesh cannot have 6 triangles for every vertex

Loop Subdivision via Edge Operations

First, split edges of original mesh in any order:



Next, flip new edges that touch a new & old vertex:



(Don’t forget to update vertex positions!)

Images cribbed from Keenan Crane, cribbed from Denis Zorin

Continuity of Loop Subdivision Surface

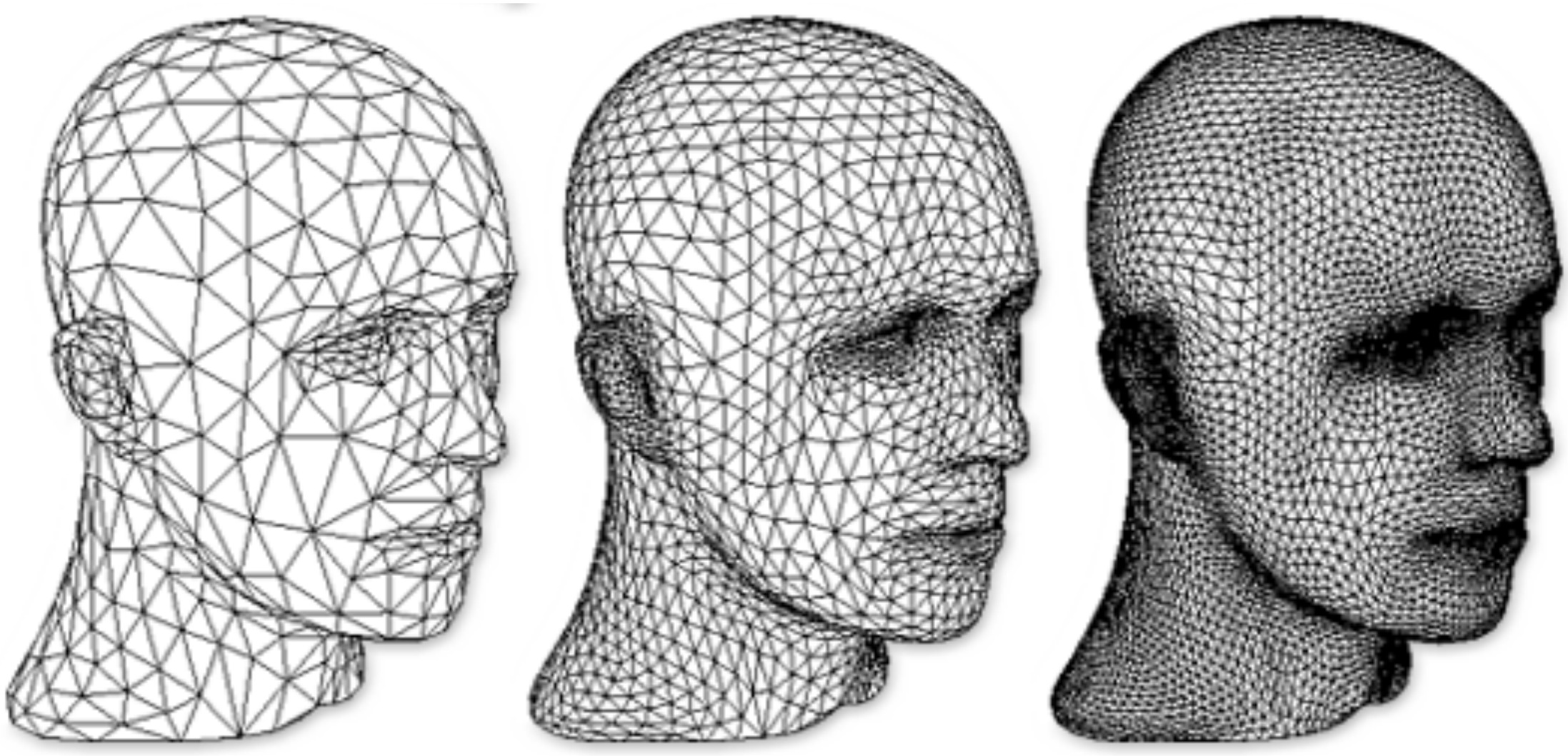
At extraordinary points

- Surface is at least C^1 continuous

Everywhere else (“ordinary” regions)

- Surface is C^2 continuous

Loop Subdivision Results



B-Splines

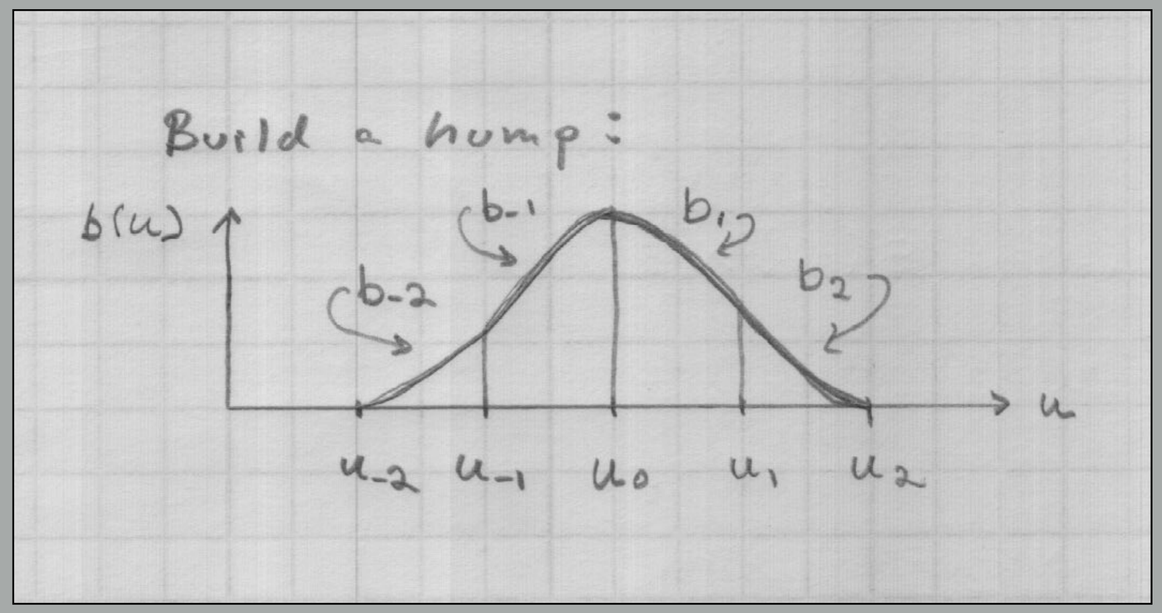
B-Splines

Goal: C^2 cubic curves with local support

- Give up interpolation
- Get convex hull property

Build basis by designing “hump” functions

B-Splines

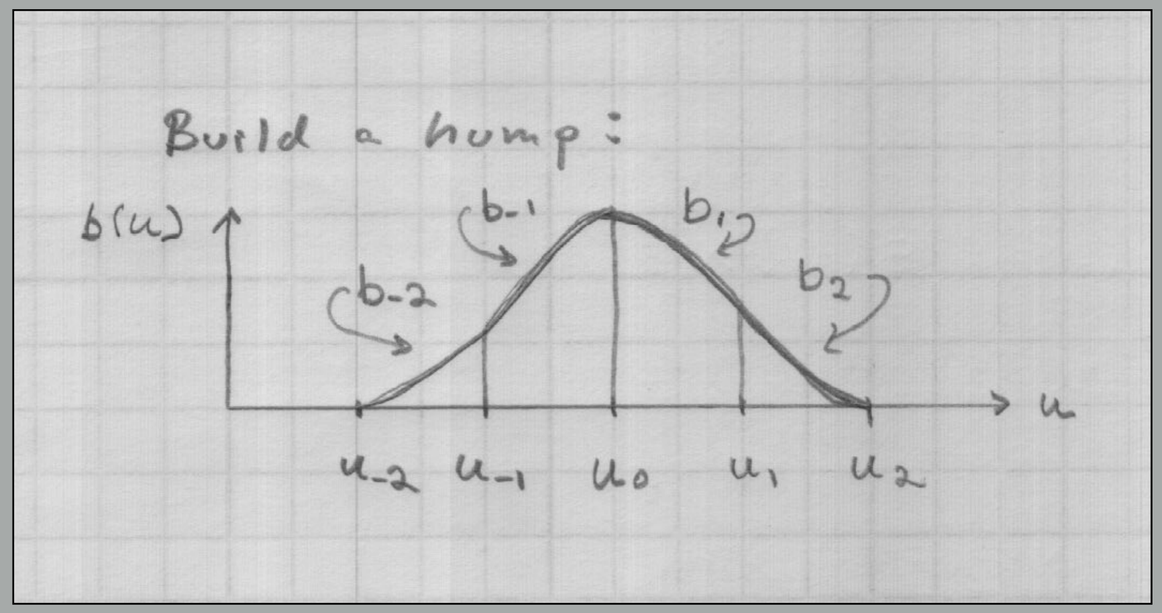


$$\mathbf{b}(u) = \begin{cases} \mathbf{b_{-2}}(u) & \text{if } u_{-2} \leq u < u_{-1} \\ \mathbf{b_{-1}}(u) & \text{if } u_{-1} \leq u < u_0 \\ \mathbf{b_{+1}}(u) & \text{if } u_0 \leq u < u_{+1} \\ \mathbf{b_{+2}}(u) & \text{if } u_{+1} \leq u \leq u_{+2} \end{cases}$$

$$b''_{-2}(u_{-2}) = b'_{-2}(u_{-2}) = b_{-2}(u_{-2}) = 0$$
$$b''_{+2}(u_{+2}) = b'_{+2}(u_{+2}) = b_{+2}(u_{+2}) = 0$$

$$b_{-2}(u_{-1}) = b_{-1}(u_{-1})$$
$$b_{-1}(u_0) = b_{+1}(u_0)$$
$$b_{+1}(u_{+1}) = b_{+2}(u_{+1})$$

B-Splines



$$\mathbf{b}(u) = \begin{cases} \mathbf{b_{-2}}(u) & \text{if } u_{-2} \leq u < u_{-1} \\ \mathbf{b_{-1}}(u) & \text{if } u_{-1} \leq u < u_0 \\ \mathbf{b_{+1}}(u) & \text{if } u_0 \leq u < u_{+1} \\ \mathbf{b_{+2}}(u) & \text{if } u_{+1} \leq u \leq u_{+2} \end{cases}$$

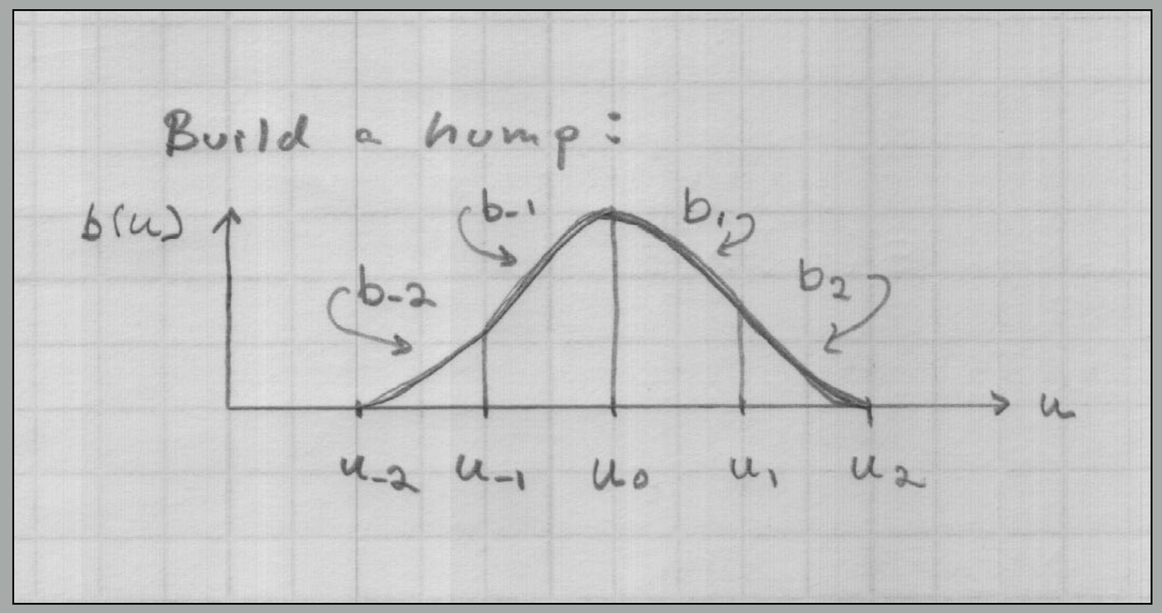
$$b''_{-2}(u_{-2}) = b'_{-2}(u_{-2}) = b_{-2}(u_{-2}) = 0 \quad \leftarrow 3 \text{ constraints}$$
$$b''_{+2}(u_{+2}) = b'_{+2}(u_{+2}) = b_{+2}(u_{+2}) = 0 \quad \leftarrow 3 \text{ constraints}$$

$$b_{-2}(u_{-1}) = b_{-1}(u_{-1})$$
$$b_{-1}(u_0) = b_{+1}(u_0) \quad \leftarrow$$
$$b_{+1}(u_{+1}) = b_{+2}(u_{+1})$$

Repeat for b' and b''
$3 \times 3 = 9$ constraints

Total 15 constraints ..... need one more

B-Splines



$$\mathbf{b}(u) = \begin{cases} \mathbf{b_{-2}}(u) & \text{if } u_{-2} \leq u < u_{-1} \\ \mathbf{b_{-1}}(u) & \text{if } u_{-1} \leq u < u_0 \\ \mathbf{b_{+1}}(u) & \text{if } u_0 \leq u < u_{+1} \\ \mathbf{b_{+2}}(u) & \text{if } u_{+1} \leq u \leq u_{+2} \end{cases}$$

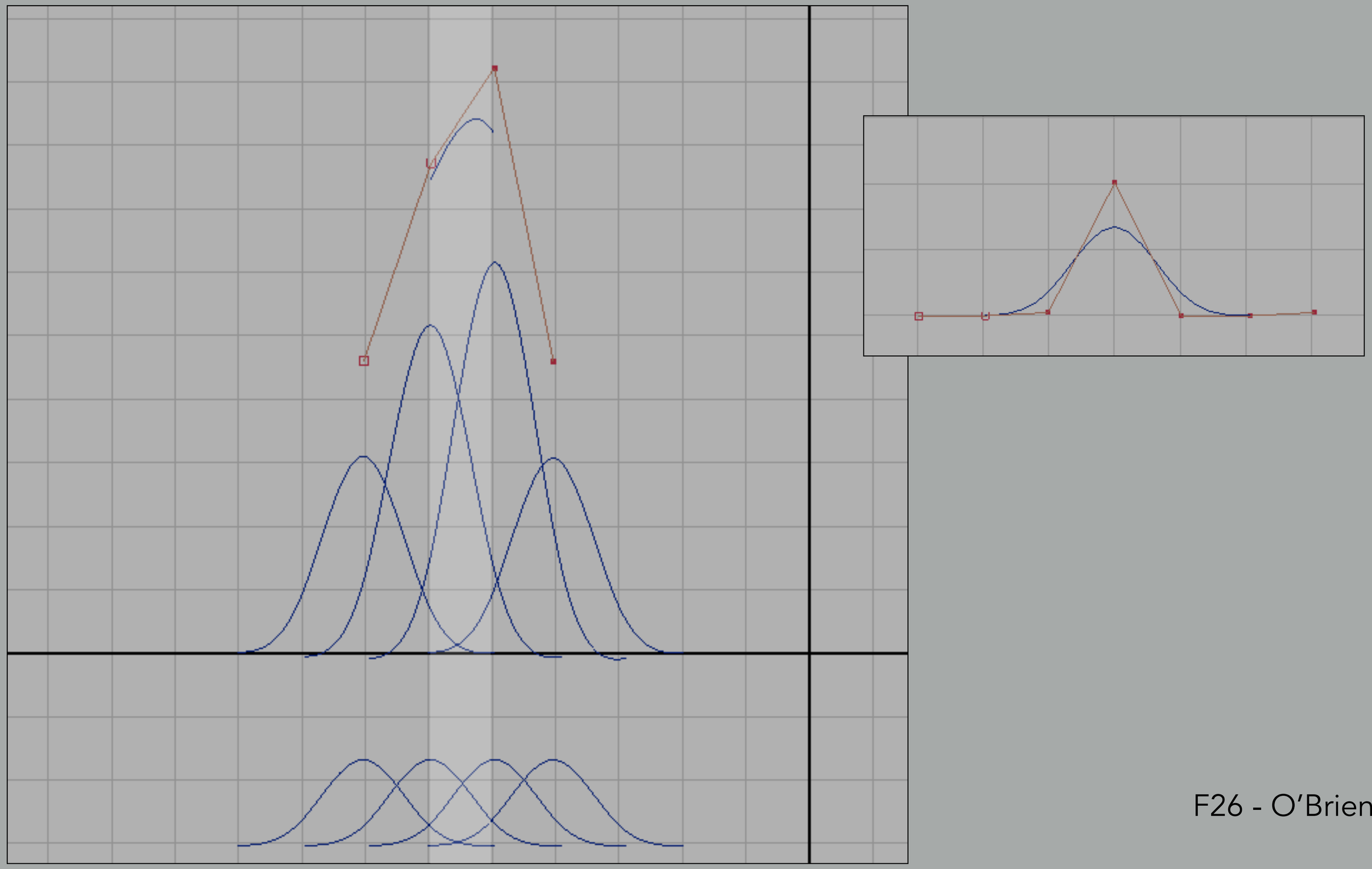
$b''_{-2}(u_{-2}) = b'_{-2}(u_{-2}) = b_{-2}(u_{-2}) = 0$ ← 3 constraints
$b''_{+2}(u_{+2}) = b'_{+2}(u_{+2}) = b_{+2}(u_{+2}) = 0$ ← 3 constraints

$b_{-2}(u_{-1}) = b_{-1}(u_{-1})$
$b_{-1}(u_0) = b_{+1}(u_0)$ ← [Repeat for b' and b'' 3×3=9 constraints
$b_{+1}(u_{+1}) = b_{+2}(u_{+1})$

$b_{-2}(u_{-2}) + b_{-1}(u_{-1}) + b_{+1}(u_0) + b_{+2}(u_{+1}) = 1$ ← 1 constraint (convex hull)

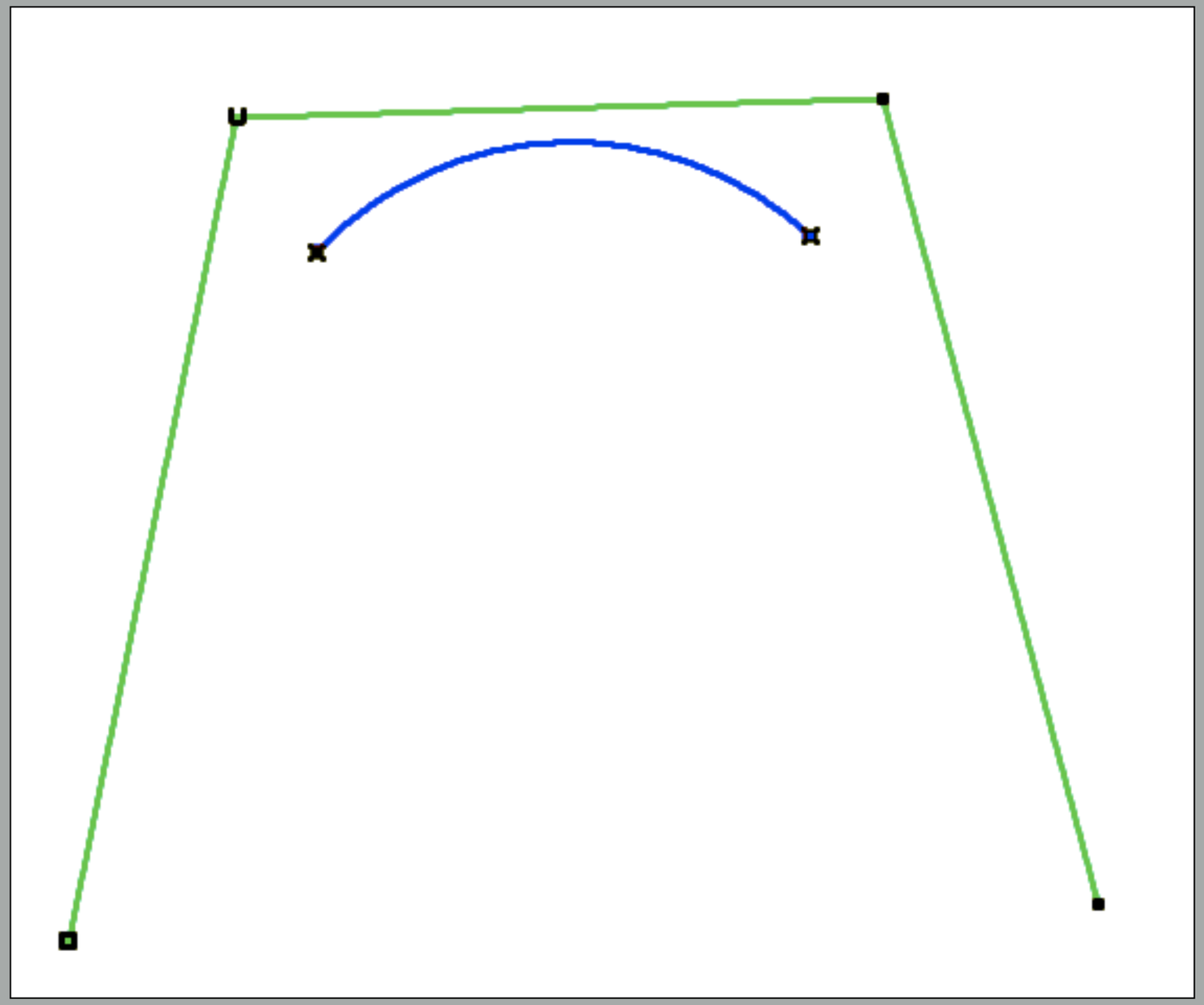
Total 16 constraints

B-Splines

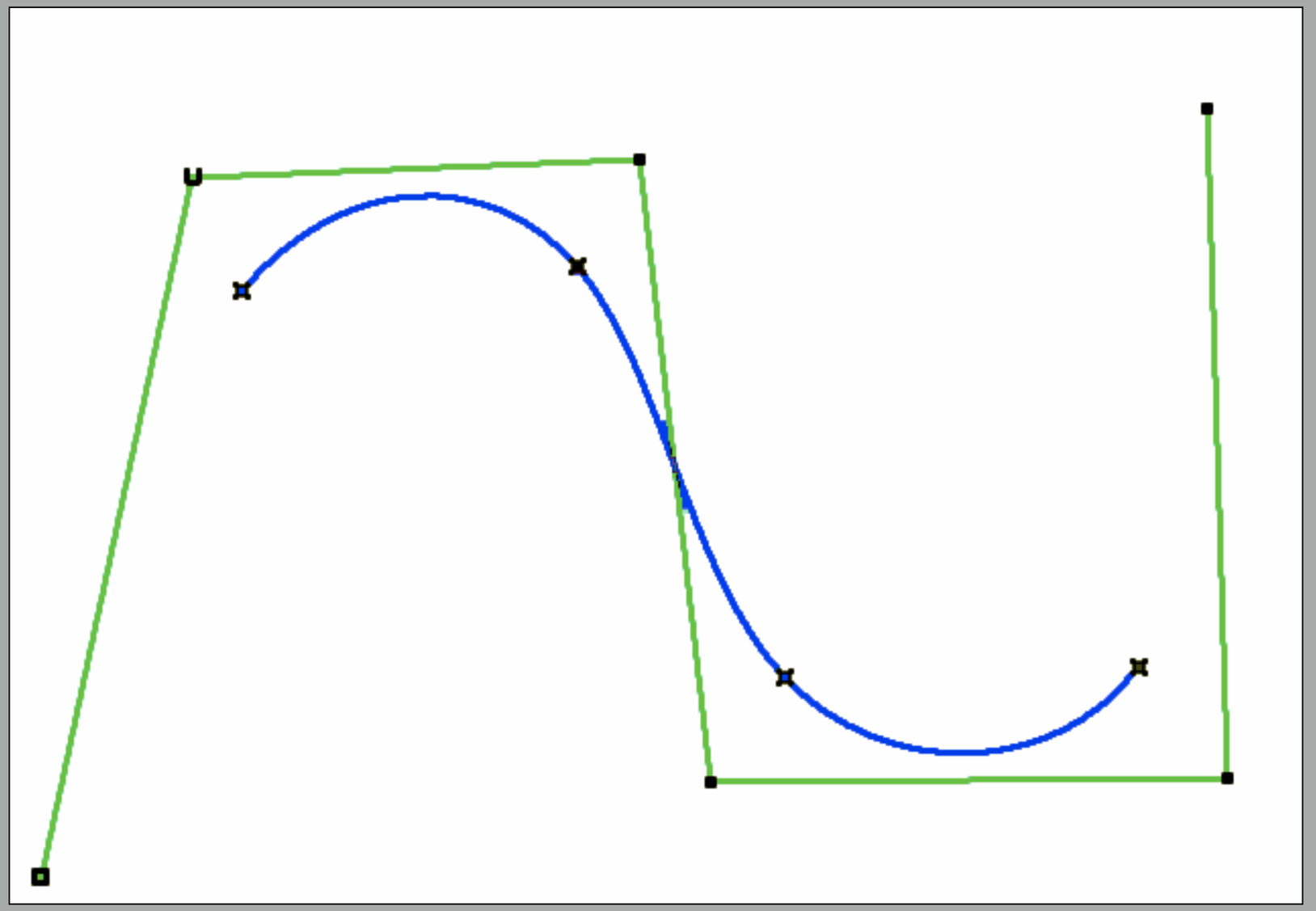


F26 - O’Brien

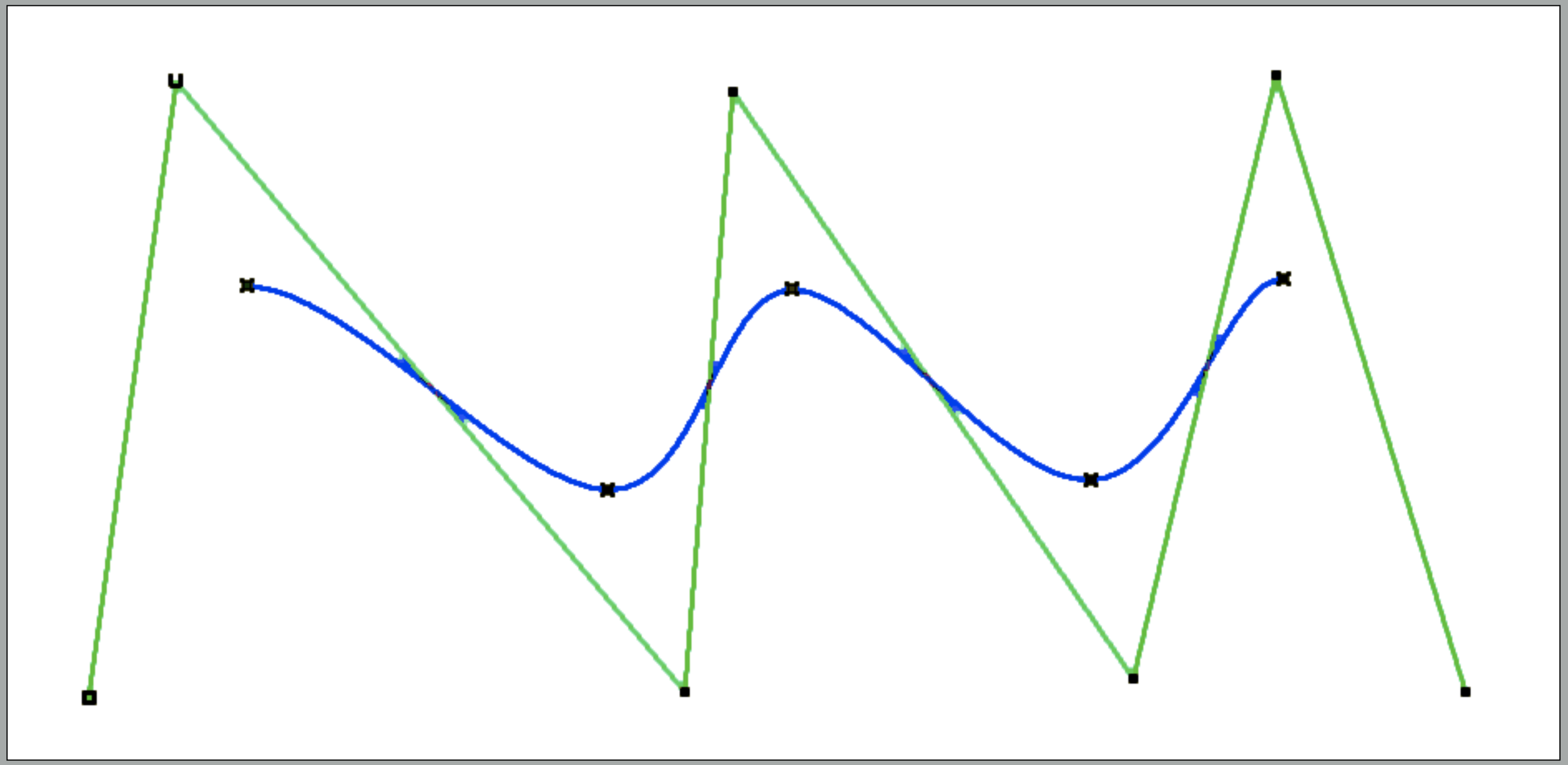
B-Splines



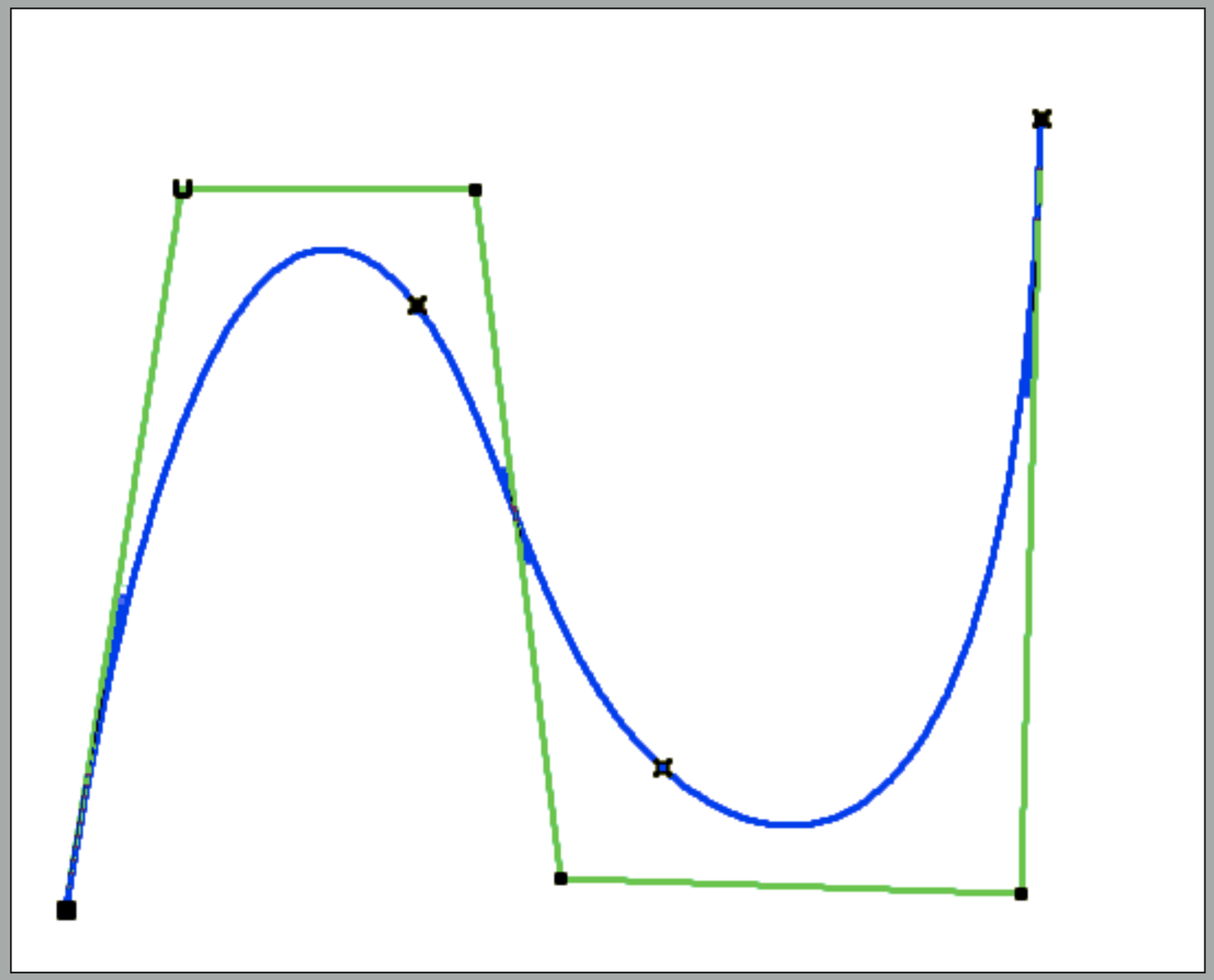
B-Splines



B-Splines



B-Splines



Example with end knots repeated

B-Splines

Build a curve w/ overlapping bumps

Continuity

- Inside bumps C^2
- Bumps "fade out" with C^2 continuity

Boundaries

- Circular
- Repeat end points
- Extra end points

B-Splines

- Notation
  - The basis functions are the $b_i(u)$
  - "Hump" functions are the concatenated function
    - Sometimes the humps are called basis... can be confusing
  - The u_i are the knot locations
  - The weights on the hump/basis functions are control points

B-Splines

- Similar construction method can give higher continuity with higher degree polynomials
- Repeating knots drops continuity
  - Limit as knots approach each other
- Still cubics, so conversion to other cubic basis is just a matrix multiplication

B-Splines

- Geometric construction
  - Due to Cox and de Boor
  - My own notation, beware if you compare w/ text

- Let hump centered on u_i be $N_{i,4}(u)$

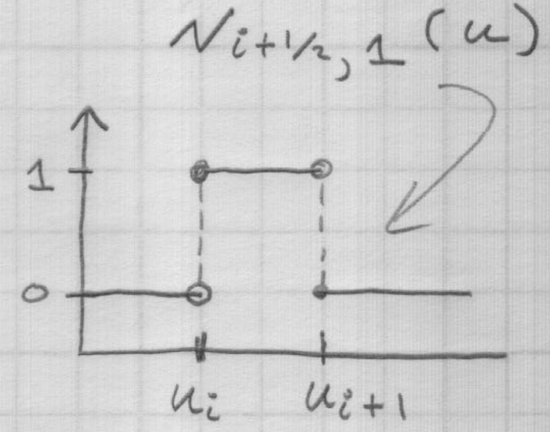
Cubic is order 4

$N_{i,k}(u)$ Is order k hump, centered at u_i

Note: i is integer if k is even
else $(i + 1/2)$ is integer

B-Splines

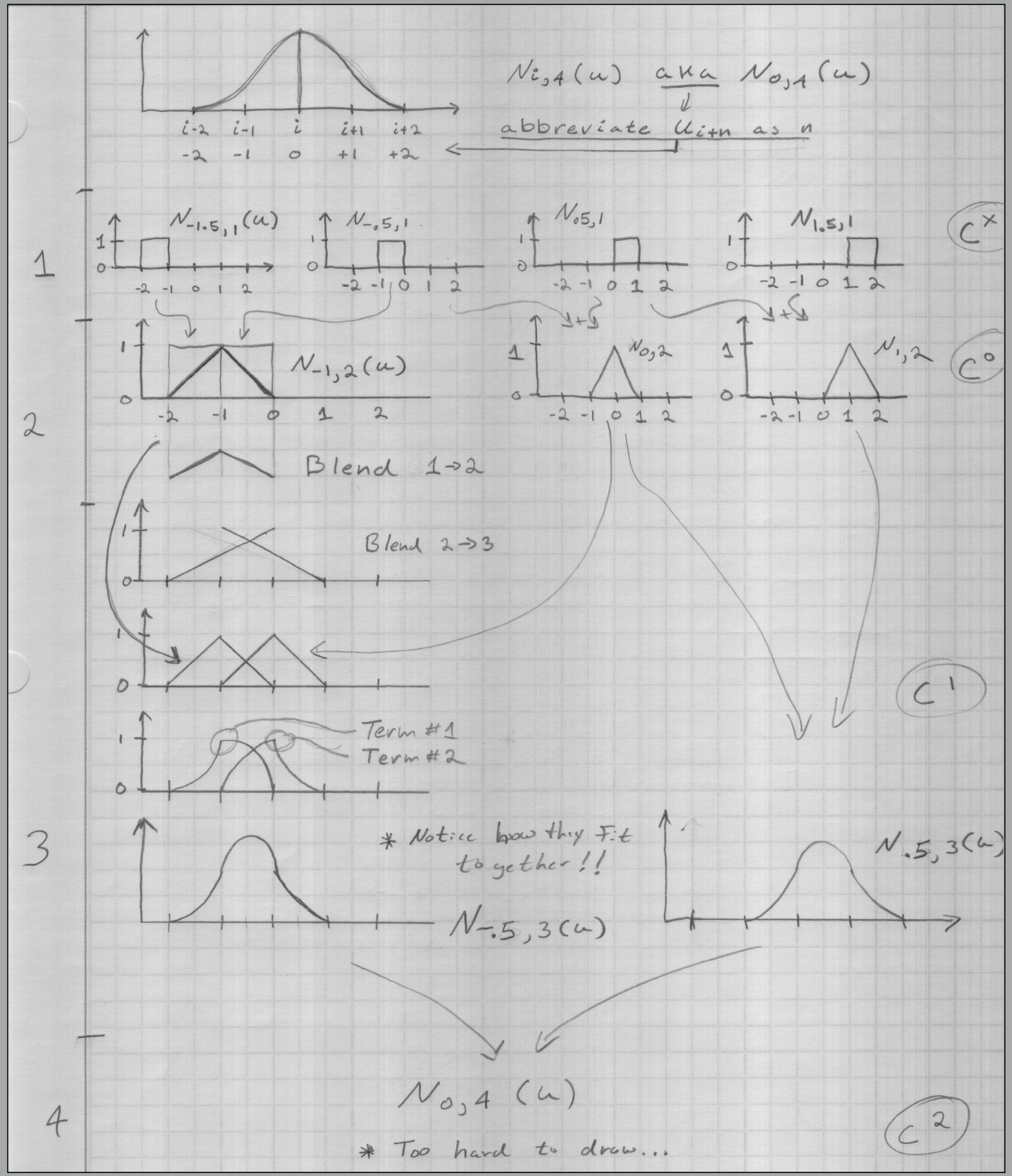
$$N_{i,1}(u) = \begin{cases} 1 & \text{if } u_{i-1/2} \le u < u_{i+1/2} \\ 0 & \text{else} \end{cases}$$



$$N_{i,k}(u) = \underbrace{\frac{(u - u_{i-k/2})\, N_{i-1/2,k-1}(u)}{u_{i+k/2-1} - u_{i-k/2}}}_{\text{"Term \#1"}} + \underbrace{\frac{(u_{i+k/2} - u)\, N_{i+1/2,k-1}(u)}{u_{i+k/2} - u_{i-k/2+1}}}_{\text{"Term \#2"}}$$

$k \ge 2$

recursive defn.


$N_{i,4}(u)$ aka $N_{0,4}(u)$
abbreviate u_{i+n} as n
i-2 i-1 i i+1 i+2
-2 -1 0 +1 +2
1
$N_{-1.5,1}(u)$
$N_{-.5,1}$
$N_{.5,1}$
$N_{1.5,1}$
C^{-1}
2
$N_{-1,2}(u)$
$N_{0,2}$
$N_{1,2}$
C^0
Blend 1→2
Blend 2→3
Term #1
Term #2
C^1
3
* Notice how they fit together!!
$N_{-.5,3}(u)$
$N_{.5,3}(u)$
$N_{0,4}(u)$
* Too hard to draw...
C^2
4

NURBS

- **N**on**u**niform **R**ational **B**-**S**plines
  - Basically B-Splines using homogeneous coordinates
  - Transform under perspective projection
  - A bit of extra control

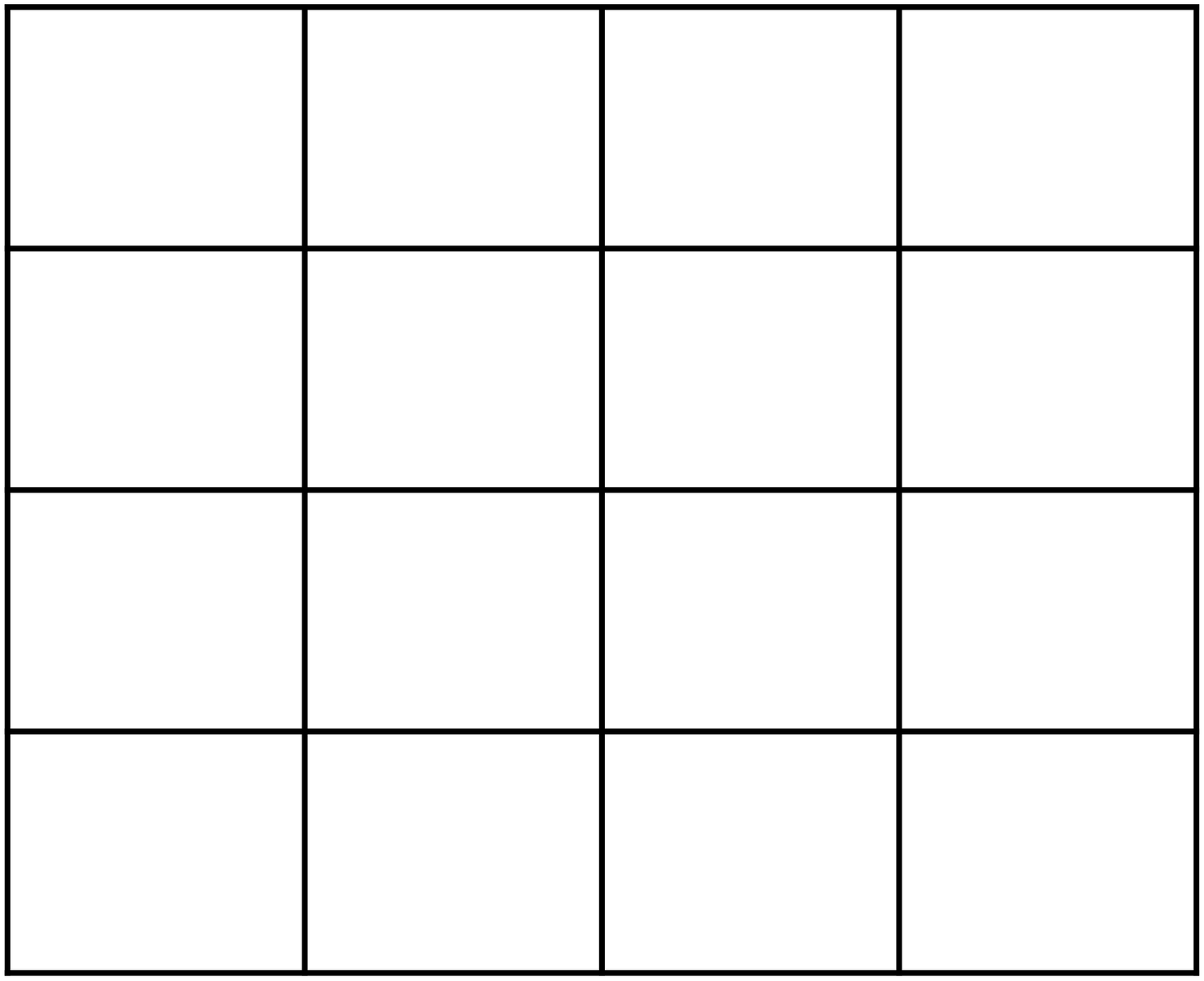
NURBS

$$\mathbf{p}_i = \begin{bmatrix} p_{ix} \\ p_{iy} \\ p_{iz} \\ p_{iw} \end{bmatrix} \qquad \mathbf{x}(u) = \frac{\sum_i \begin{bmatrix} p_{ix} \\ p_{iy} \\ p_{iz} \end{bmatrix} N_i(u)}{\sum_i p_{iw} N_i(u)}$$

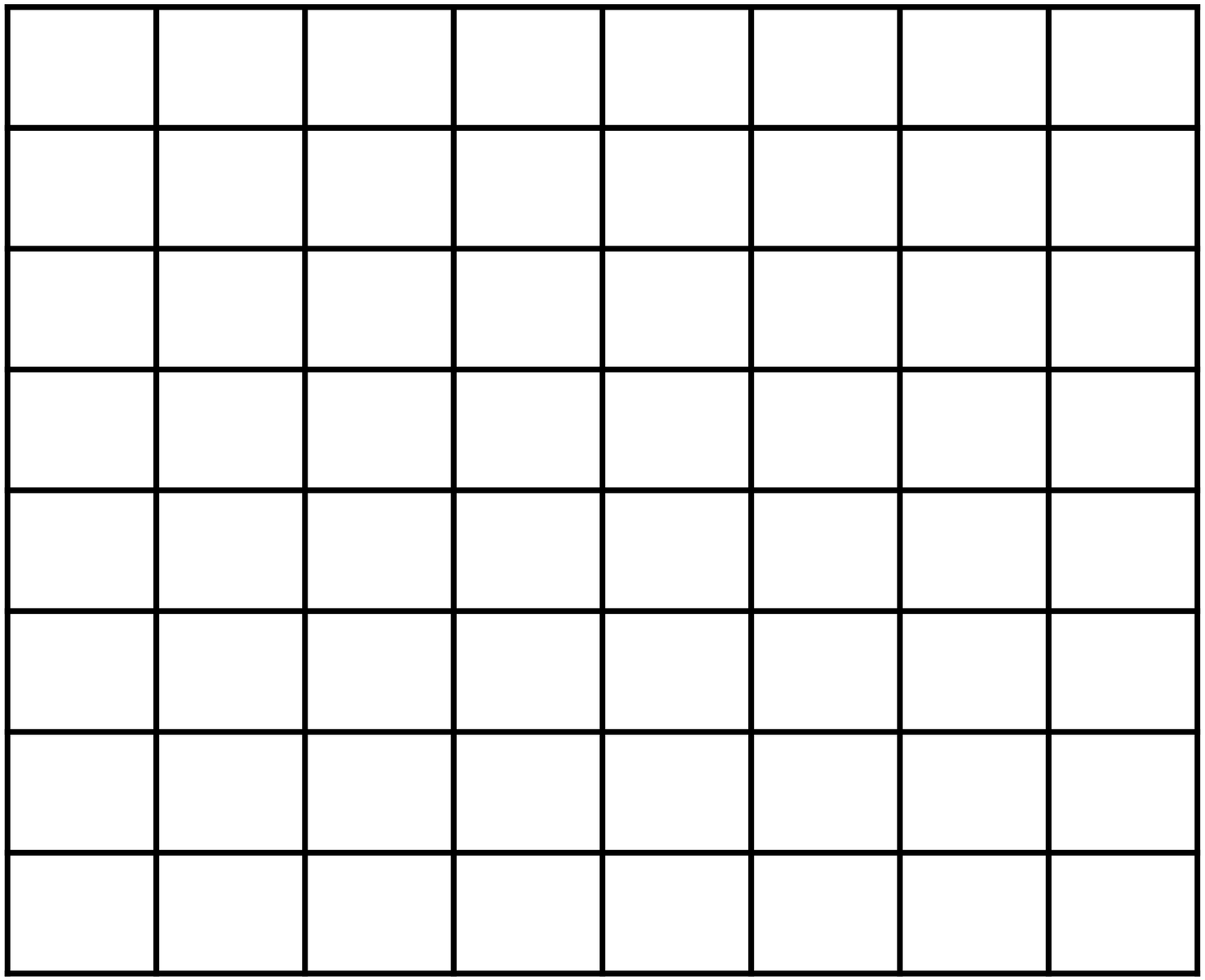
- Non-linear in the control points
- The p_{iw} are sometimes called "thumb weights" because increasing the value will pull curve toward corresponding control point.

Catmull-Clark Subdivision (Generalization of B-Splines)

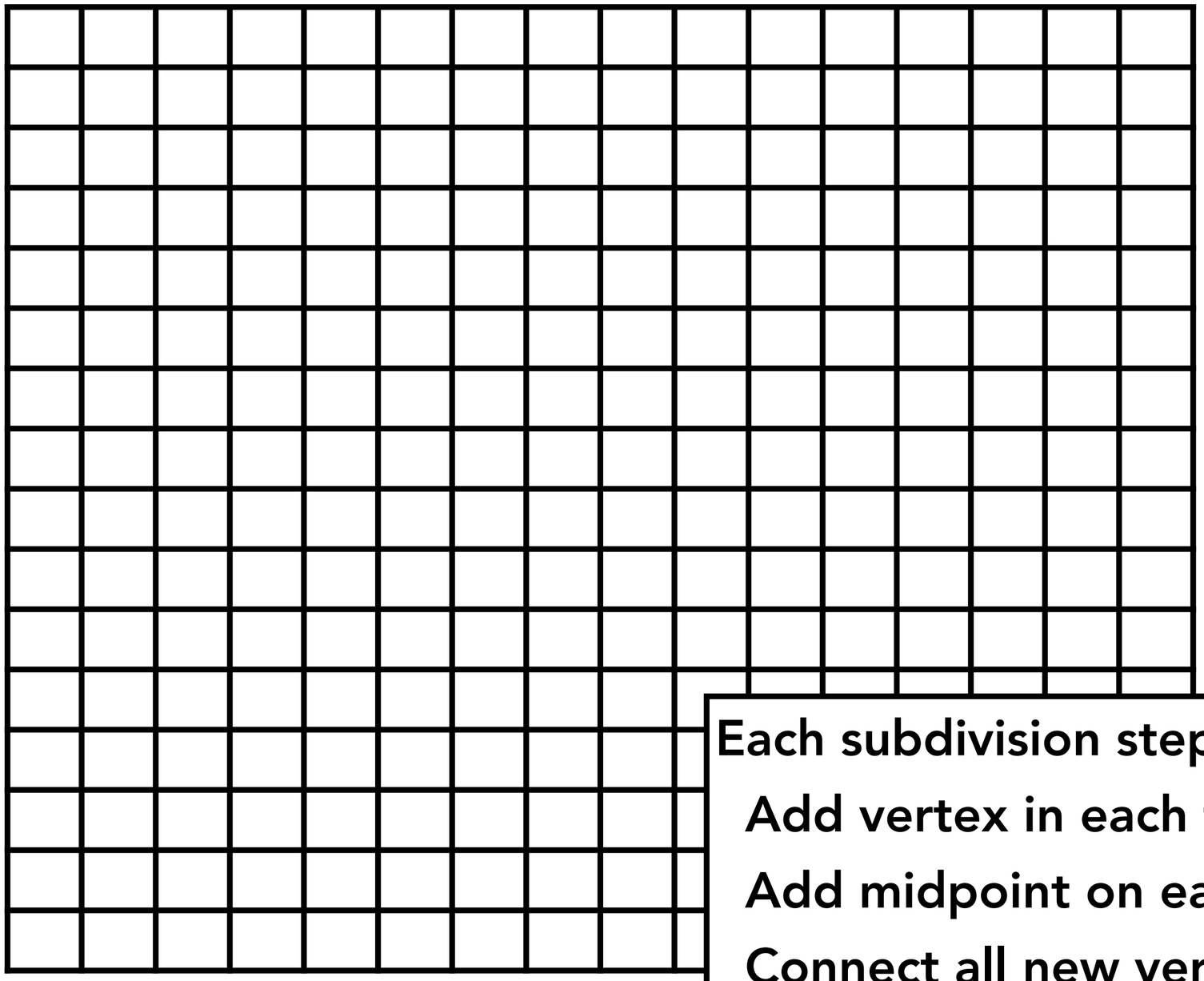
B-Spline Subdivision (Regular Quad Mesh)



B-Spline Subdivision (Regular Quad Mesh)



B-Spline Subdivision (Regular Quad Mesh)

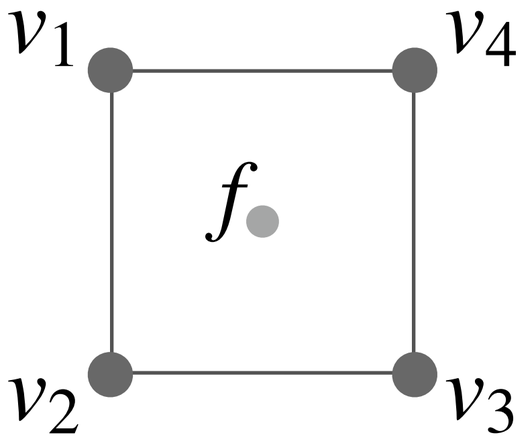


Each subdivision step:

- **Add vertex in each face**
- **Add midpoint on each edge**
- **Connect all new vertices**

B-Spline Vertex Update Rules (Quad Mesh)

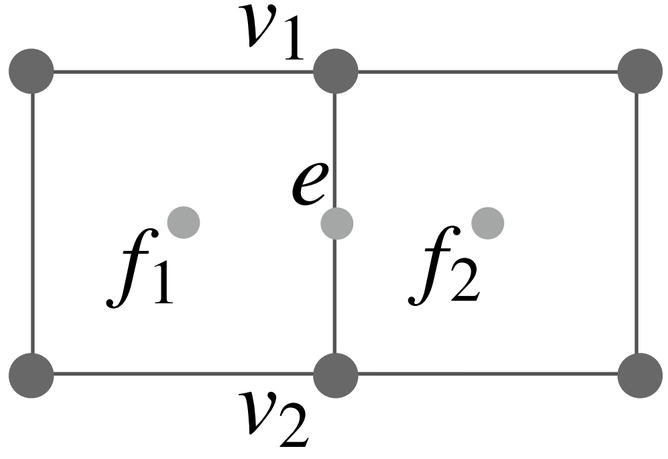
Face point

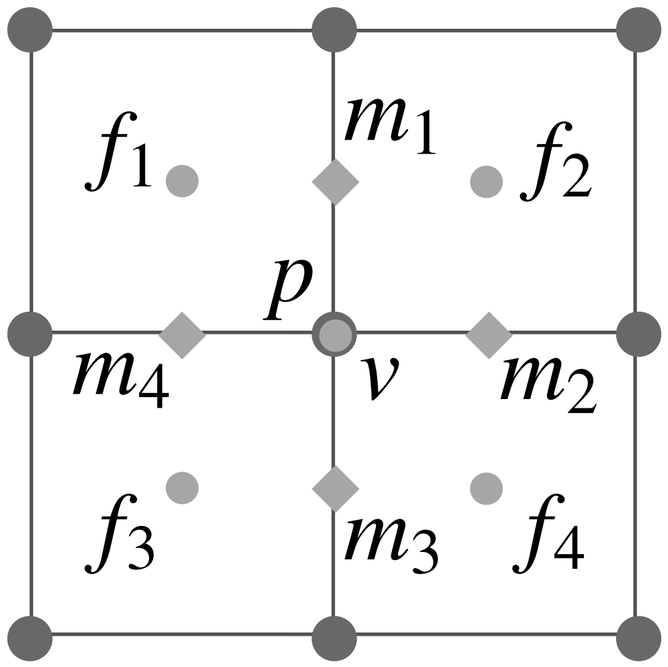


$$f = \frac{v_1 + v_2 + v_3 + v_4}{4}$$

$$e = \frac{v_1 + v_2 + f_1 + f_2}{4}$$

Edge point





Vertex point

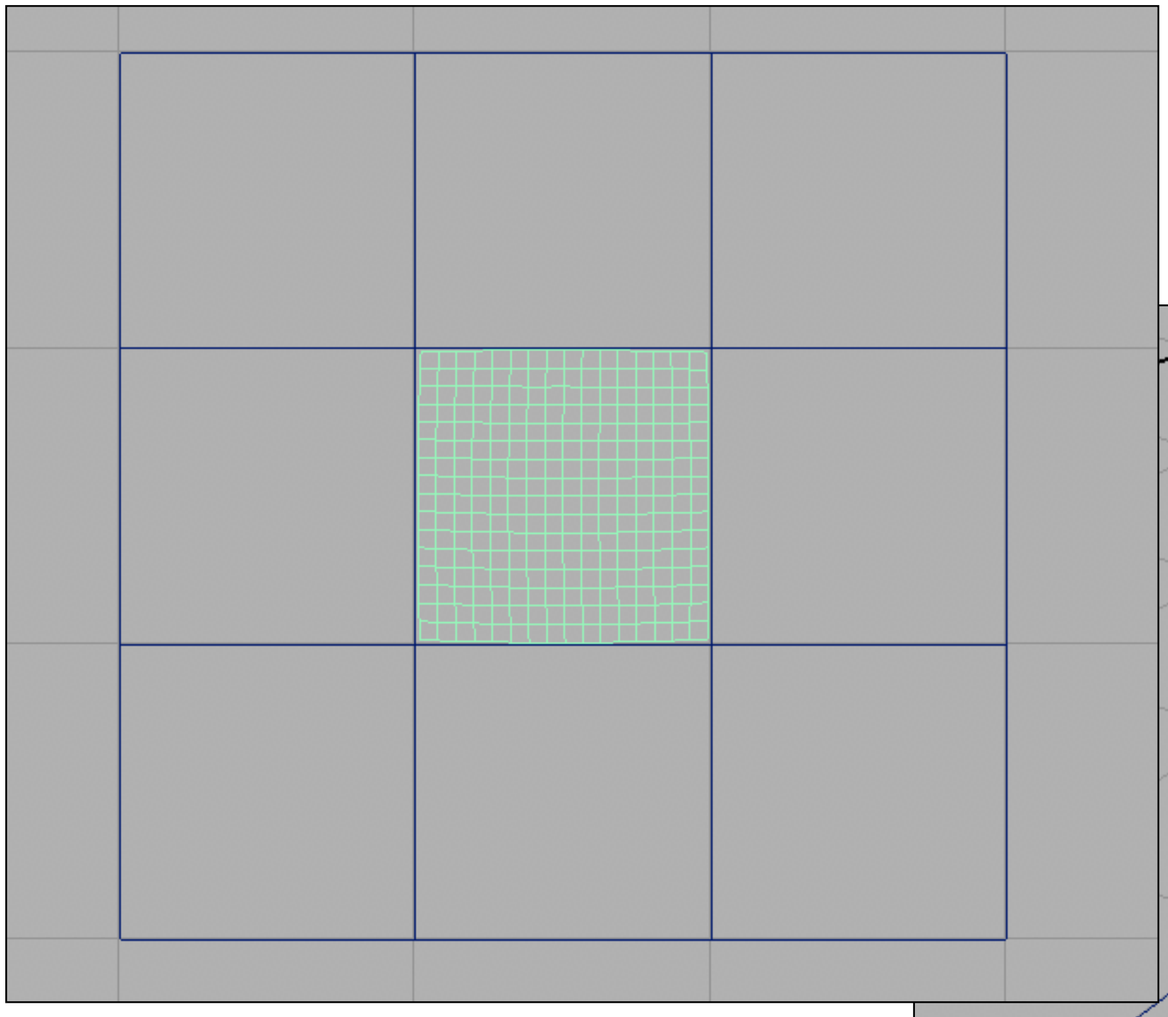
$$v = \frac{f_1 + f_2 + f_3 + f_4 + 2(m_1 + m_2 + m_3 + m_4) + 4p}{16}$$

m midpoint of edge, not "edge point"

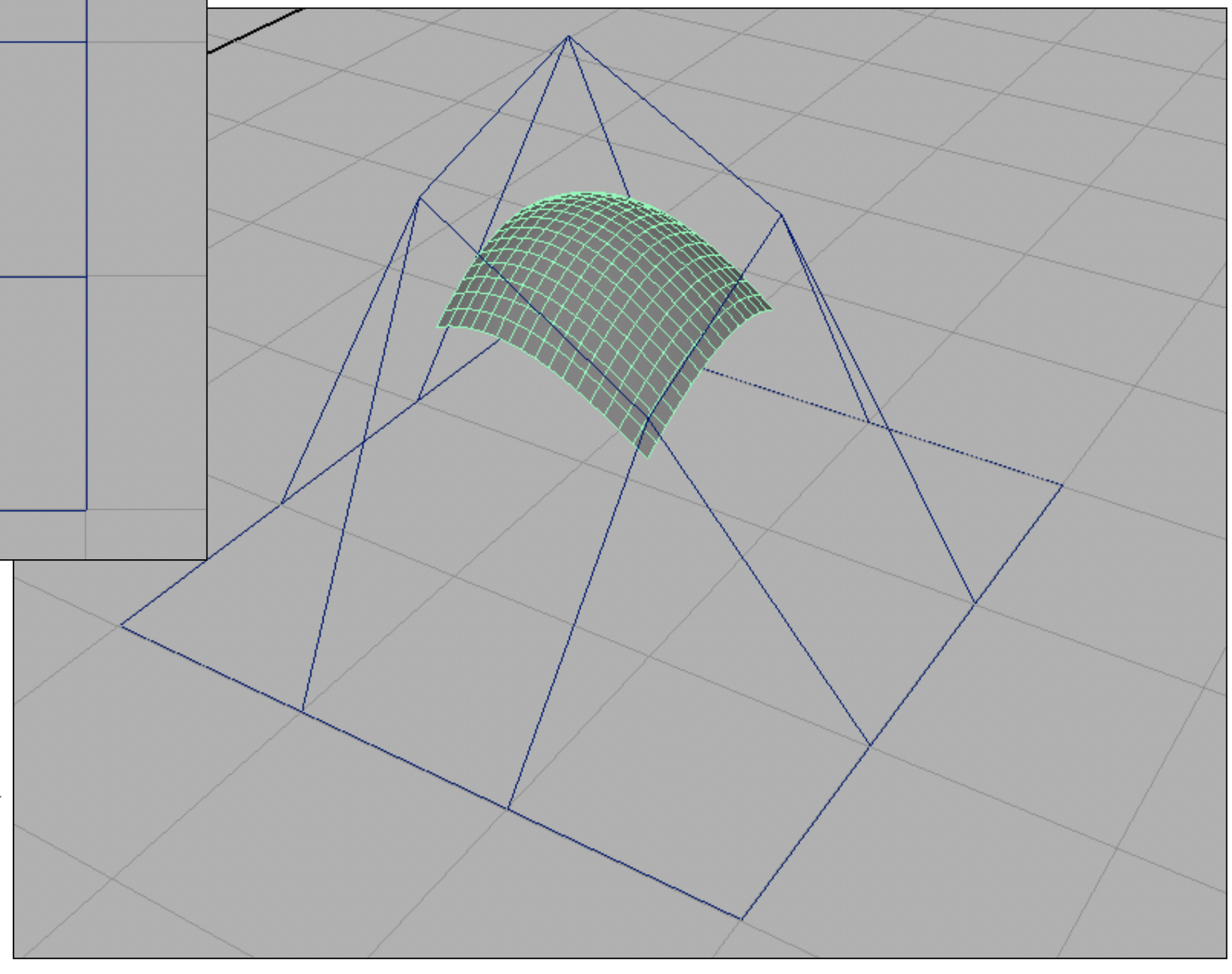
p old "vertex point"

These rules are B-Spline subdivision.

Regular B-Spline Subdivision

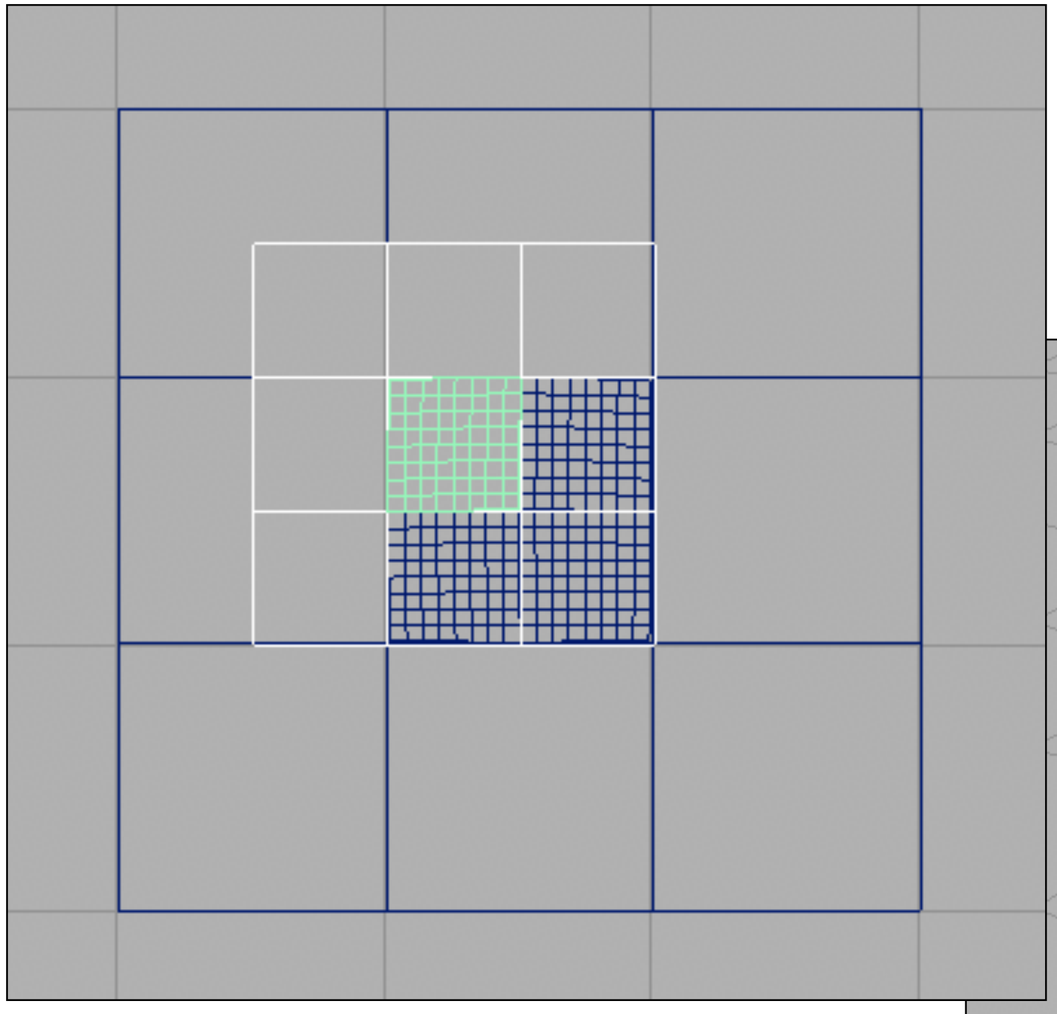


← Orthographic top-down view

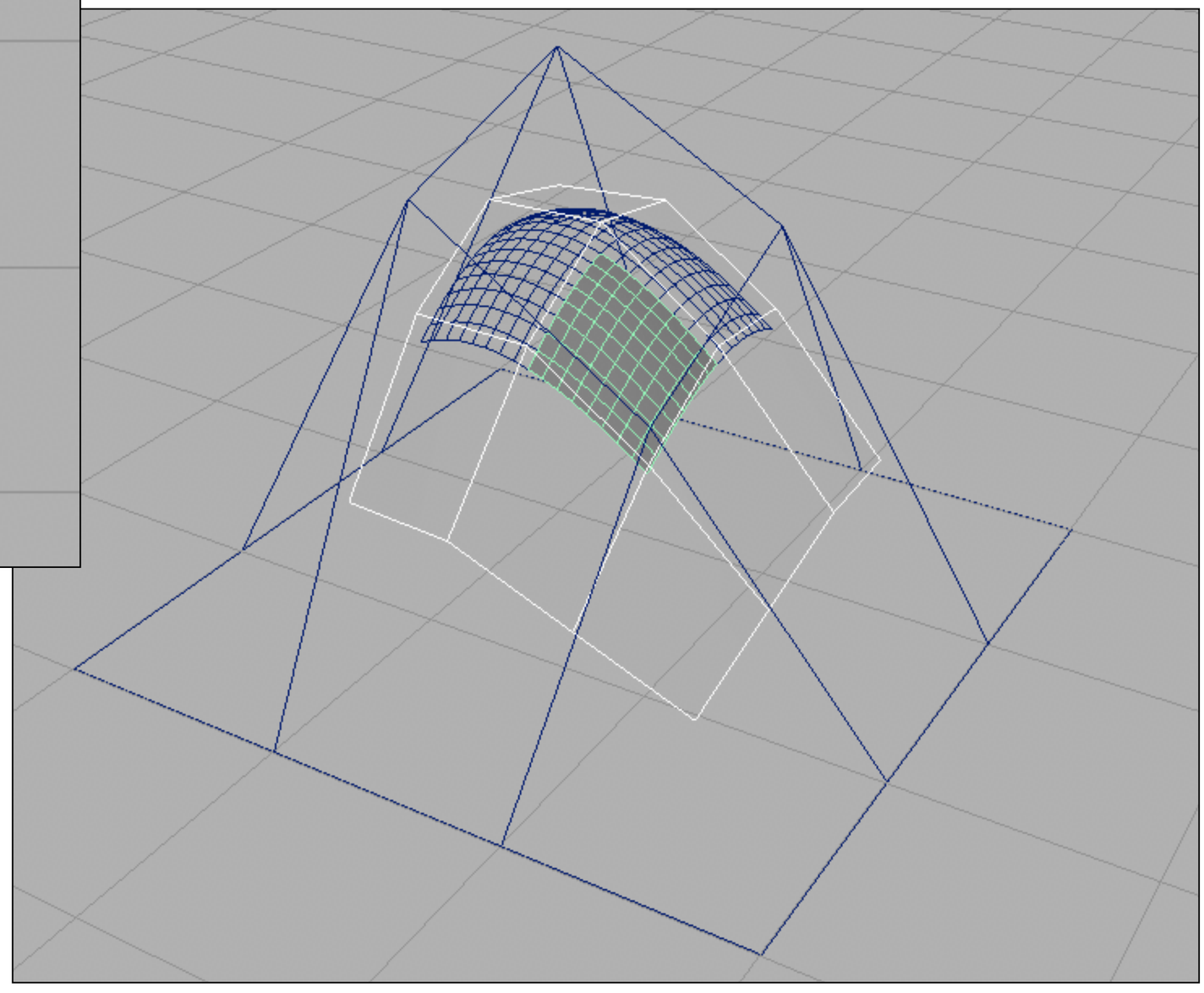


3D Perspective view →

Regular B-Spline Subdivision

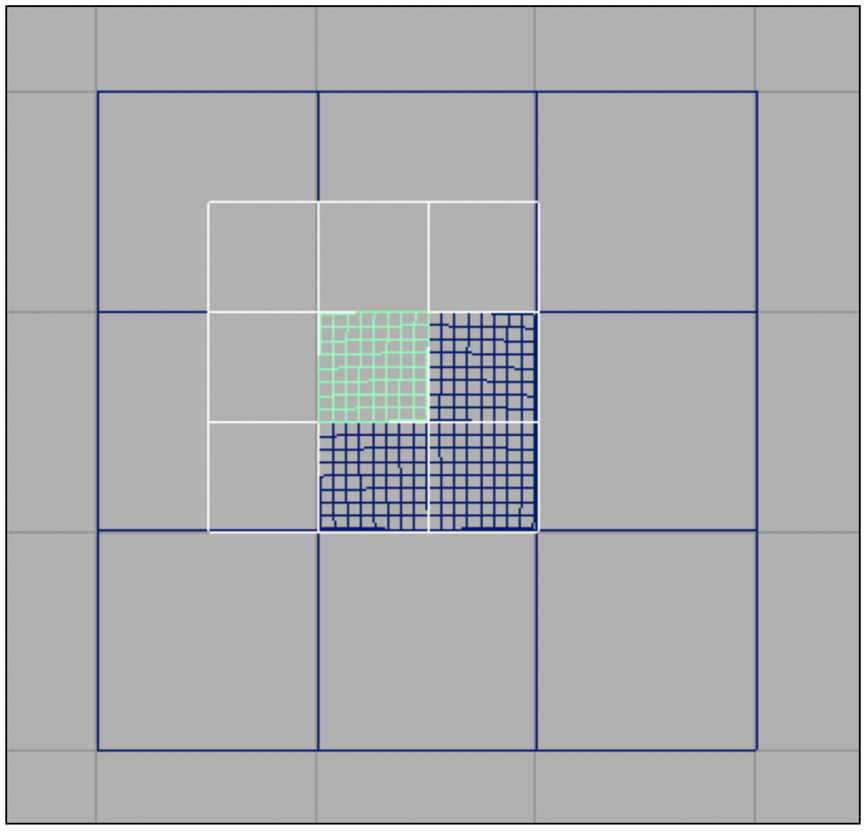


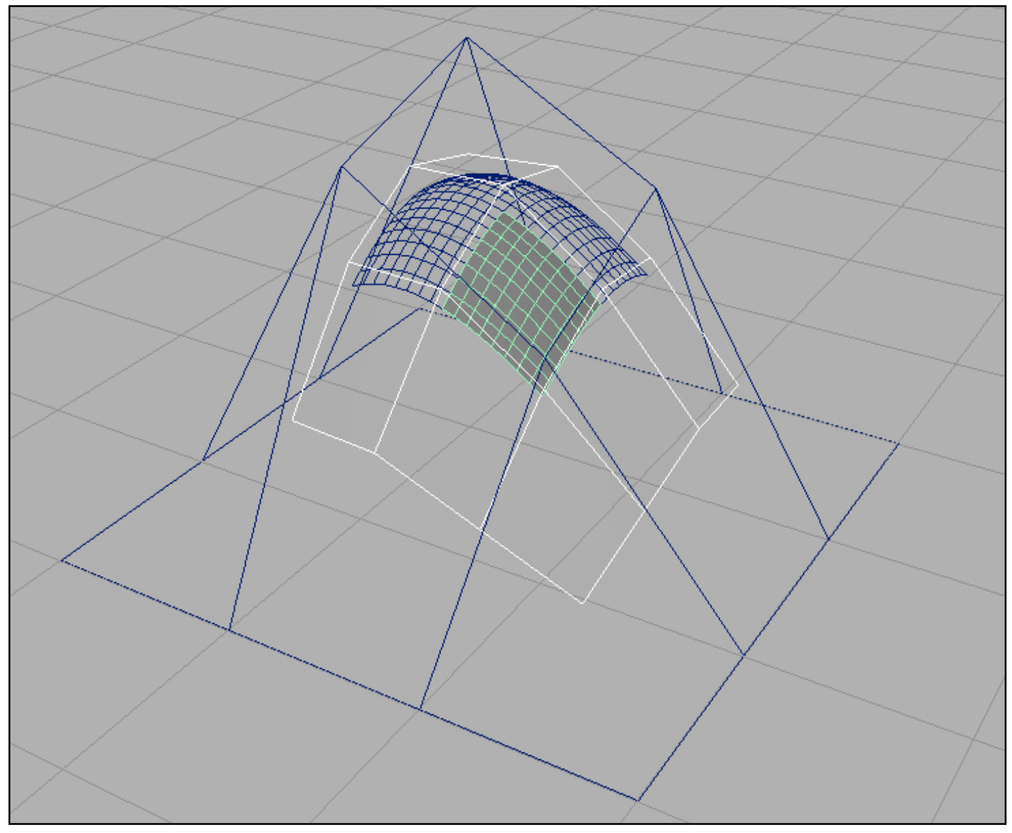
← Orthographic top-down view



3D Perspective view →

Regular B-Spline Subdivision



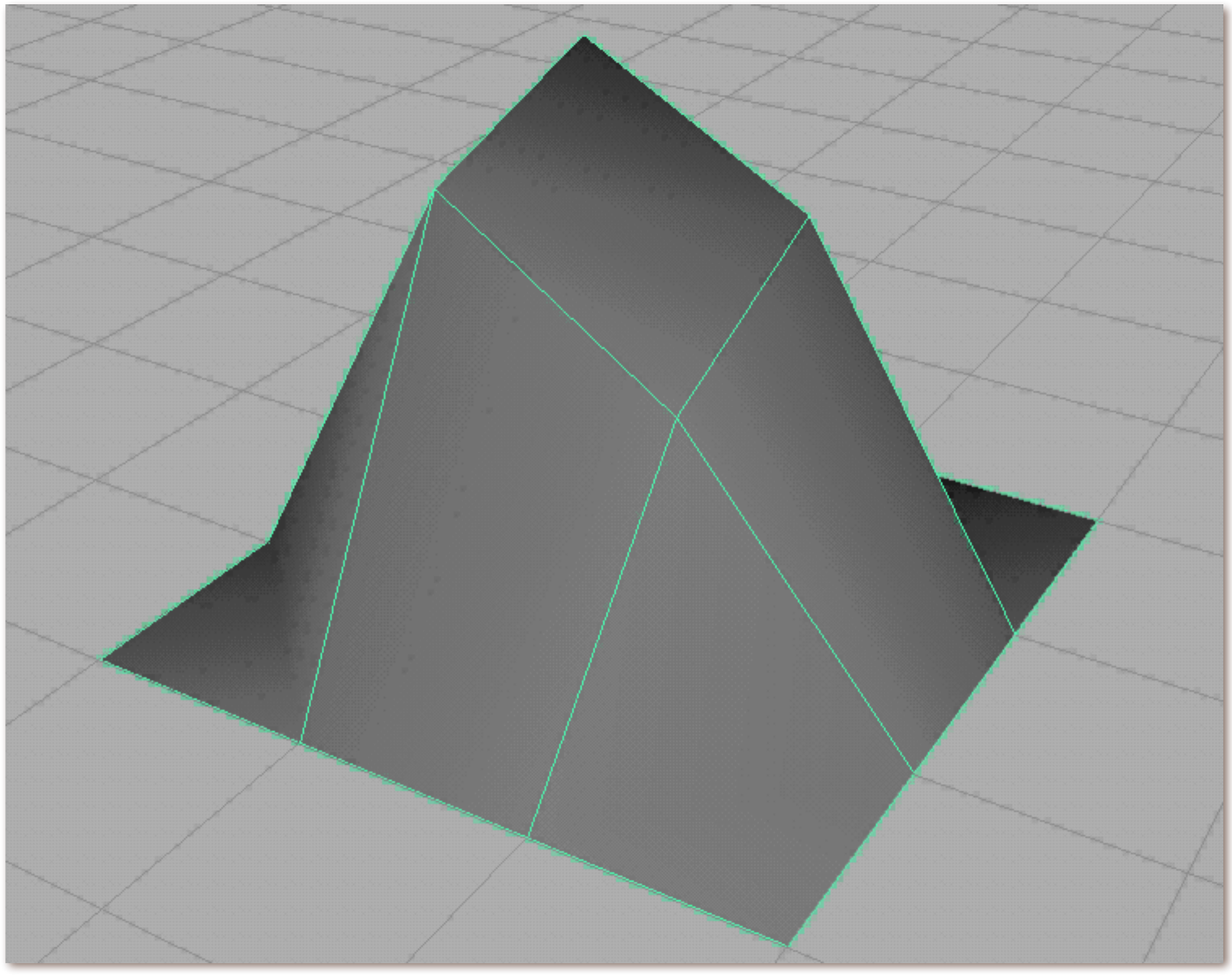


$$\mathbf{x}(u,v) = [1, u, u^2, u^3]\beta_B \, \mathbf{P} \, \beta_B^T [1, v, v^2, v^3]^T$$

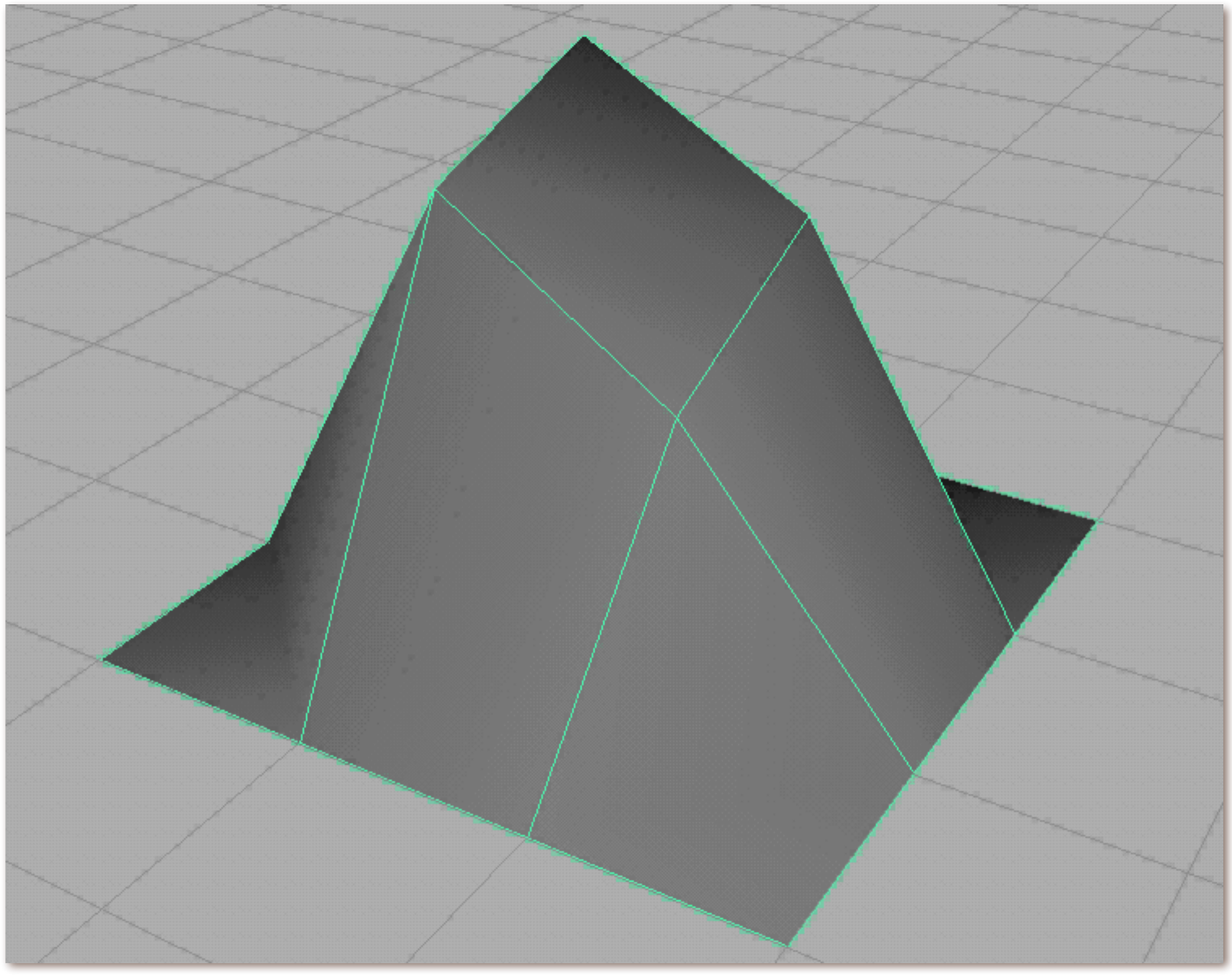
$$\mathbf{P}_{11} = \mathbf{H}_{B1} \mathbf{P} \mathbf{H}_{B1}^T$$

Same setup as Bézier, different H matrices.

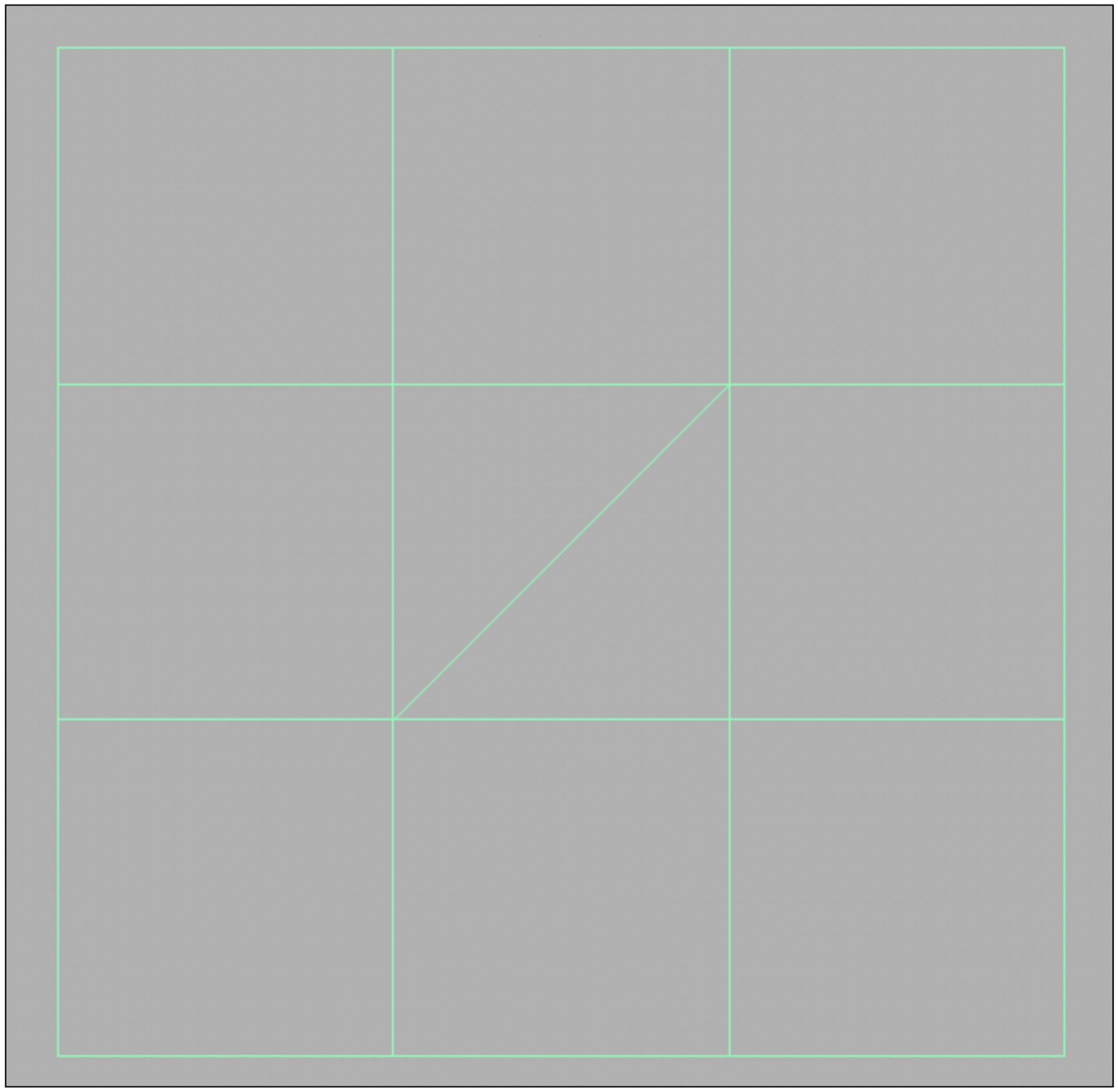
Regular B-Spline Subdivision



Regular B-Spline Subdivision

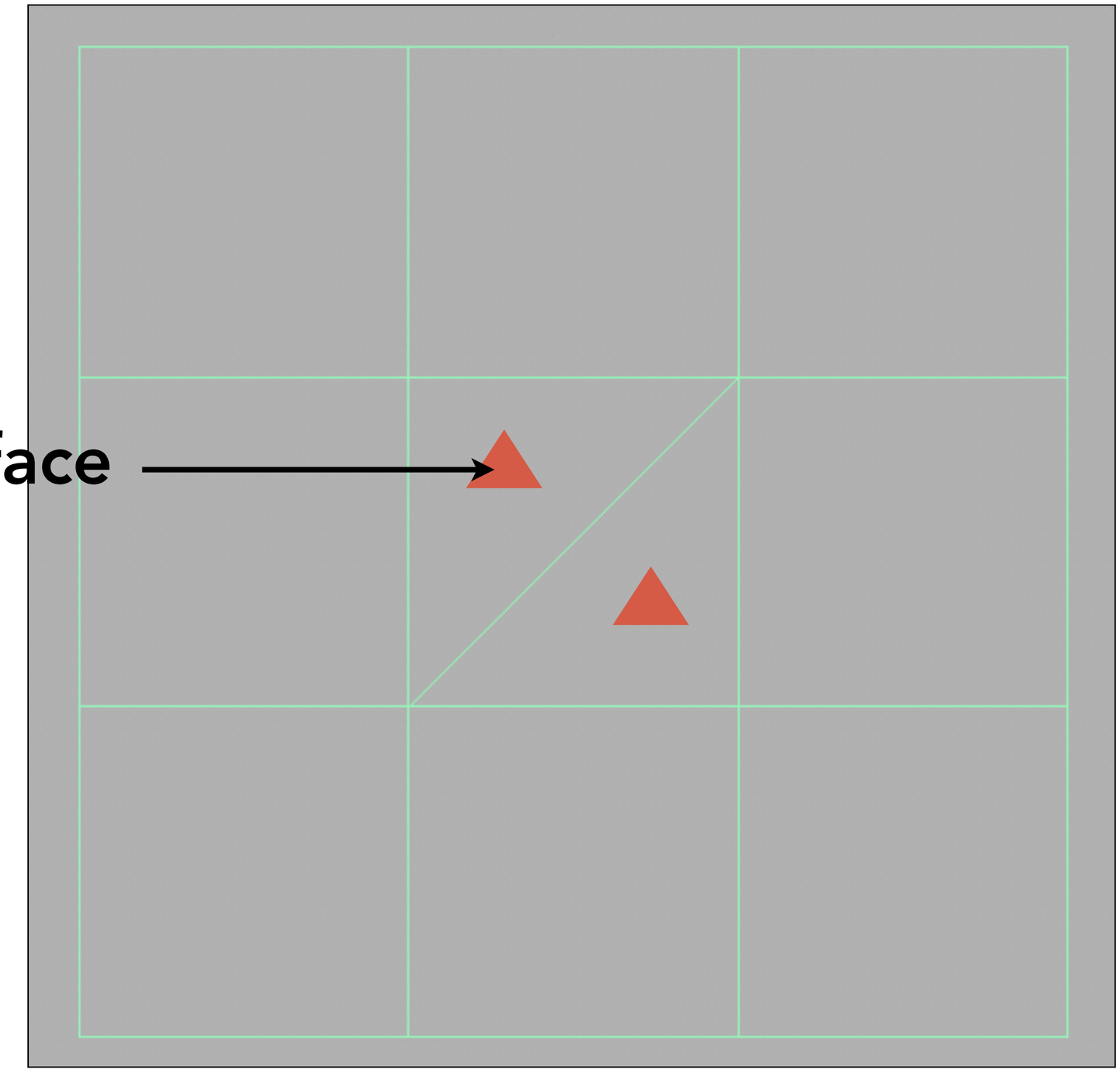


Catmull-Clark Subdivision (General Mesh)

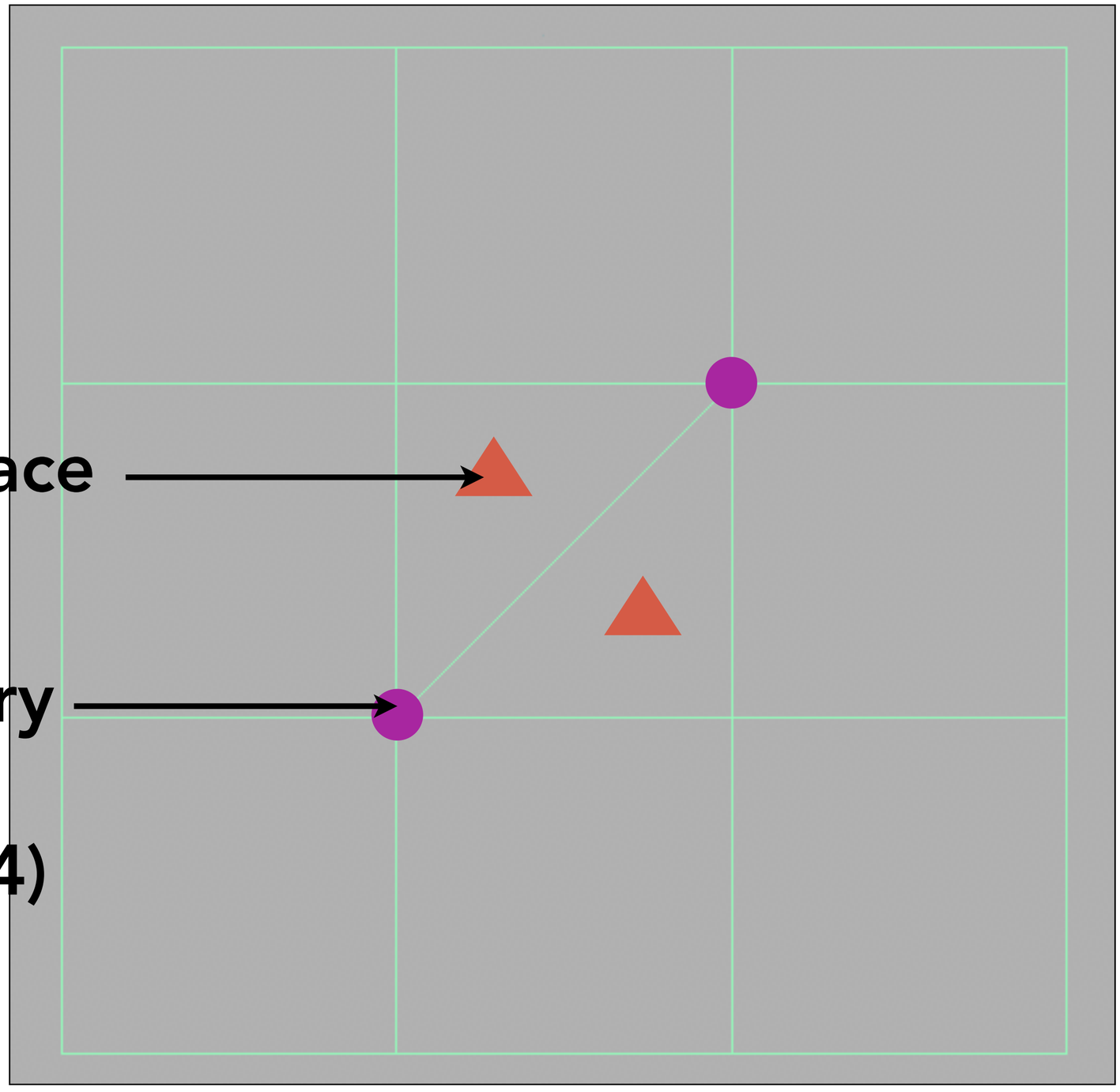


Catmull-Clark Subdivision (General Mesh)

Non-quad face



Catmull-Clark Subdivision (General Mesh)



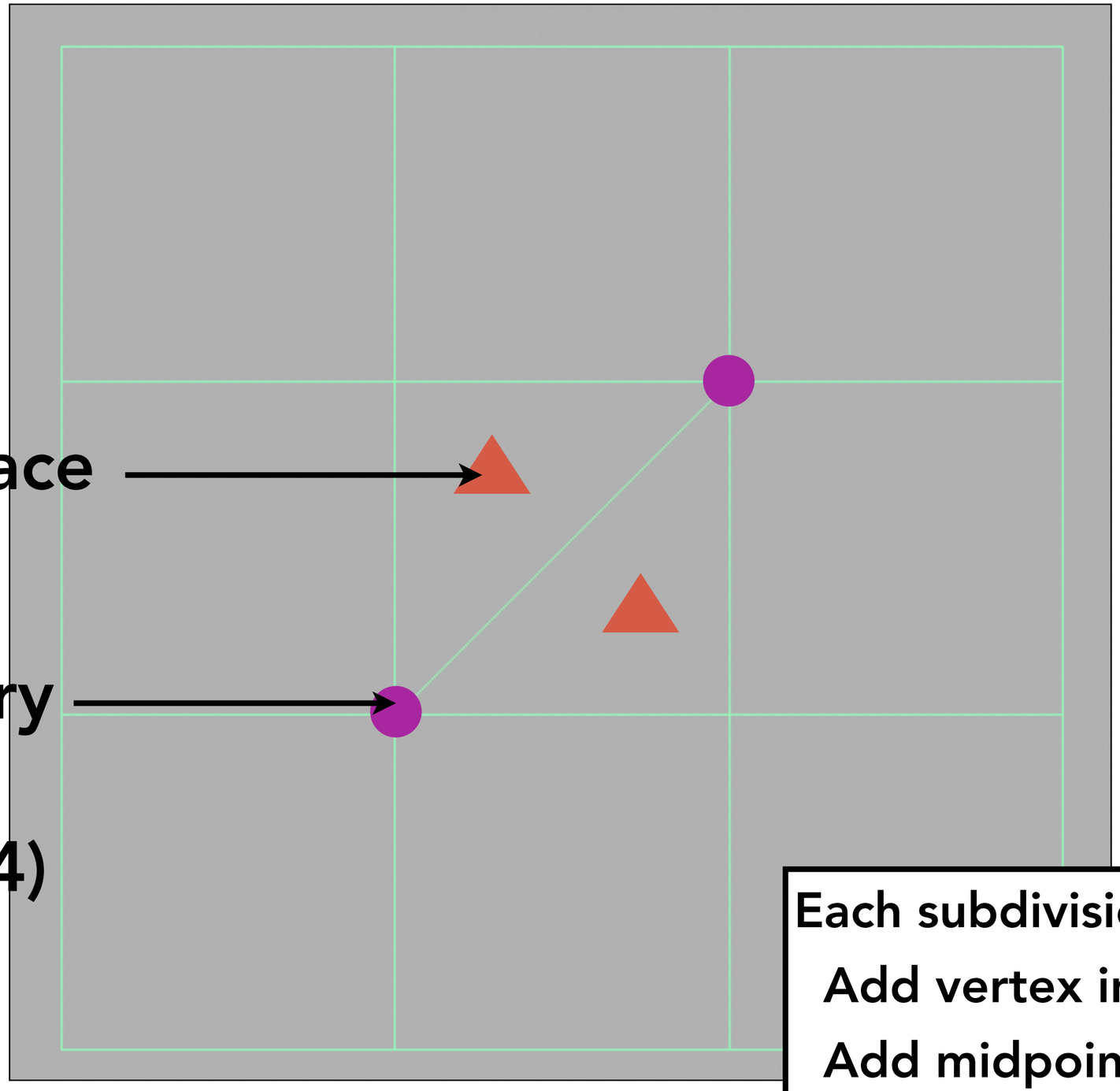
Non-quad face

Extraordinary vertex (valence != 4)

Catmull-Clark Subdivision (General Mesh)

Non-quad face

Extraordinary vertex (valence != 4)

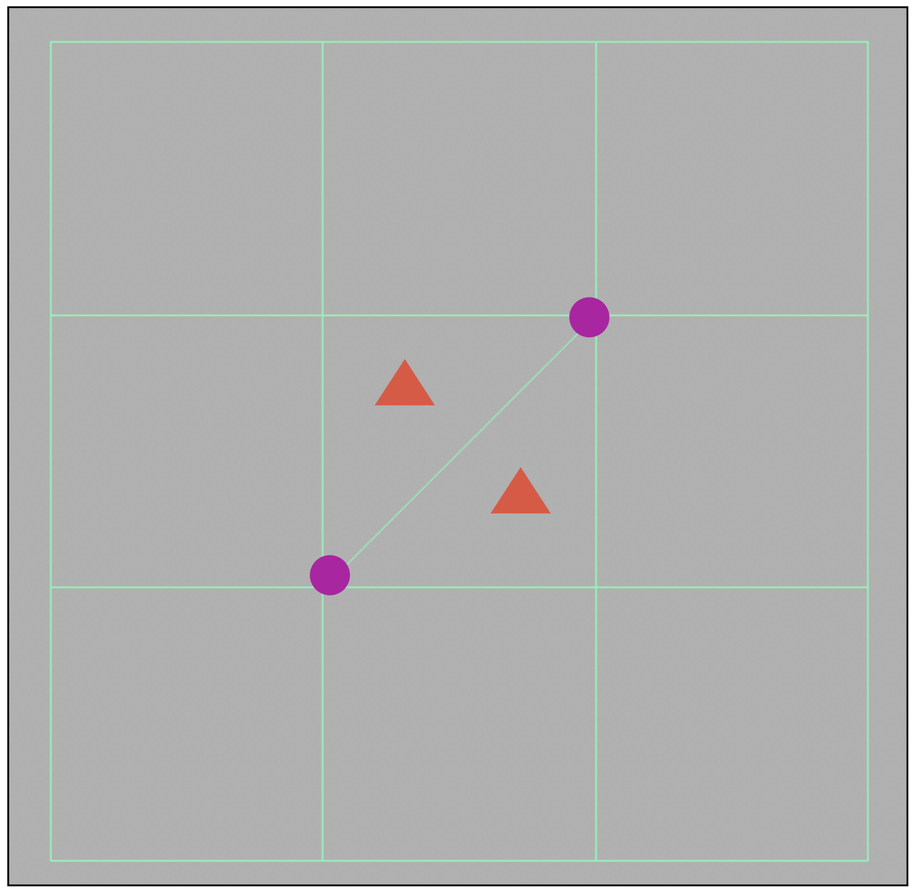


Each subdivision step:

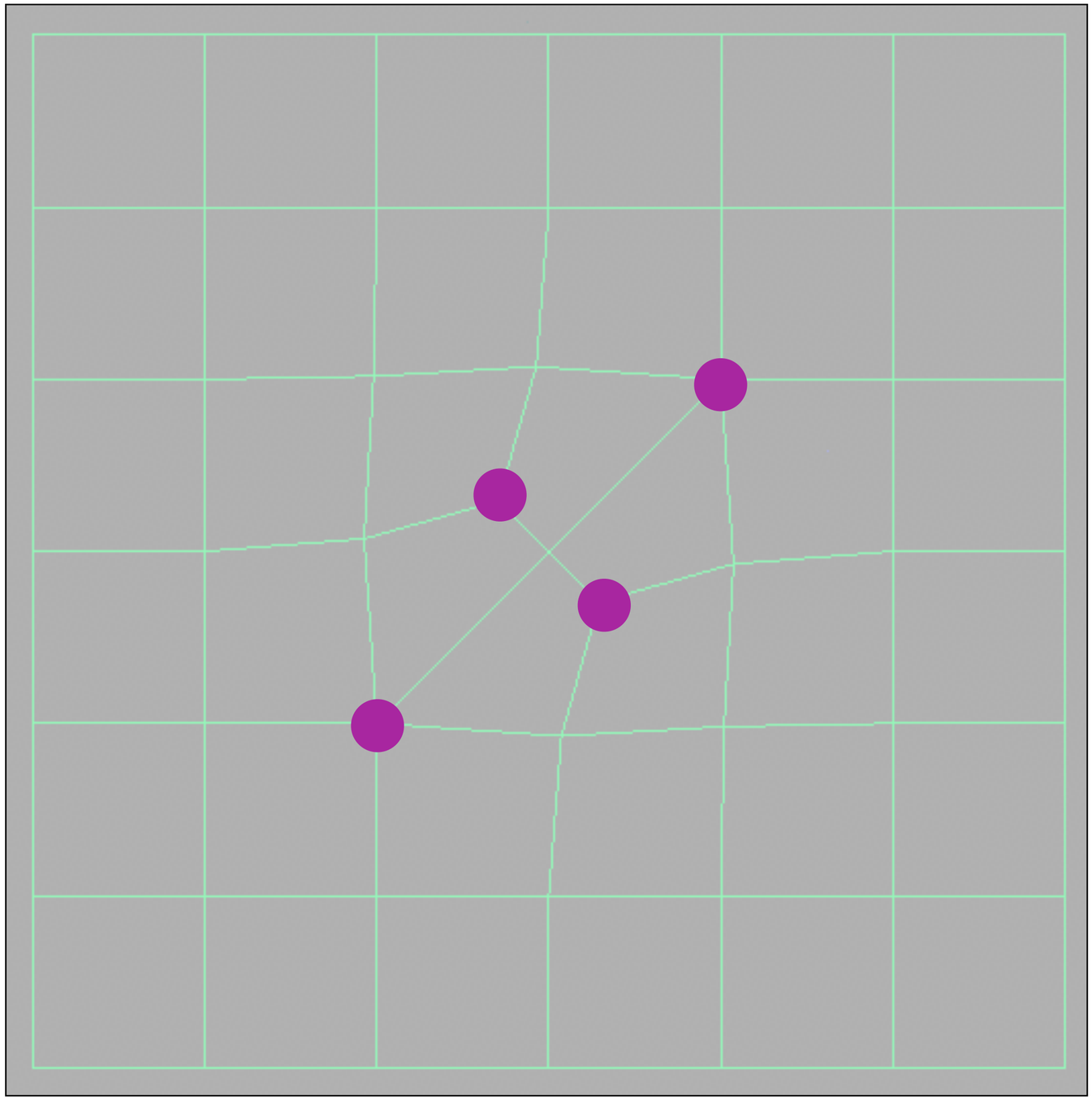
- Add vertex in each face
- Add midpoint on each edge
- Connect all new vertices

Catmull-Clark Subdivision (General Mesh)

Before

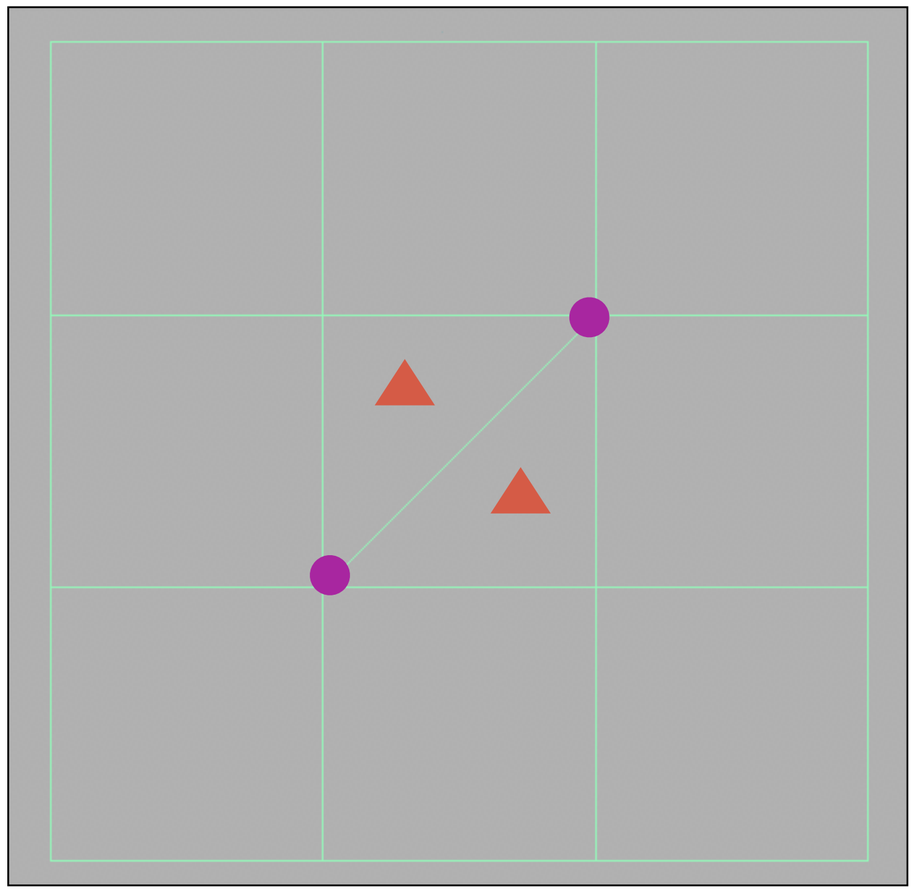


After

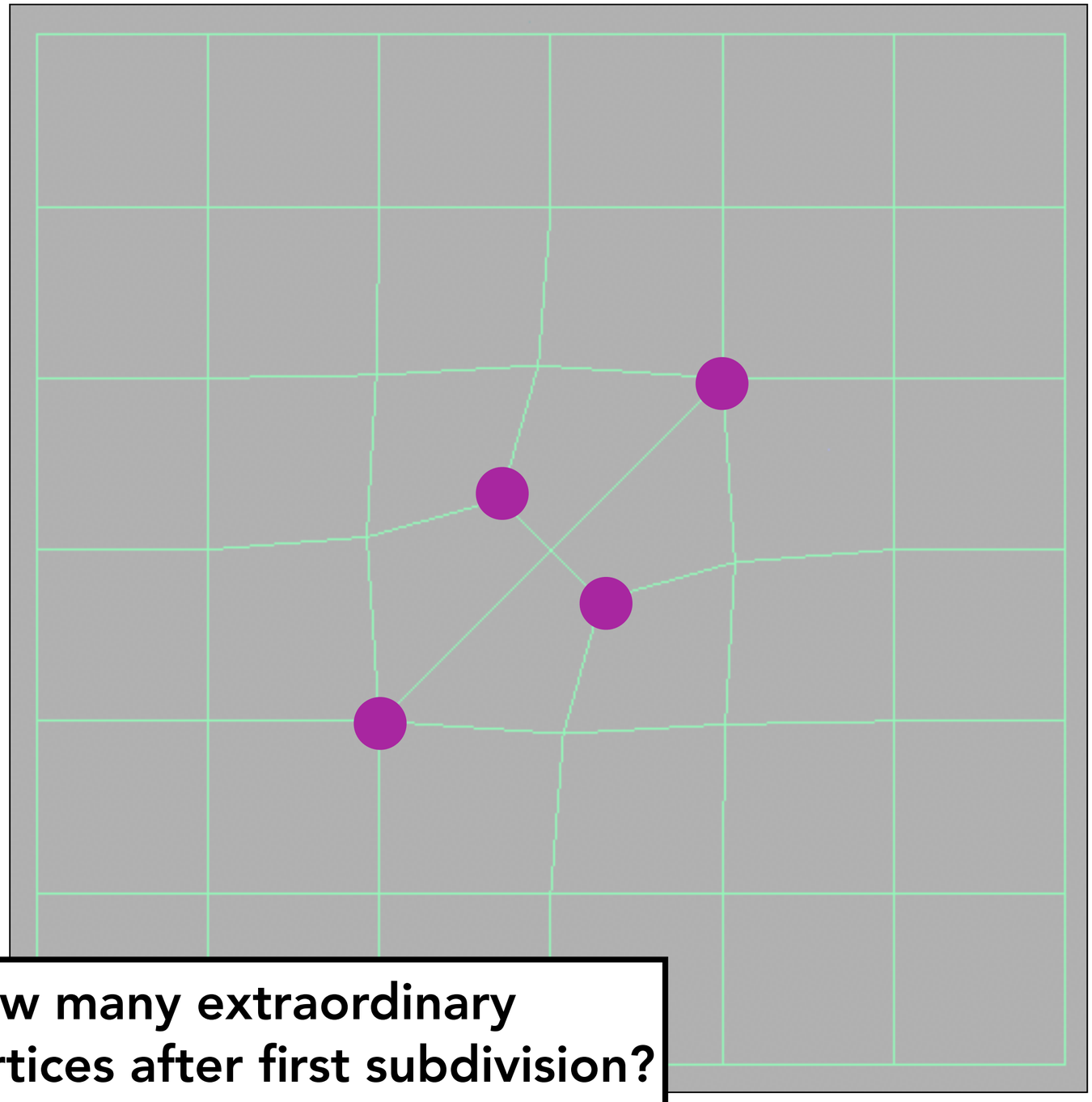


Catmull-Clark Subdivision (General Mesh)

Before



After

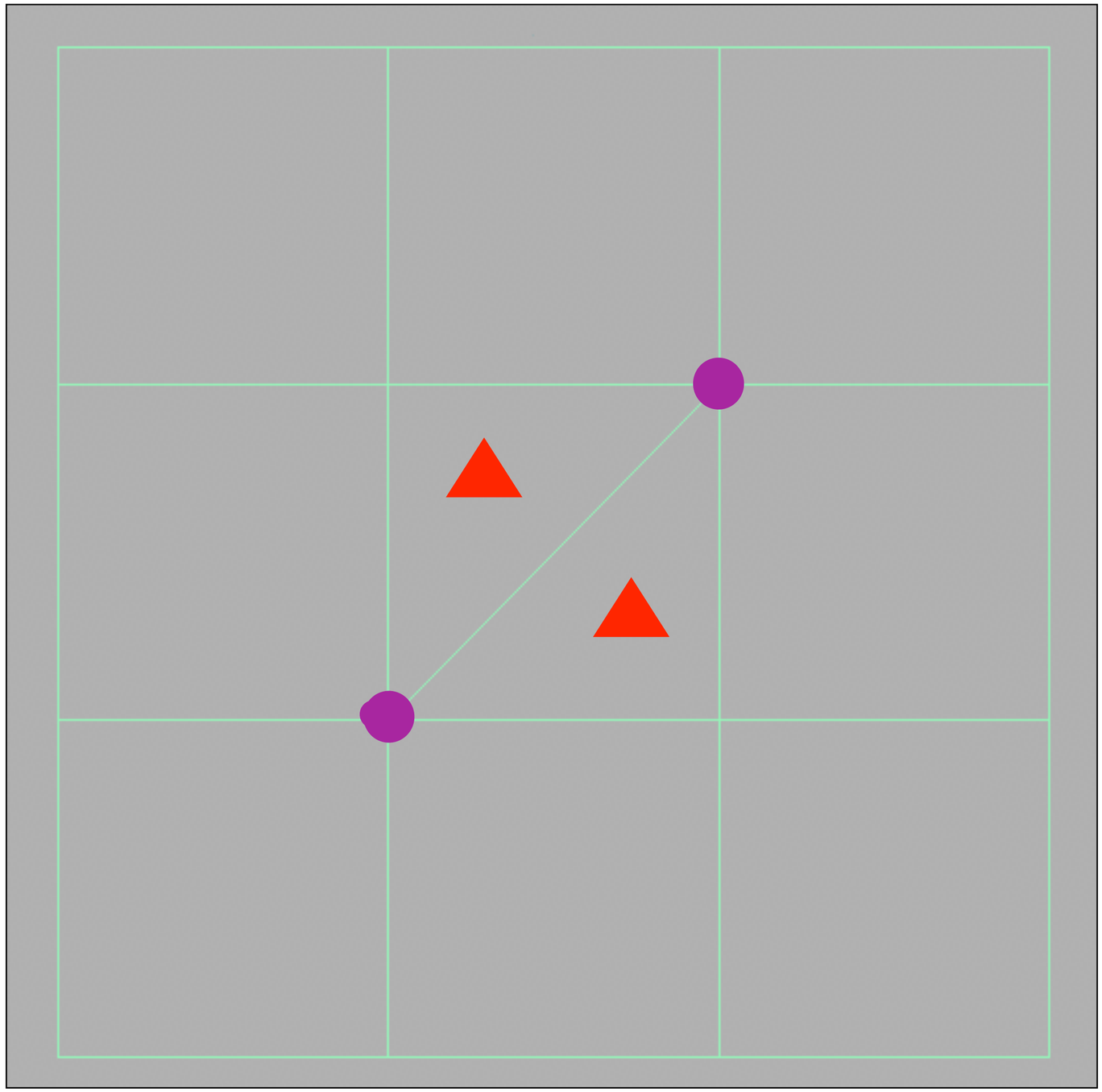


How many extraordinary vertices after first subdivision?

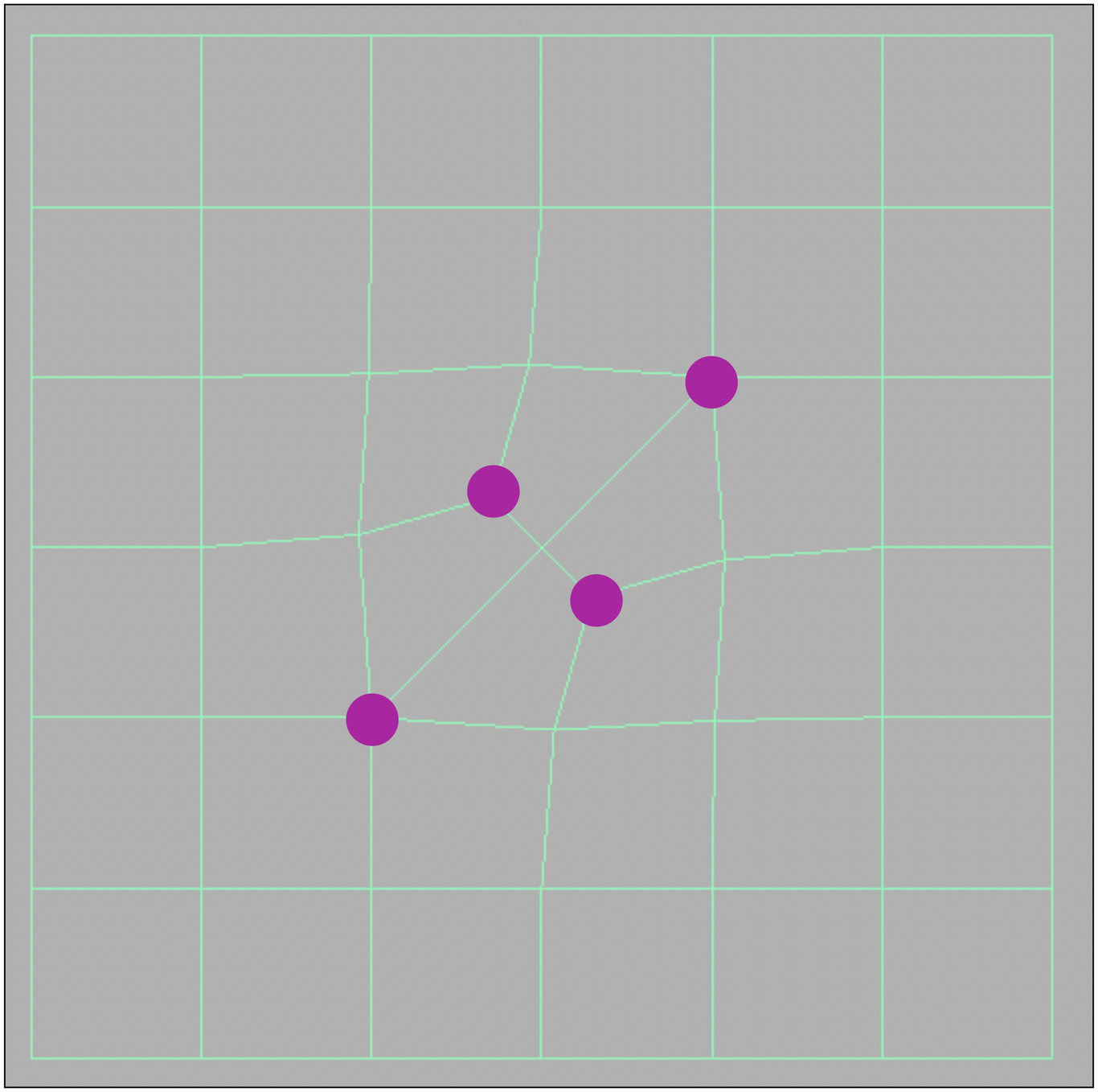
What are their valences?

How many non-quad faces?

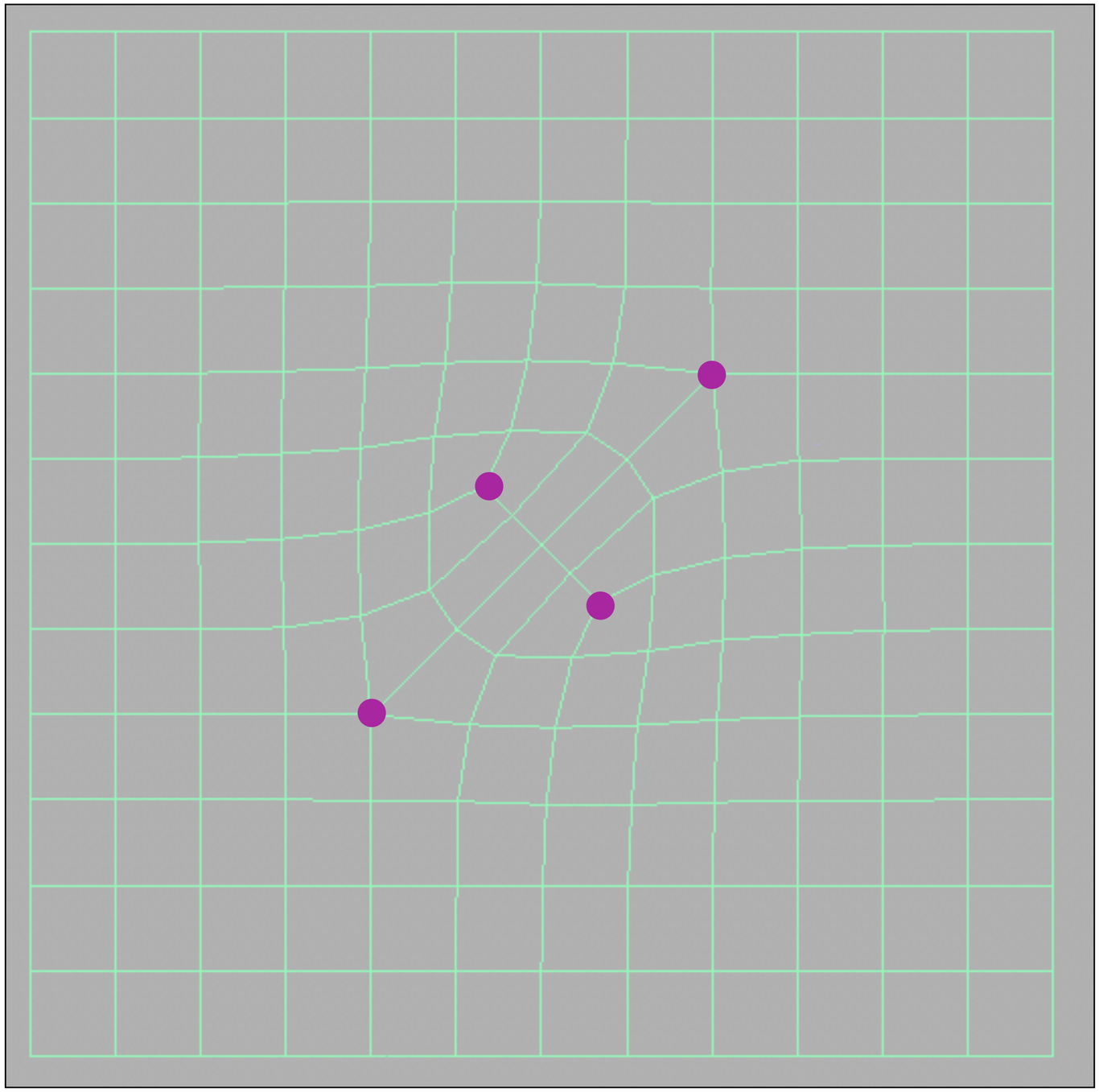
Catmull-Clark Subdivision (General Mesh)



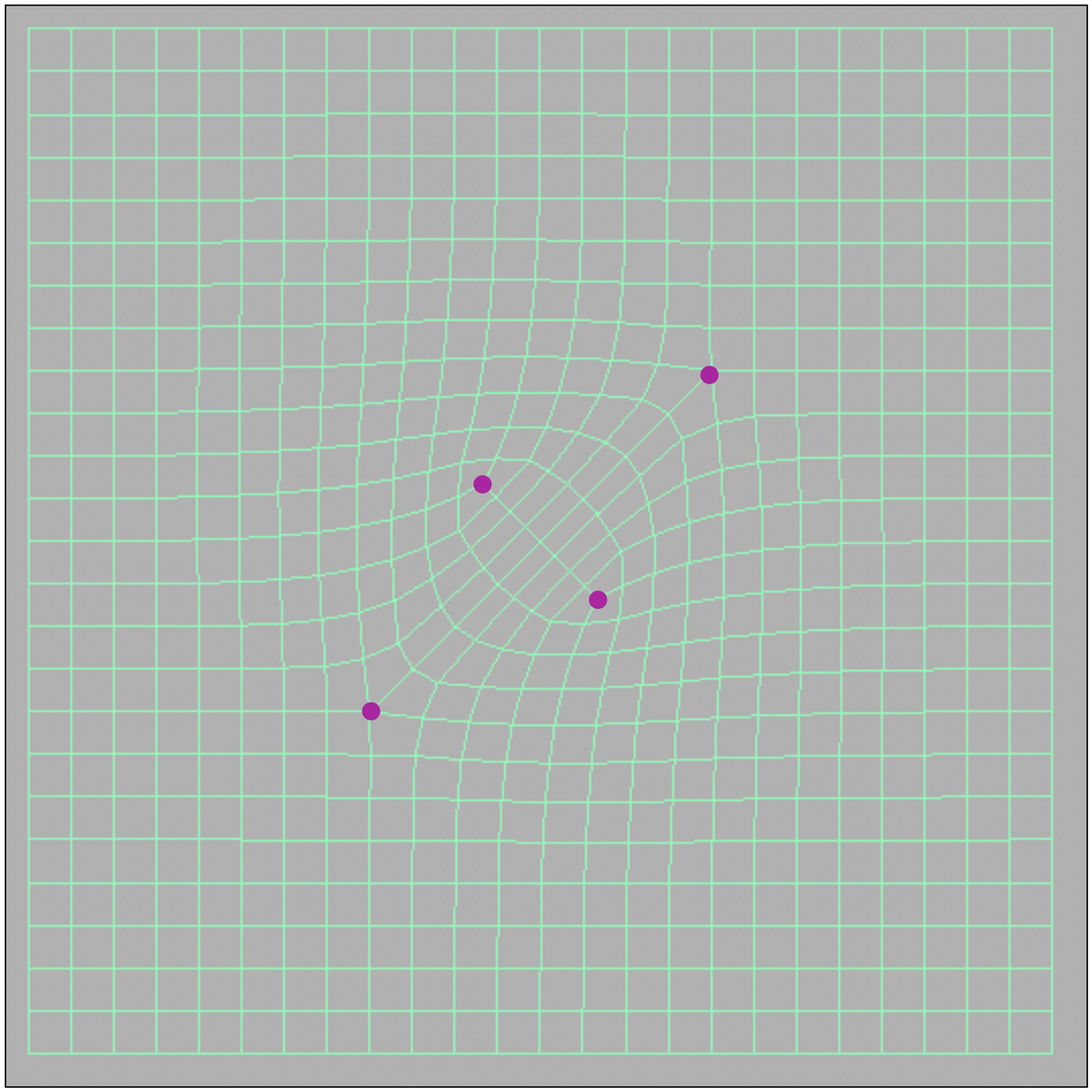
Catmull-Clark Subdivision (General Mesh)



Catmull-Clark Subdivision (General Mesh)



Catmull-Clark Subdivision (General Mesh)



Catmull-Clark Vertex Update Rules (General Mesh)

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$f =$ average of surrounding vertices

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$$e = \frac{f_1 + f_2 + v_1 + v_2}{4}$$

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$$f = \text{average of surrounding vertices}$$

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$$v = \frac{\bar{f}}{n} + \frac{2\bar{m}}{n} + \frac{p(n-3)}{n}$$

$\bar{m} =$ average of adjacent midpoints
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$n =$ valence of vertex
$p =$ old "vertex" point

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$\bar{m} =$ average of adjacent midpoints
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$n =$ valence of vertex
$p =$ old "vertex" point

These rules reduce to earlier quad rules for ordinary vertices / faces

Continuity of Catmull-Clark Surface

At extraordinary points

- Surface is at least C^1 continuous

Everywhere else (“ordinary” regions)

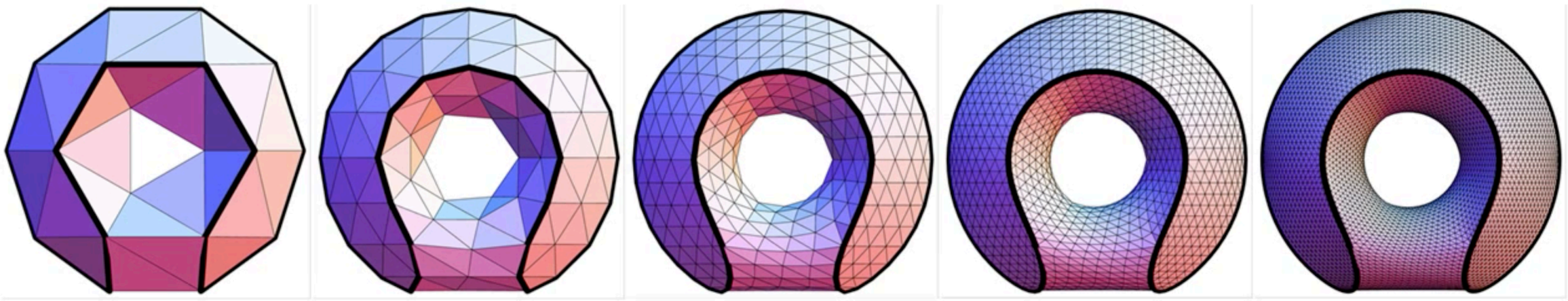
- Surface is C^2 continuous

What About Sharp Creases?

From Pixar Short, “Geri’s Game”
Hand is modeled as a Catmull Clark surface with creases between skin and fingernail

What About Sharp Creases?

Loop with Sharp Creases



Catmull-Clark with Sharp Creases

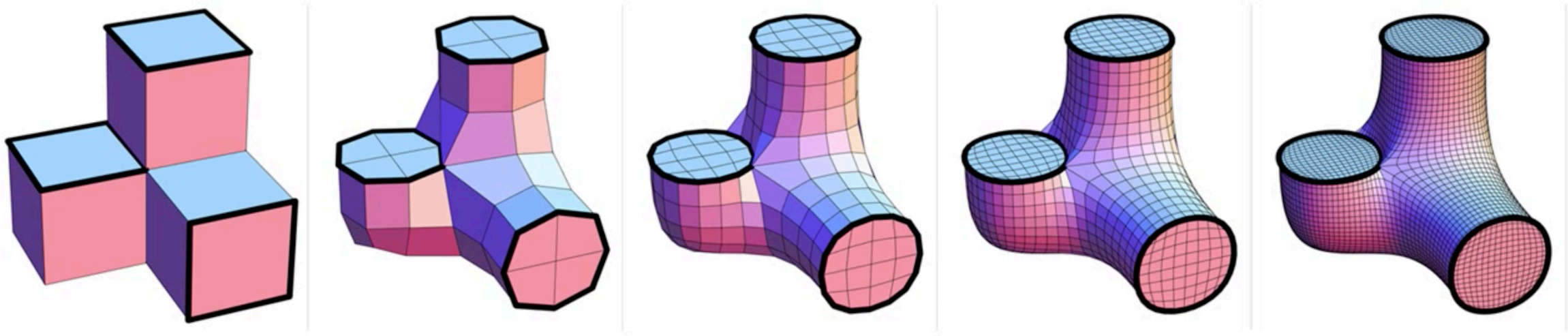
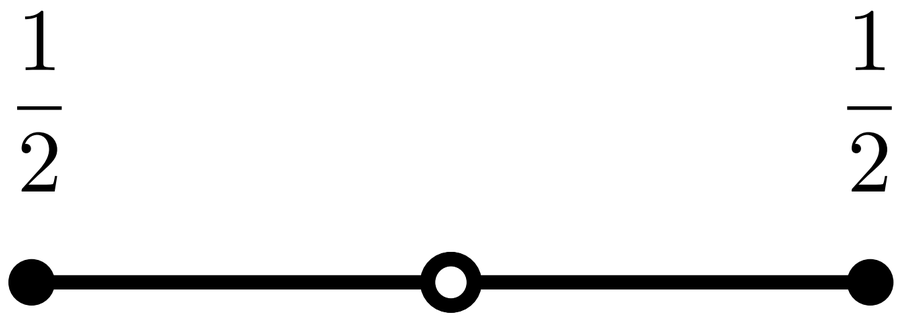


Figure from: Hakenberg et al. Volume Enclosed by Subdivision Surfaces with Sharp Creases

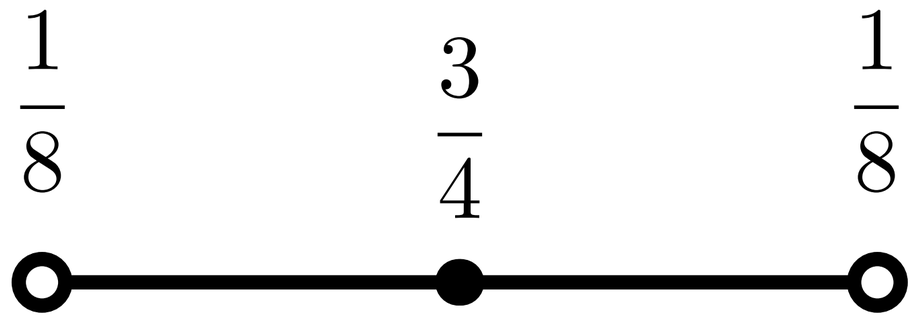
Creases + Boundaries

Can create creases in subdivision surfaces by marking certain edges as “sharp”. Boundary edges can be handled the same way

- Use different subdivision rules for vertices along these “sharp” edges



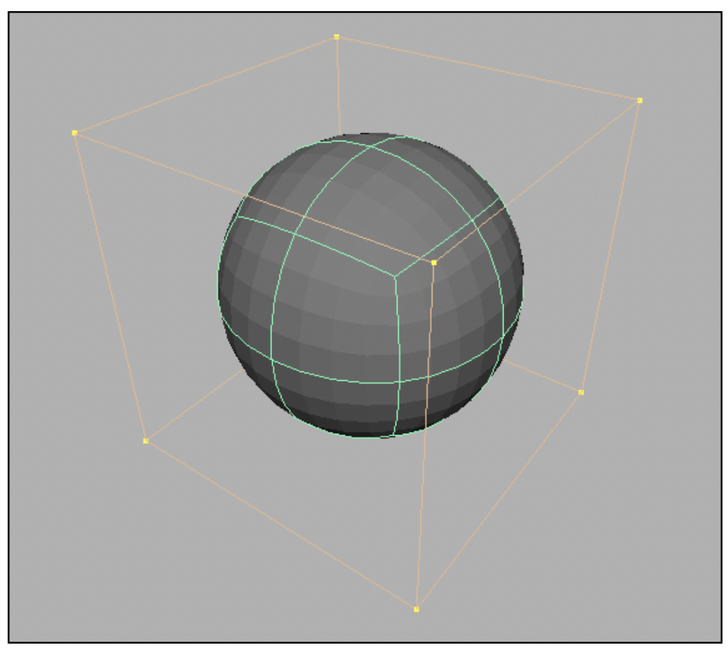
Insert new midpoint vertex, weights as shown

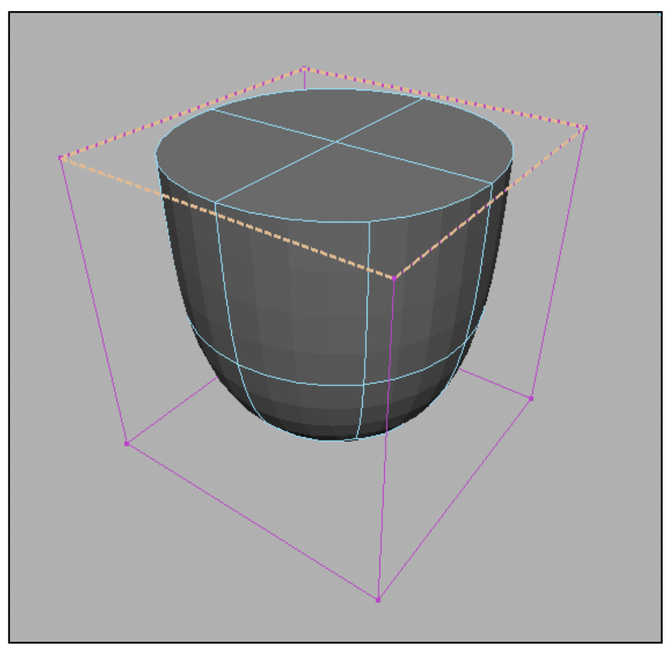


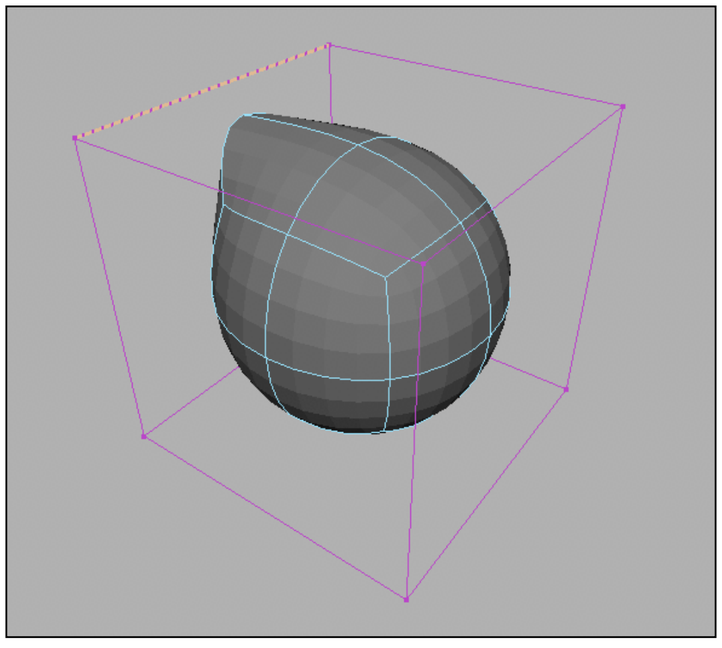
Update existing vertices, weights as shown

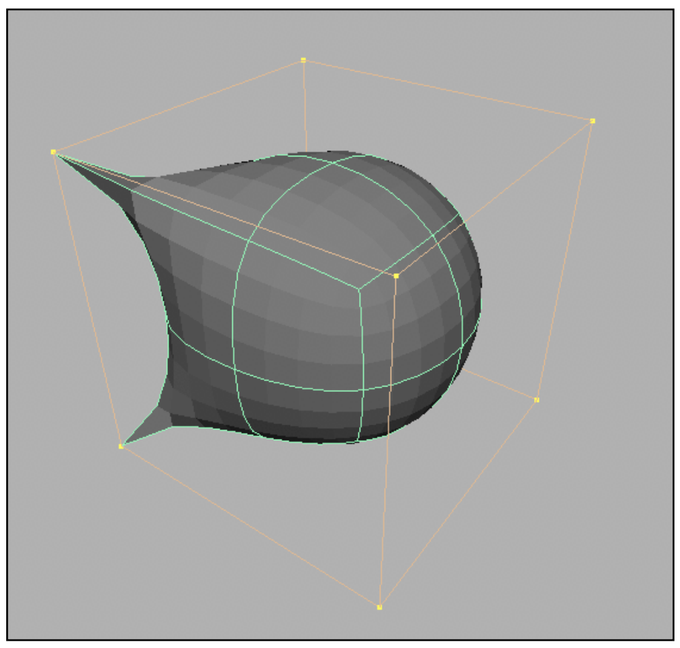
What About Sharp Creases?

Sharp vertices, edges, and faces





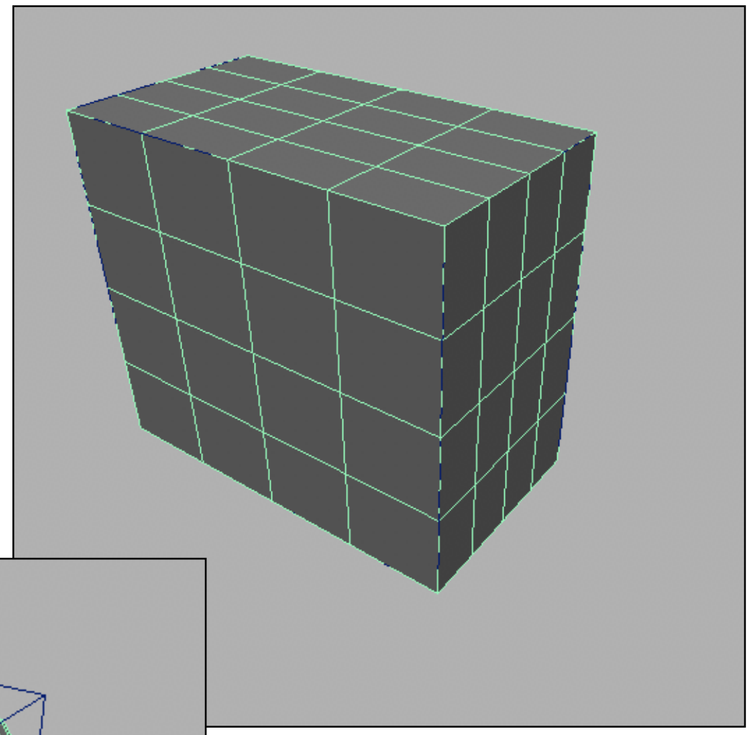


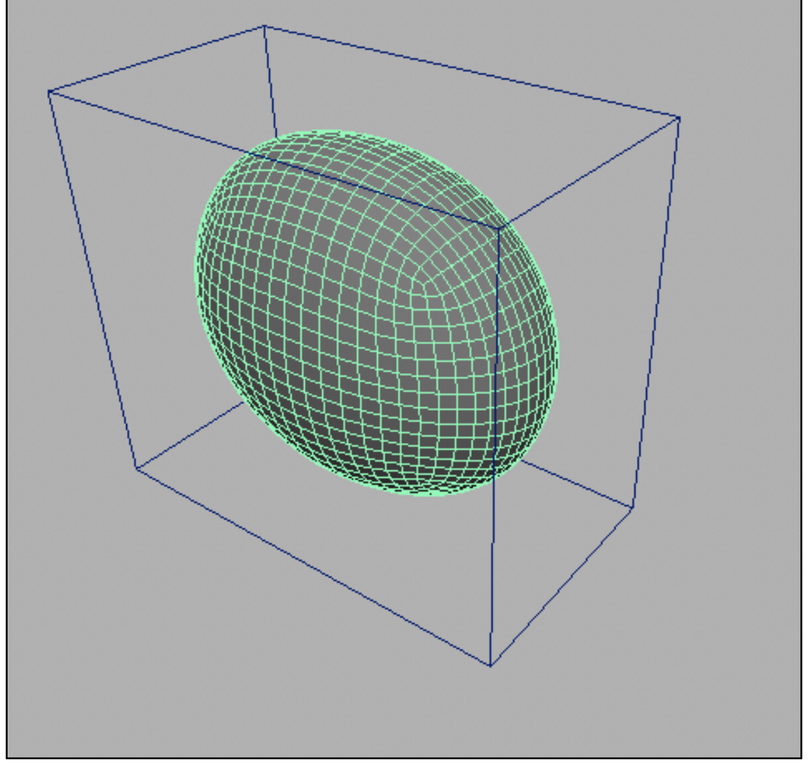


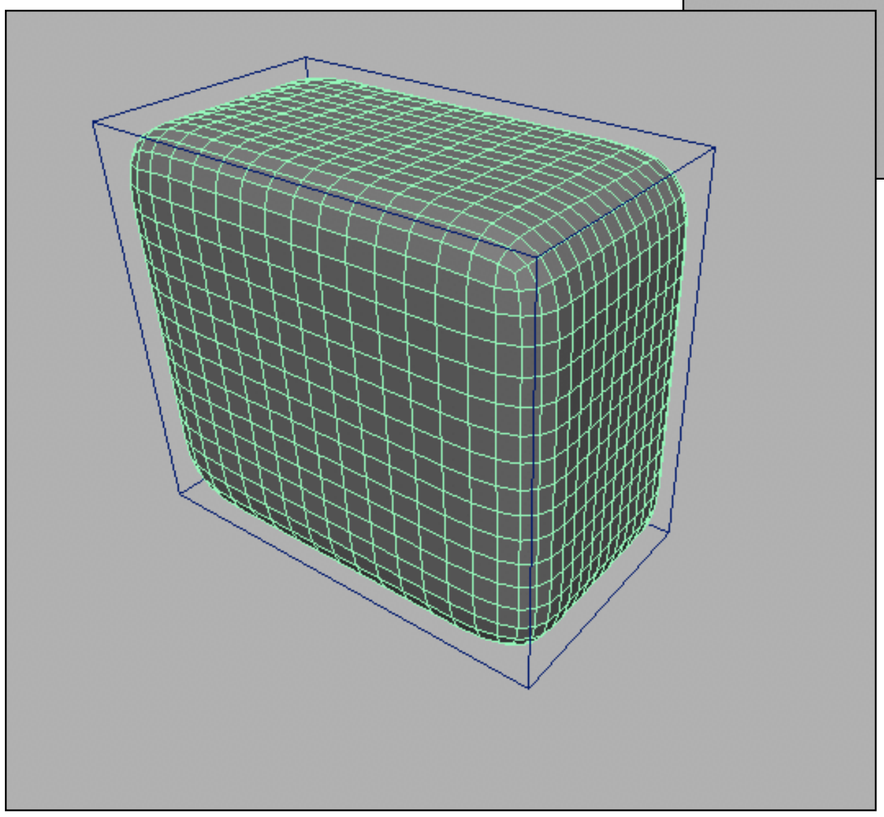
What About Controlled Smoothing?

Partial sharpness

Use sharp rule for N levels, then use smooth rules.







Subdivision in Action ("Geri's Game", Pixar)

Subdivision used for entire character:

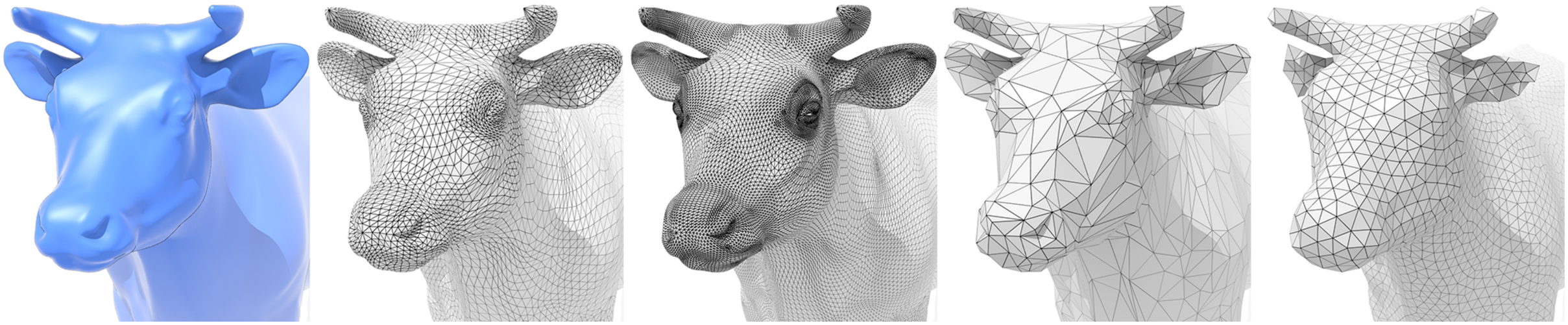
- Hands and head
- Clothing, tie, shoes

Subdivision in Action (Pixar's "Geri's Game")

Subdivision in Action (Pixar's "Geri's Game")

Global Mesh Operations: Geometry Processing

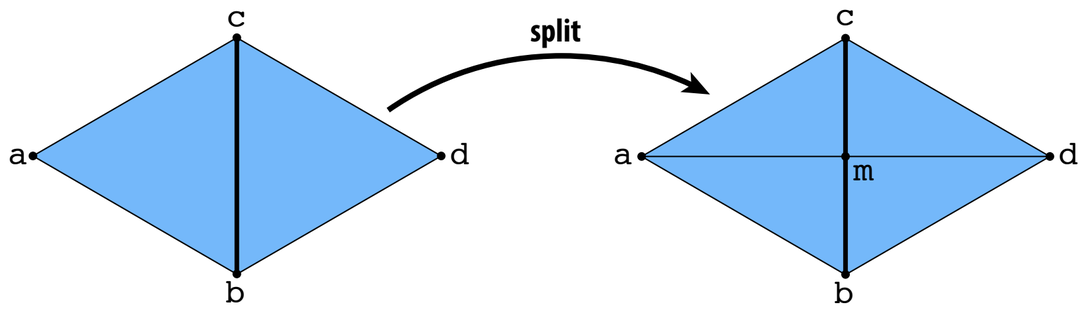
- Mesh subdivision
- Mesh simplification
- Mesh regularization



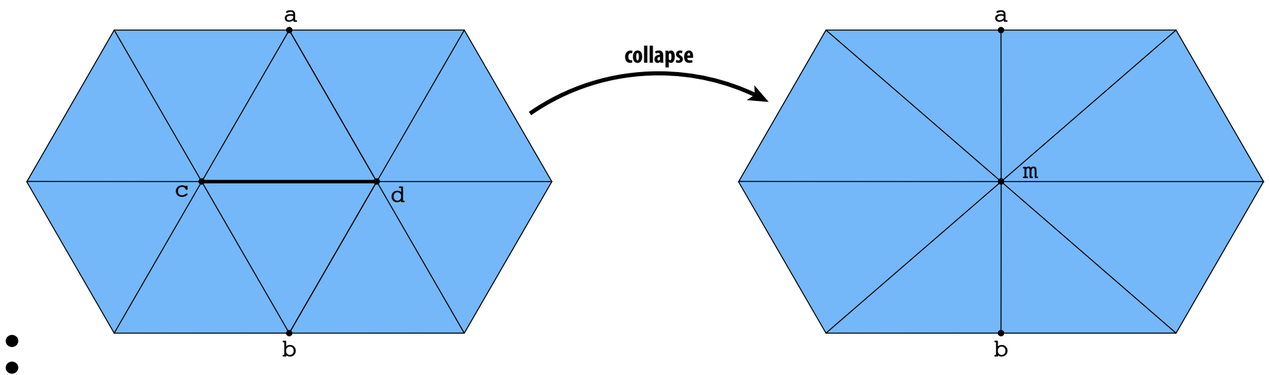
Mesh Simplification

How Do We Resample Meshes? (Reminder)

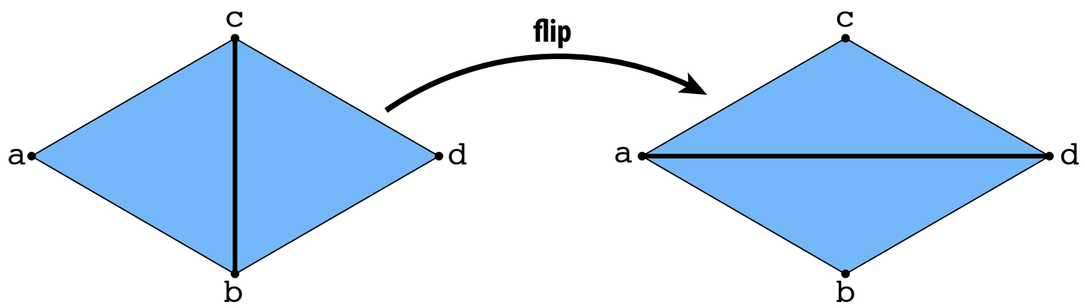
Edge split is (local) upsampling:



Edge collapse is (local) downsampling:



Edge flip is (local) resampling:



Still need to intelligently decide which edges to modify!

Mesh Simplification

Goal: reduce number of mesh elements while maintaining overall shape

30,000 triangles

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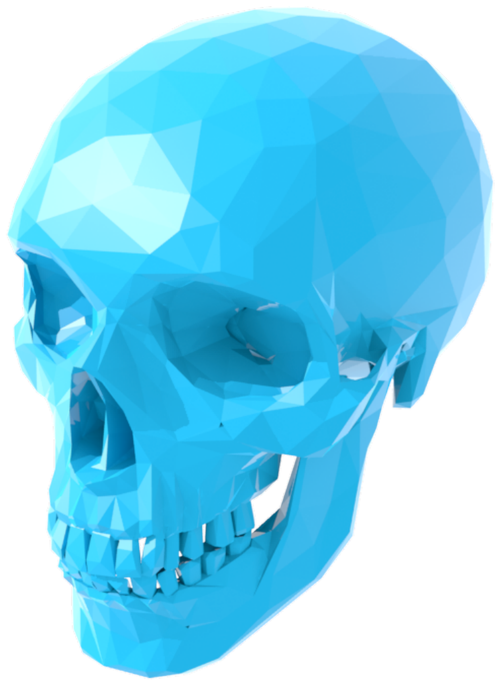
Goal: reduce number of mesh elements while maintaining overall shape

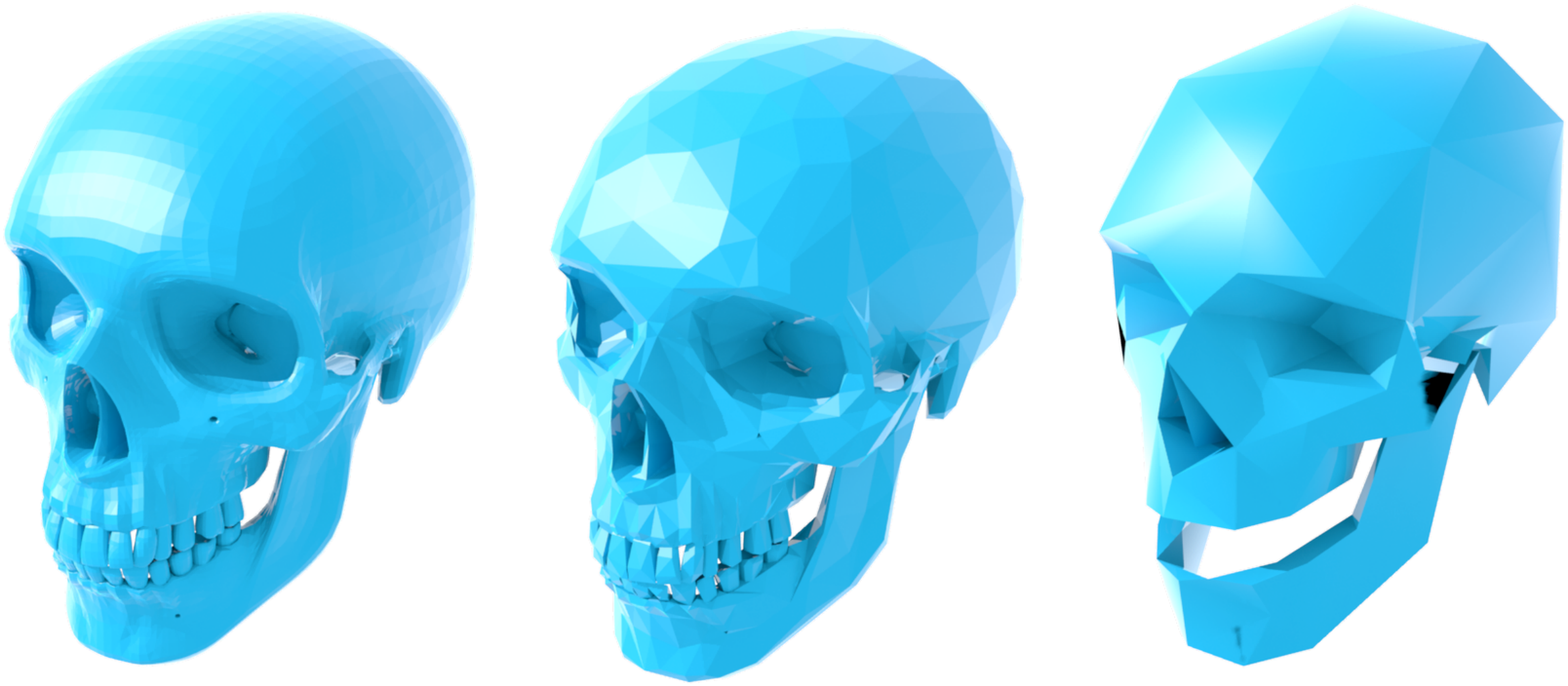
30,000 triangles



3,000

Mesh Simplification

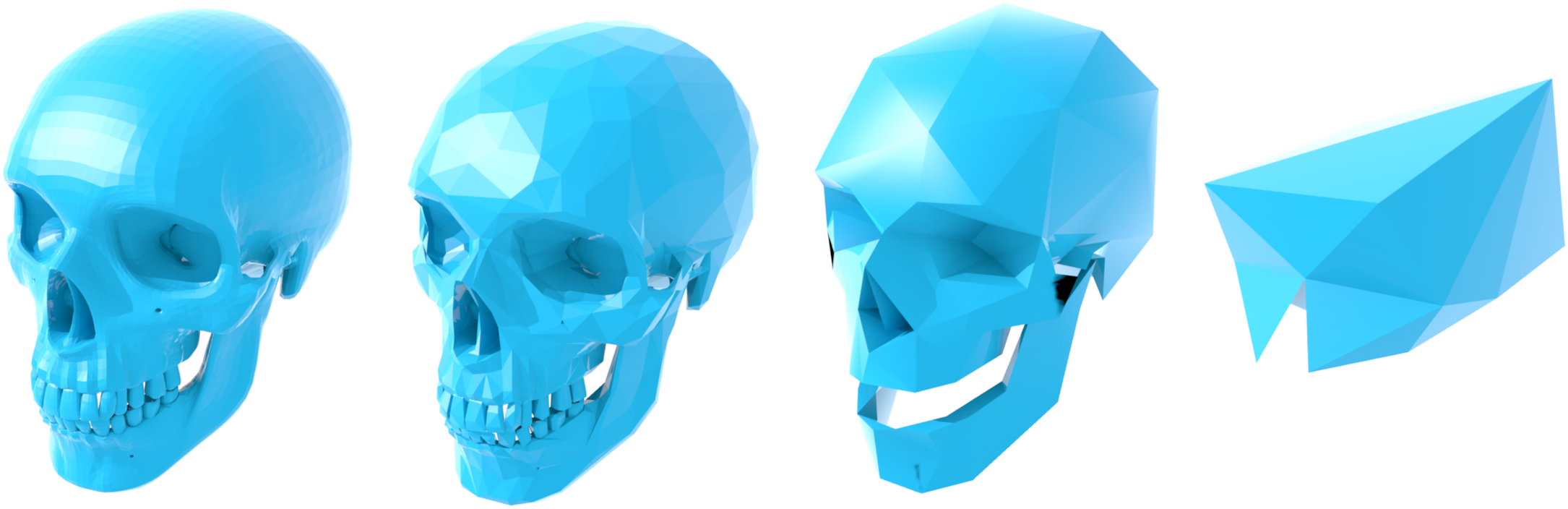
Goal: reduce number of mesh elements while maintaining overall shape



30,000 triangles **3,000** **300**

Mesh Simplification

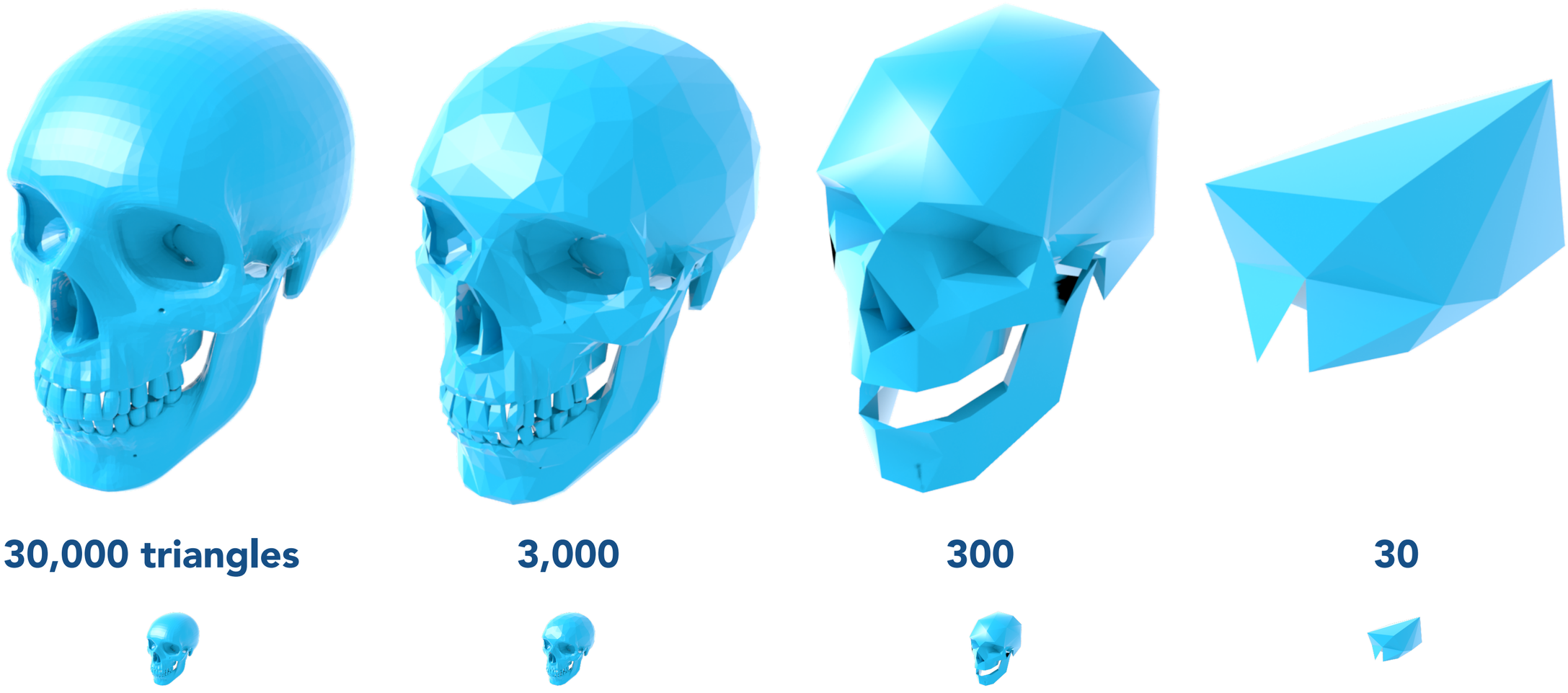
Goal: reduce number of mesh elements while maintaining overall shape



30,000 triangles **3,000** **300** **30**

Mesh Simplification

Goal: reduce number of mesh elements while maintaining overall shape



30,000 triangles | **3,000** | **300** | **30**

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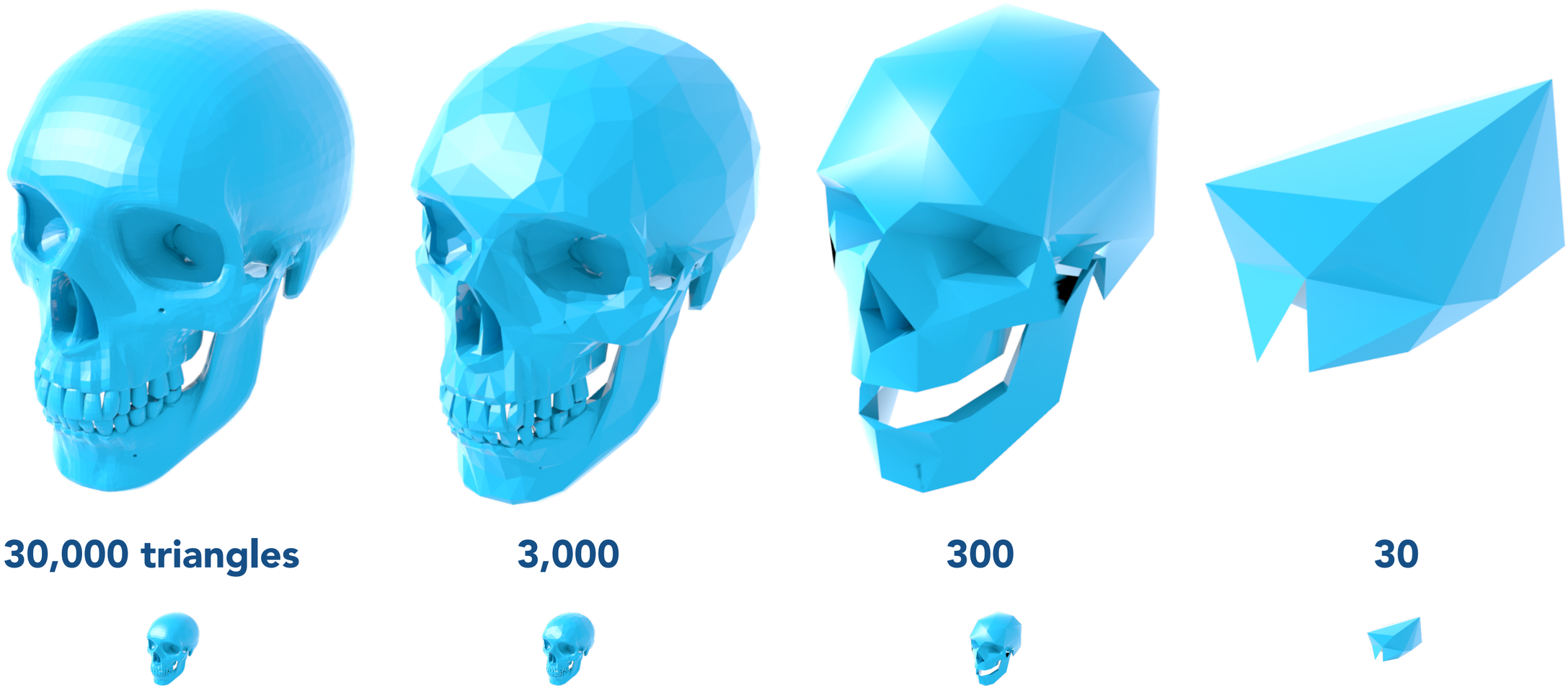
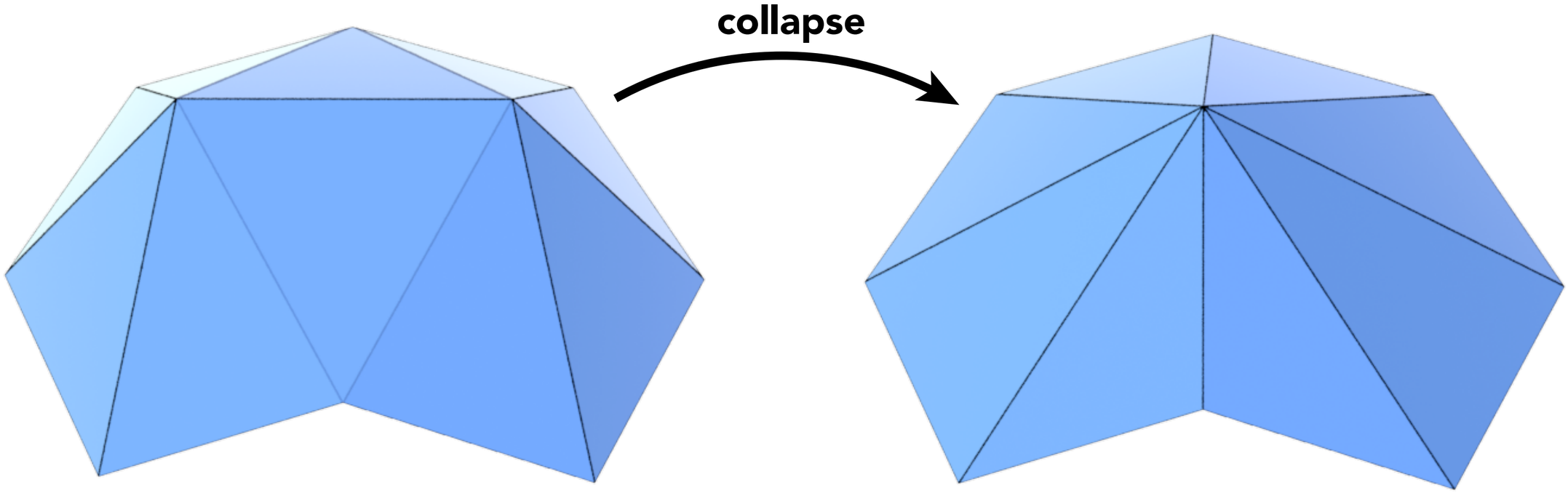


30,000 triangles | **3,000** | **300** | **30**

How to compute?

Estimate: Error Introduced by Collapsing An Edge?

- How much geometric error for collapsing an edge?



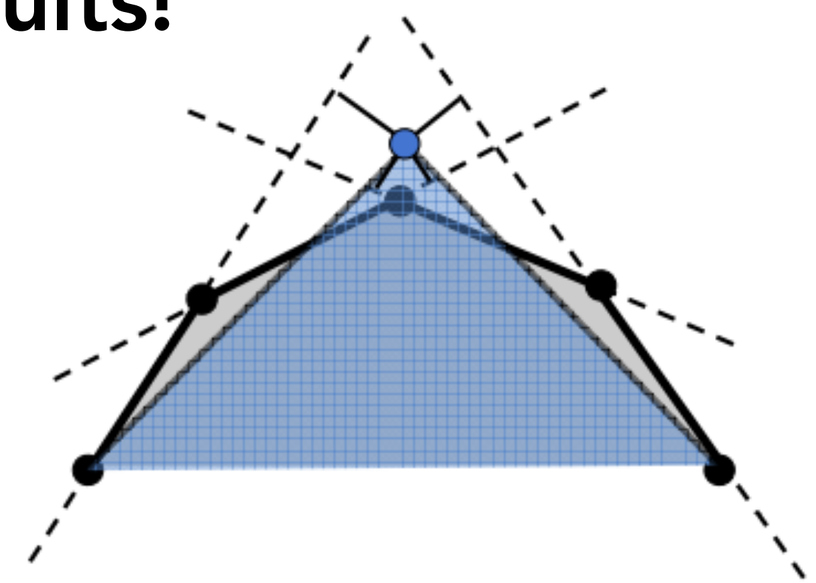
Sketch of Quadric Error Mesh Simplification

Simplification via Quadric Error

Iteratively collapse edges

Which edges? Assign score with quadric error metric*

- approximate distance to surface as sum of distances to planes containing triangles
- iteratively collapse edge with smallest score
- greedy algorithm... great results!



* (Garland & Heckbert 1997)

Quadric Error Matrix

Key idea:

- 4x4 (“quadric”) symmetric matrix encodes distance to plane

For plane ax + by + cz + d = 0

- Distance of query point (x, y, z) from plane is given by u^TQu:
  - u := (x, y, z, 1)^T is the query point in homogeneous coordinates
  - And Q is a symmetric matrix as follows:

$$Q = \begin{bmatrix} a^2 & ab & ac & ad \\ ab & b^2 & bc & bd \\ ac & bc & c^2 & cd \\ ad & bd & cd & d^2 \end{bmatrix}$$

  - Q contains 10 unique coefficients (small storage)

Quadric Error Matrix: Derivation

- Suppose in coordinates we have
  - a query point (x,y,z)
  - a normal (a,b,c)
  - an offset d := $-(x_p,y_p,z_p) \bullet (a,b,c)$

$$Q = \begin{bmatrix} a^2 & ab & ac & ad \\ ab & b^2 & bc & bd \\ ac & bc & c^2 & cd \\ ad & bd & cd & d^2 \end{bmatrix}$$

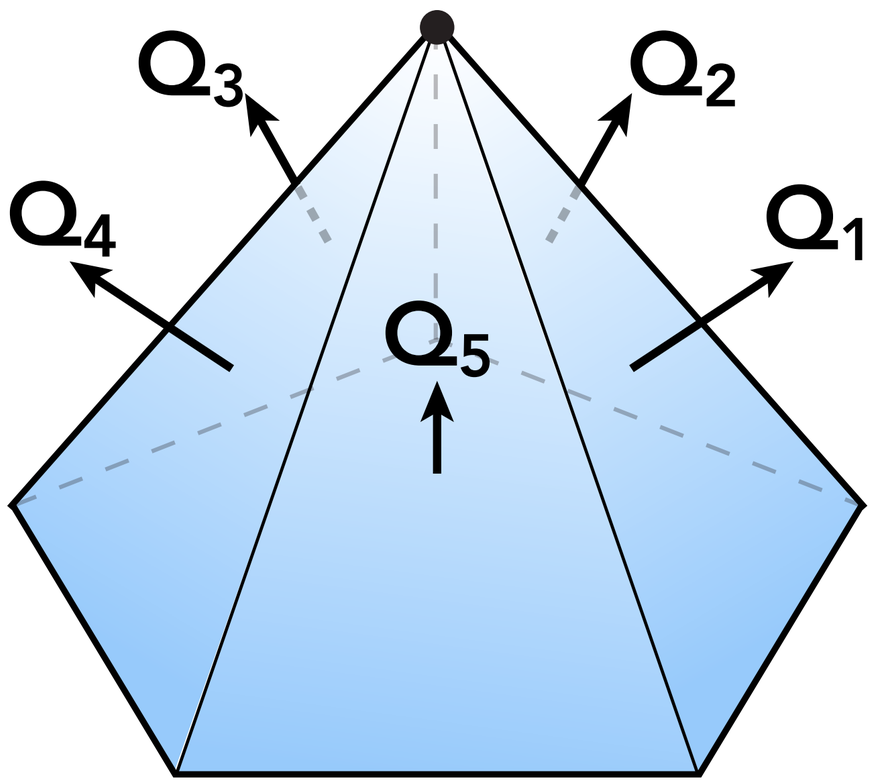
- Then in homogeneous coordinates, let
  - u := (x,y,z,1)
  - v := (a,b,c,d)
- Signed distance to plane is then
  $D = uv^T = vu^T = ax+by+cz+d$
- Squared distance is $D^2 = (uv^T)(vu^T) = u\,(v^Tv)\,u^T := u^TQu$

Quadric Error At Vertex

Approximate distance to vertex's triangles as sum of distances to each triangle's plane.

Encode this as a single quadric matrix for the vertex that is the sum of quadric error matrices for all triangles

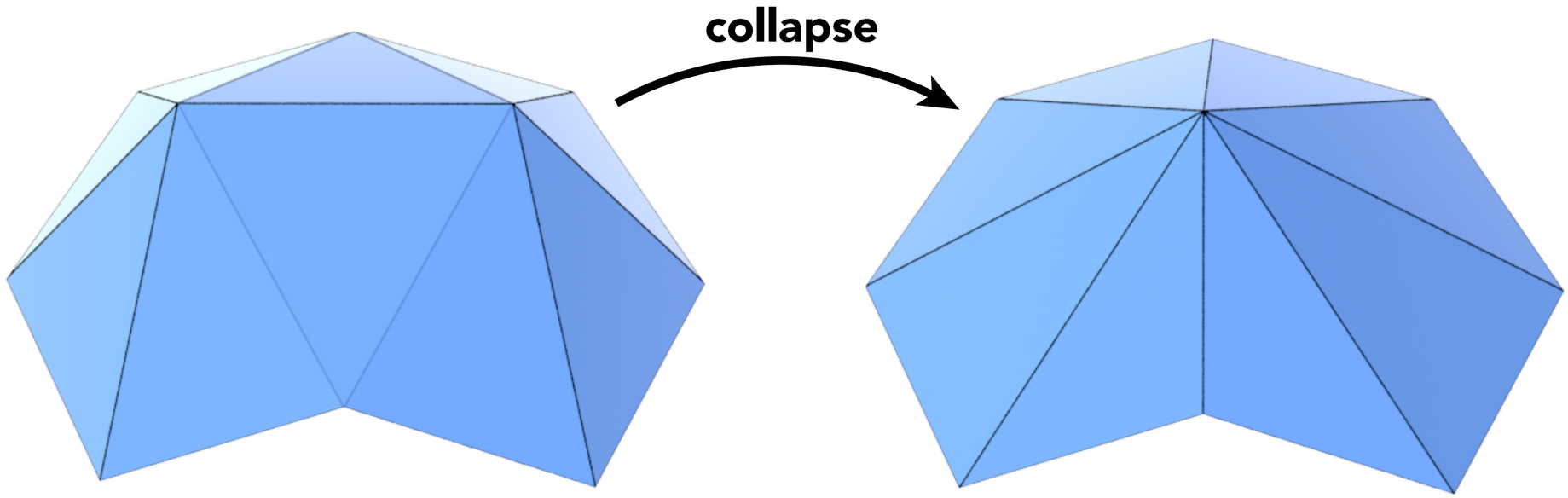
Q_V



$$Q_V = \sum_{i=1}^{N} Q_i$$

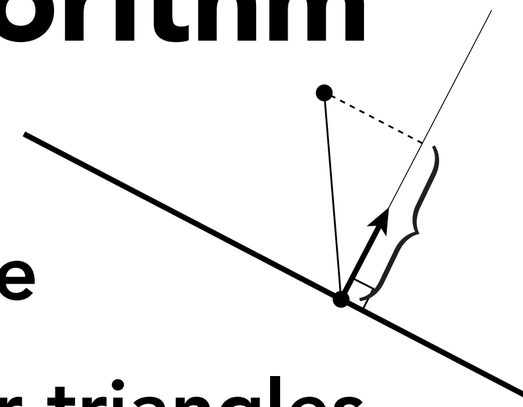
Quadric Error of Edge Collapse

- How much does it cost to collapse an edge?
- Idea: compute edge midpoint, measure quadric error

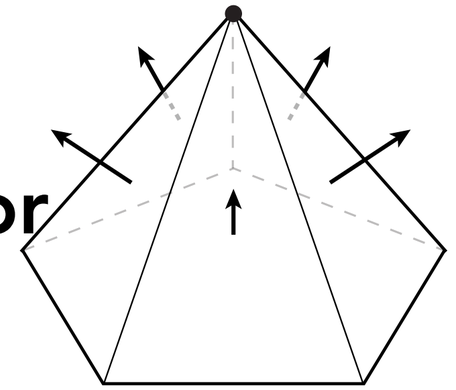


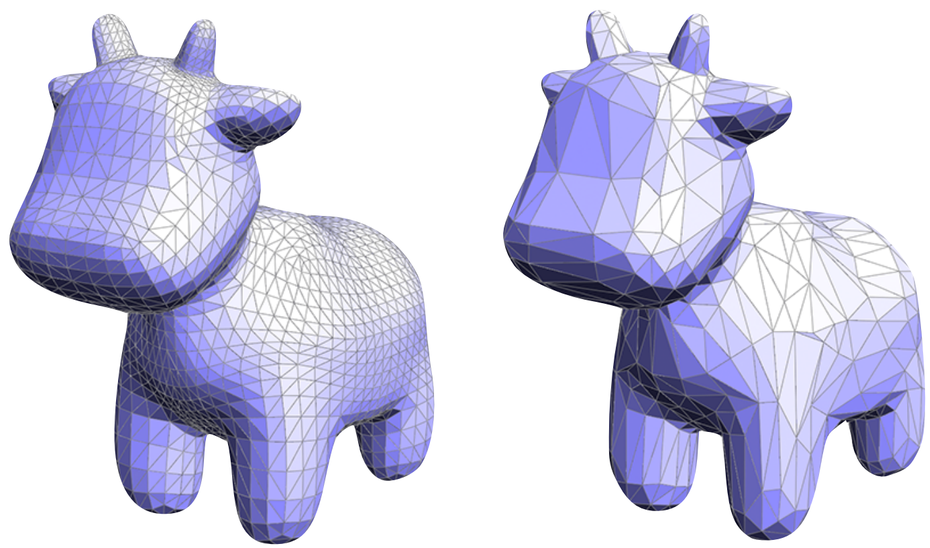
- Better idea: choose point that minimizes quadric error
- More details: Garland & Heckbert 1997.

Quadric Error Simplification: Algorithm

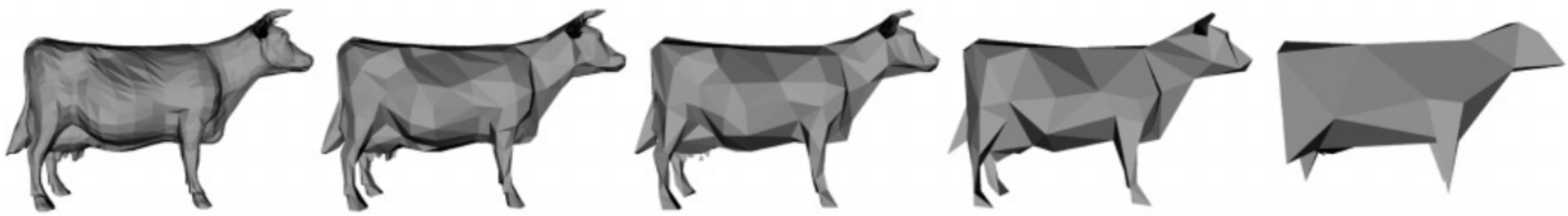


- Compute quadric error matrix Q for each triangle
- Set Q at each vertex to sum of Qs from neighbor triangles
- Set Q at each edge to sum of Qs at endpoints
- Find point at each edge minimizing quadric error
- Until we reach target # of triangles:
  - collapse edge (i,j) with smallest cost to get new vertex m
  - add Q_i and Q_j to get quadric Q_m at vertex m
  - update cost of edges touching vertex m

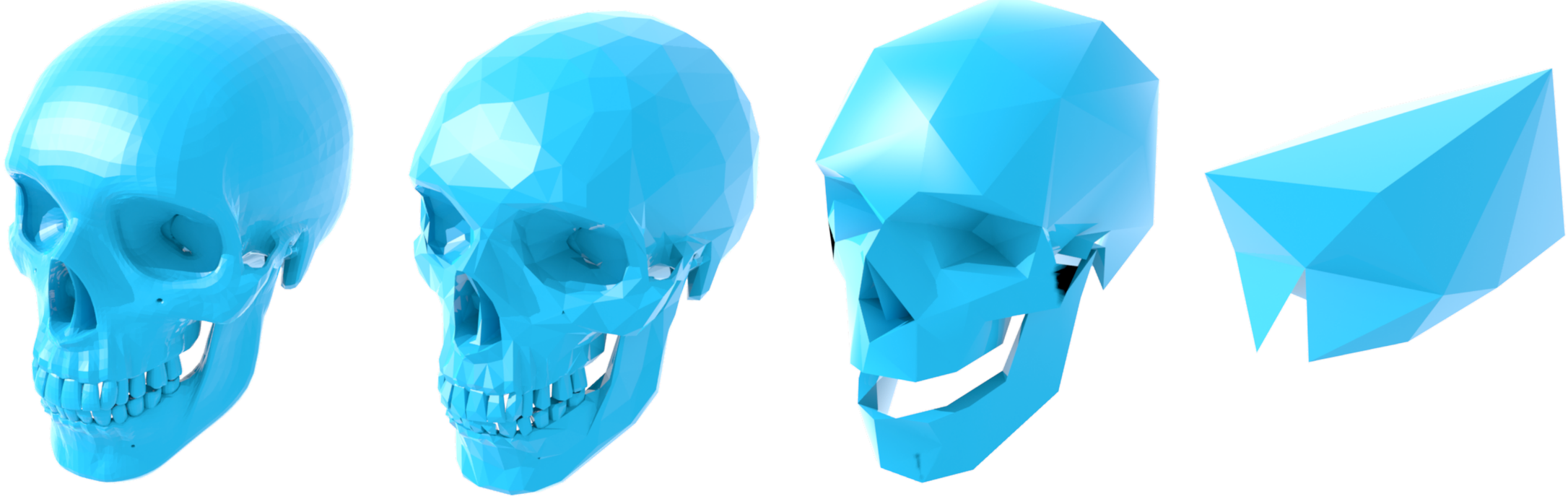




Quadric Error Mesh Simplification



5,804 994 532 248 64



30,000 triangles **3,000** **300** **30**

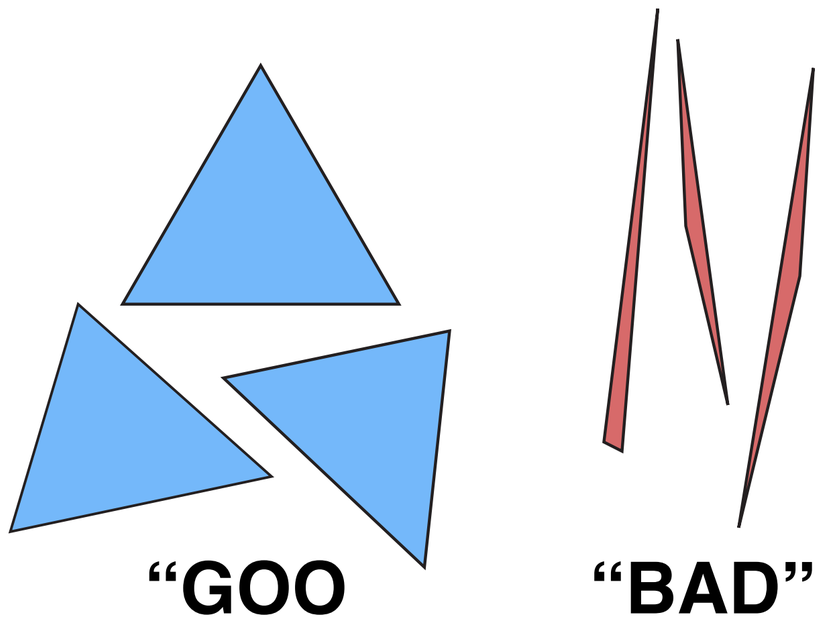
Garland and Heckbert '97

Mesh Regularization

What Makes a “Good” Triangle Mesh?

What Makes a “Good” Triangle Mesh?

One rule of thumb: triangle shape



What Makes a “Good” Triangle Mesh?

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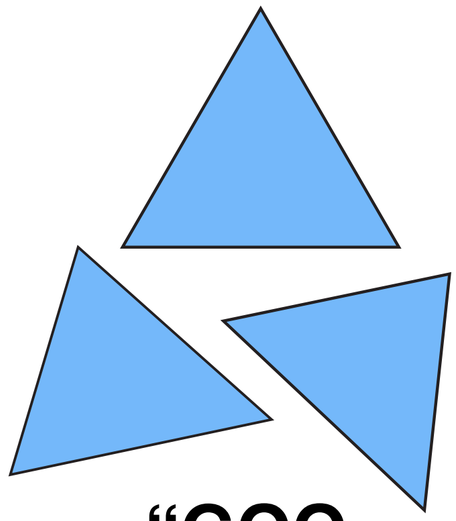
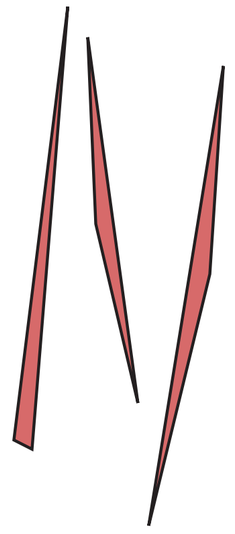


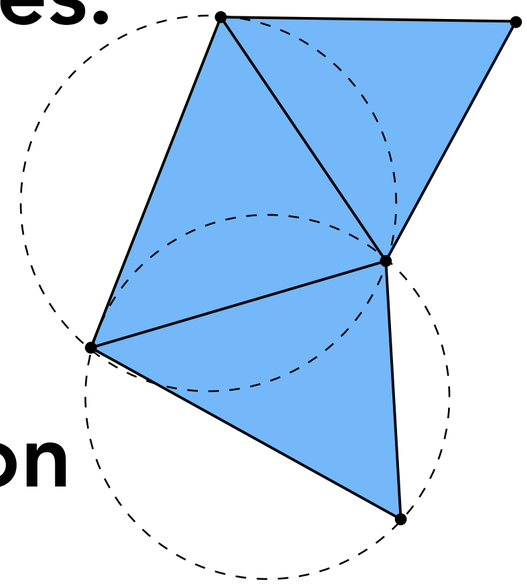
“GOO



“BAD”

More specific condition: Delaunay

- “Circumcircle interiors contain no vertices.”

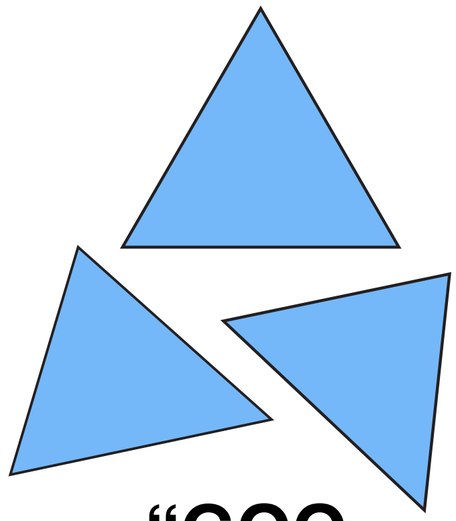


Not always a good condition, but often*

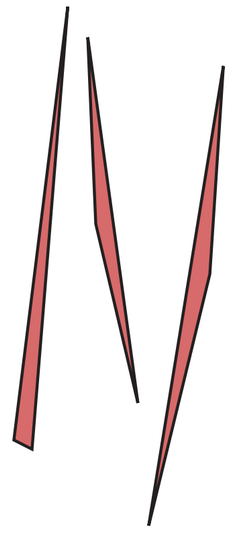
- Good for simulation
- Not always best for shape approximation

What Makes a “Good” Triangle Mesh?

One rule of thumb: triangle shape



“GOO



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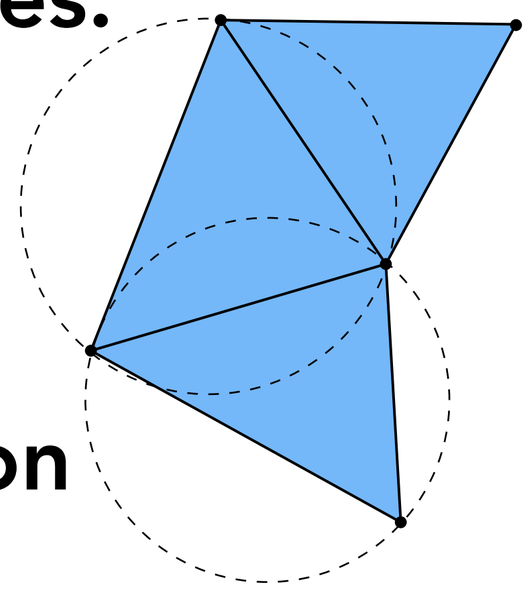
More specific condition: Delaunay

- “Circumcircle interiors contain no vertices.”



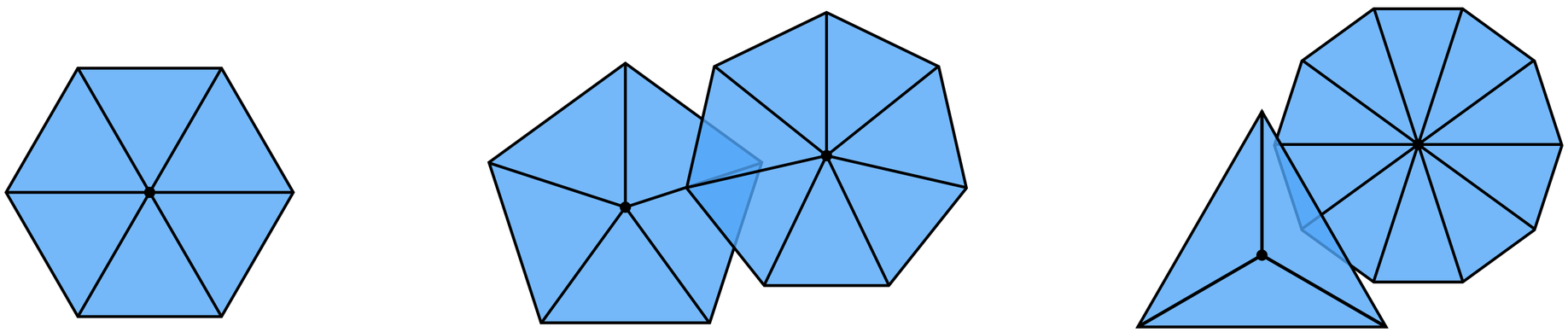
Not always a good condition, but often*

- Good for simulation
- Not always best for shape approximation

*See Shewchuk, “What is a Good Linear Element”

What Else Constitutes a Good Mesh?

Rule of thumb: regular vertex degree
Triangle meshes: ideal is every vertex with valence 6:



“GOOD “OK” “BAD”

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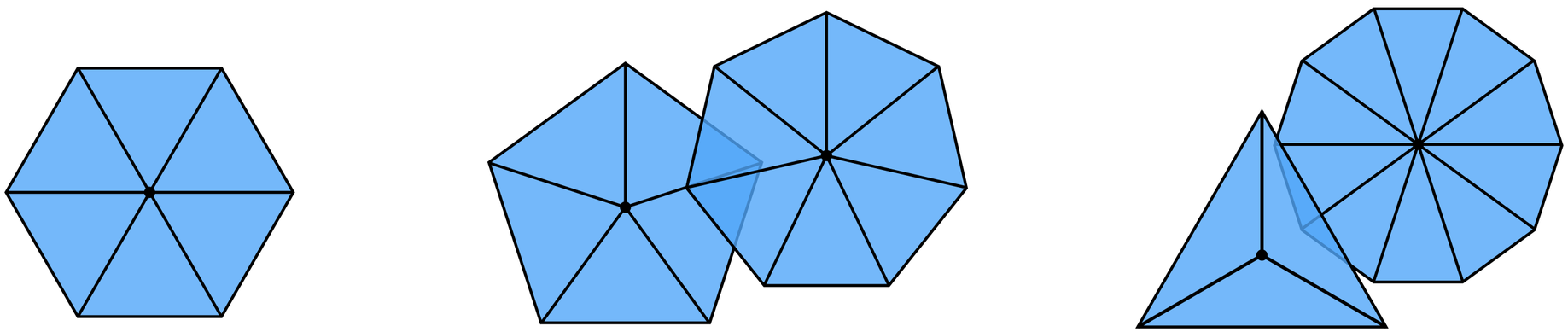
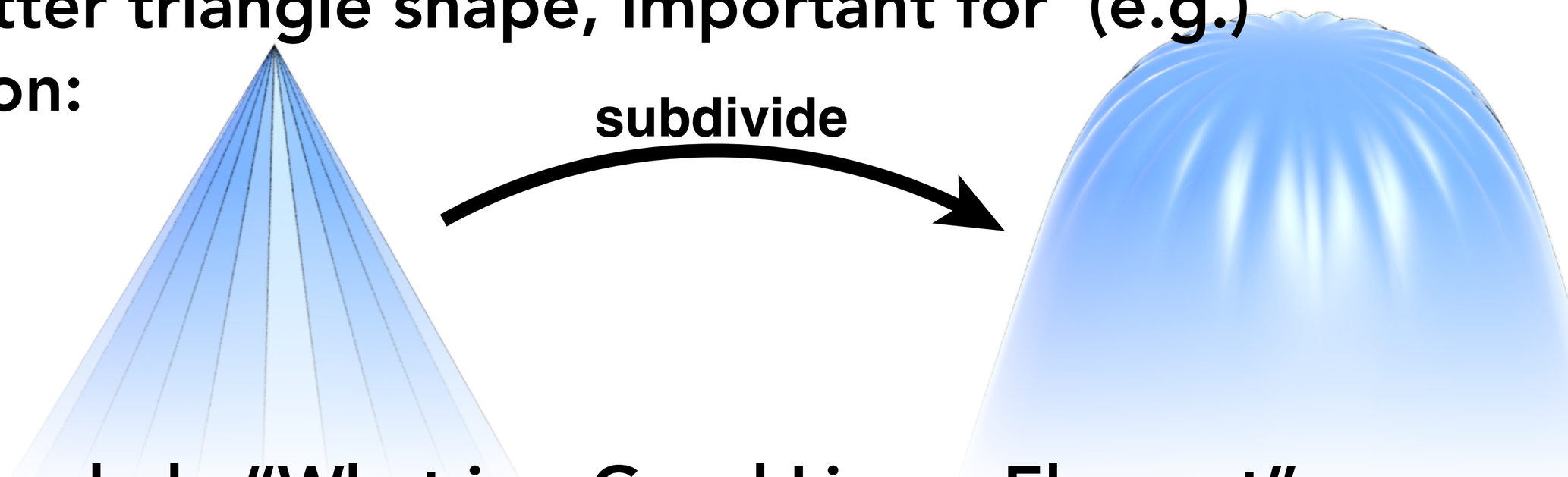


“GOOD “OK” “BAD”

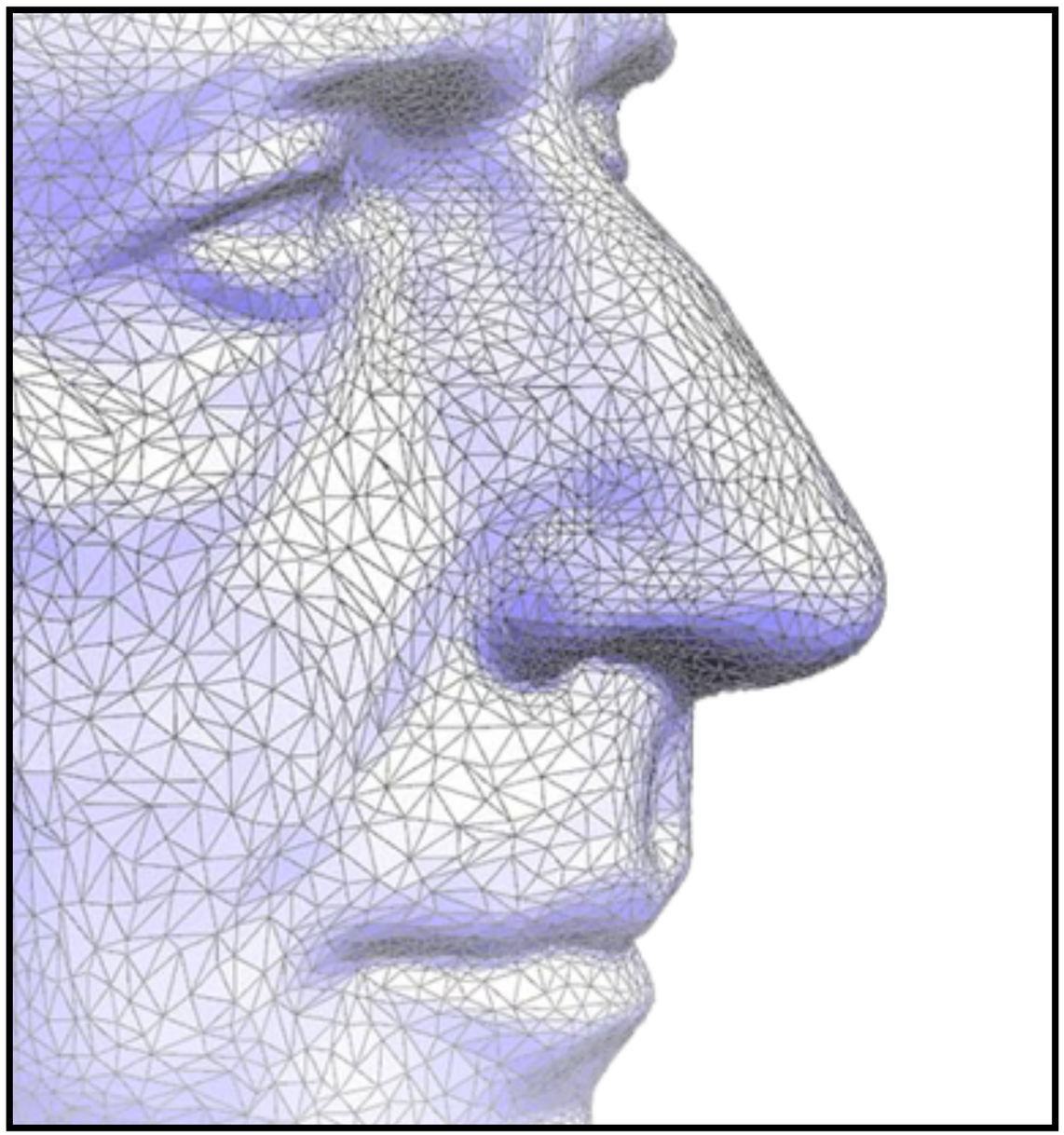
Why? Better triangle shape, important for (e.g.) subdivision:

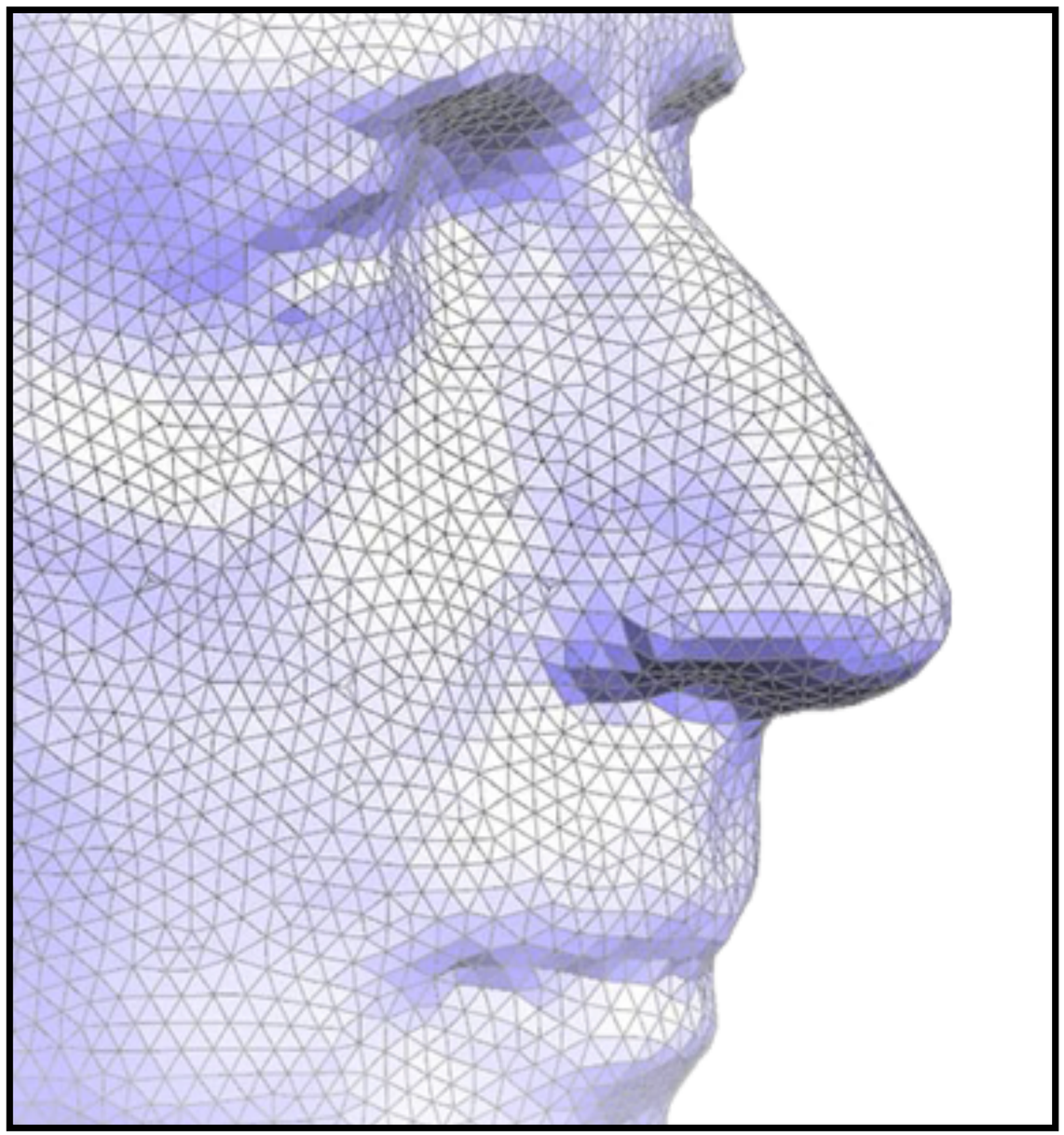


*See Shewchuk, “What is a Good Linear Element”

Isotropic Remeshing

Try to make triangles uniform in shape and size

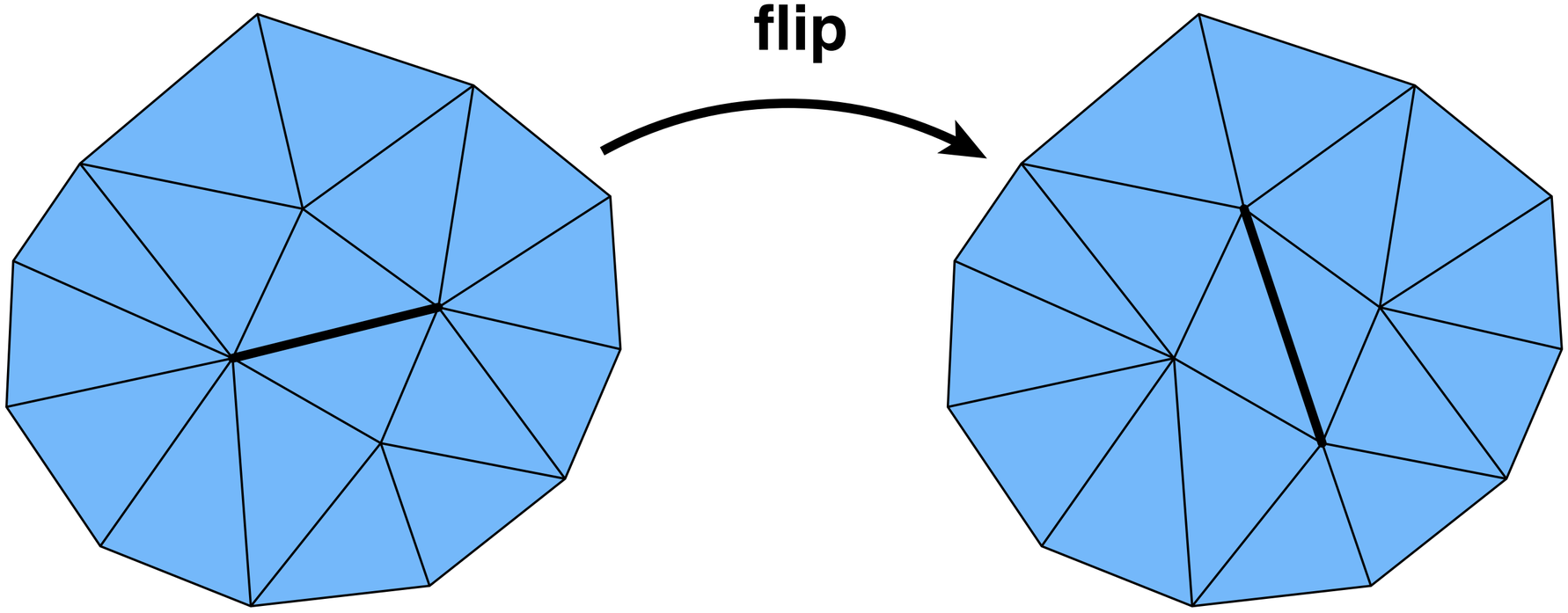




How Do We Improve Degree?

Edge flips!

If total deviation from degree 6 gets smaller, flip it!



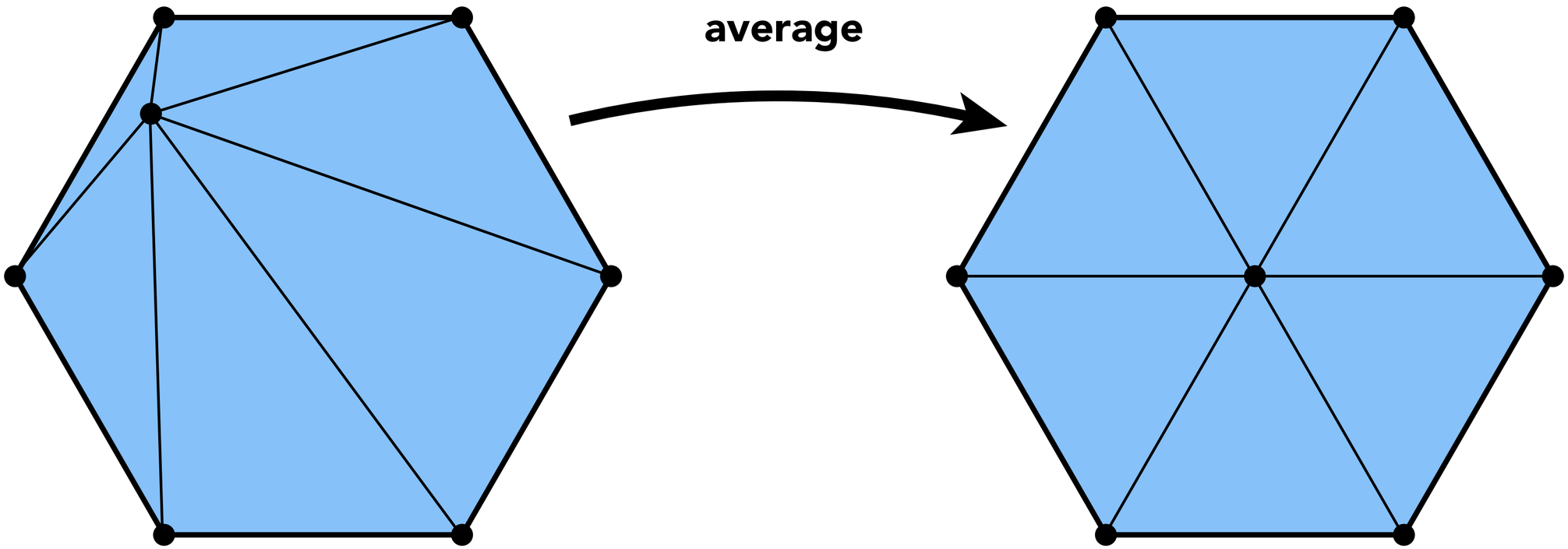
Iterative edge flipping acts like “discrete diffusion” of degree

No (known) guarantees; works well in practice

How Do We Make Triangles “More Round”?

Delaunay doesn’t mean equilateral triangles

Can often improve shape by centering vertices:

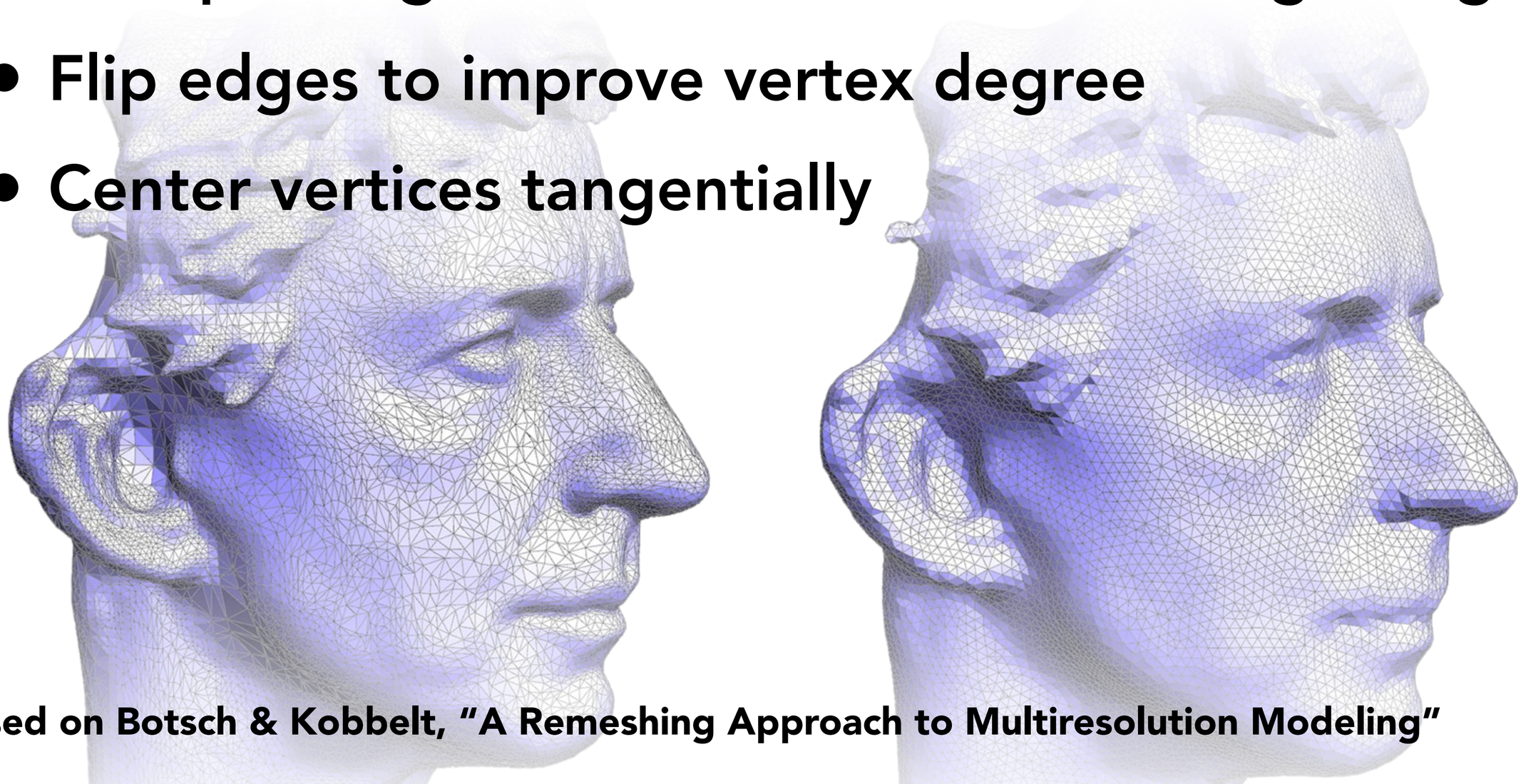


[Crane, “Digital Geometry Processing with Discrete Exterior Calculus”]

Isotropic Remeshing Algorithm*

Repeat four steps:

- Split edges over 4/3rds mean edge legth
- Collapse edges less than 4/5ths mean edge length
- Flip edges to improve vertex degree
- Center vertices tangentially



*Based on Botsch & Kobbelt, “A Remeshing Approach to Multiresolution Modeling”

Acknowledgments

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